AI VIETNAM
All-in-One Course
(TA Session)

# Big Data

## Extra

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Truong-Binh Duong – STA

Year

# Outline

- Introduction
- Hadoop
- Spark
- Question

# Getting Started

## ❖ Objectives

**In this session, we will discuss about:**

- Introduction to big data.
- Introduction to Hadoop.
- How to install virtual machine.
- How to install and use Hadoop on virtual machine.
- Introduction to spark and pyspark.

# Introduction

# Introduction

## ❖ Getting Started


| UNIT | VALUE |
|---|---|
| bit | 1 bit |
| byte (B) | 8 bits |
| kilobyte (KB) | 1024 bytes |
| megabyte (MB) | 1024 kilobytes |
| gigabyte (GB) | 1024 megabytes |
| terabyte (TB) | 1024 gigabytes |
| petabyte (PB) | 1024 terabytes |
| exabyte (EB) | 1024 petabytes |
| zettabyte (ZB) | 1024 exabytes |
| yottabyte (YB) | 1024 zettabytes |

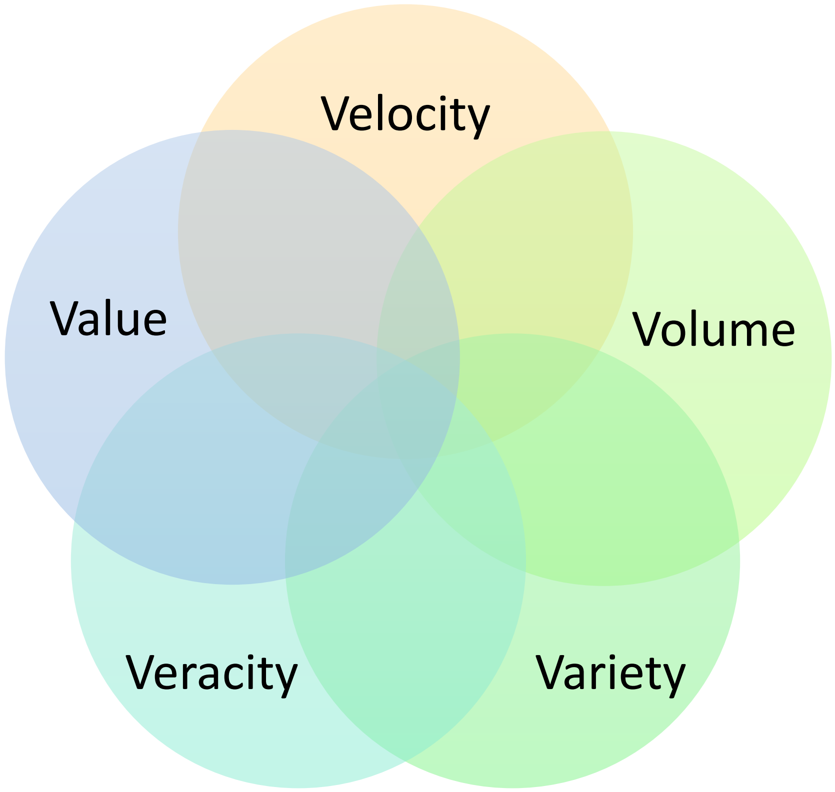
# Introduction

## ❖ Big Data Definition


**Big Data:** A collection of datasets that is large, complex that make it very difficult to process using traditional data processing applications.

# Introduction

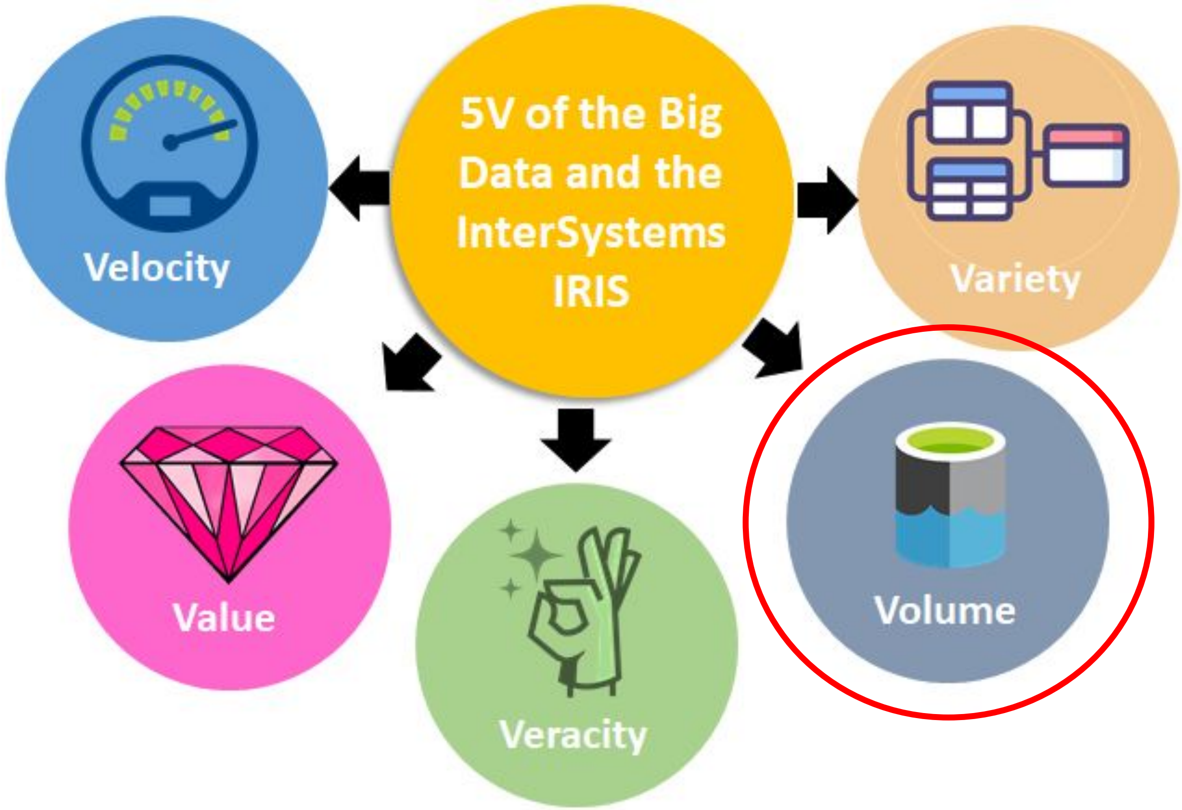
## ❖ Big Data Characteristics

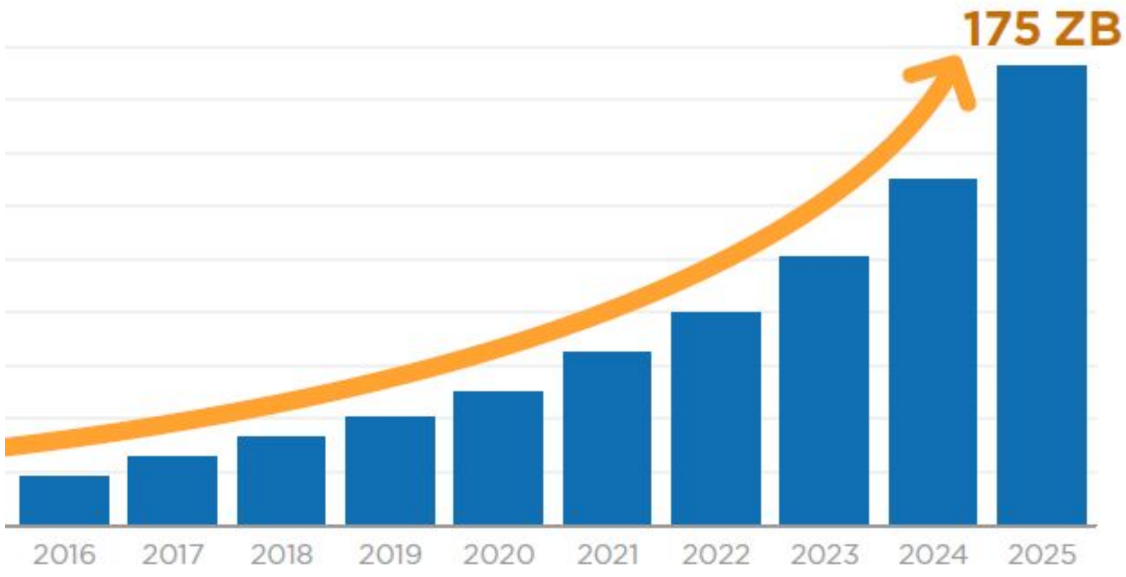



5V of Big Data

# Introduction

## ❖ Big Data Characteristics




The amount of data generated on Internet each year

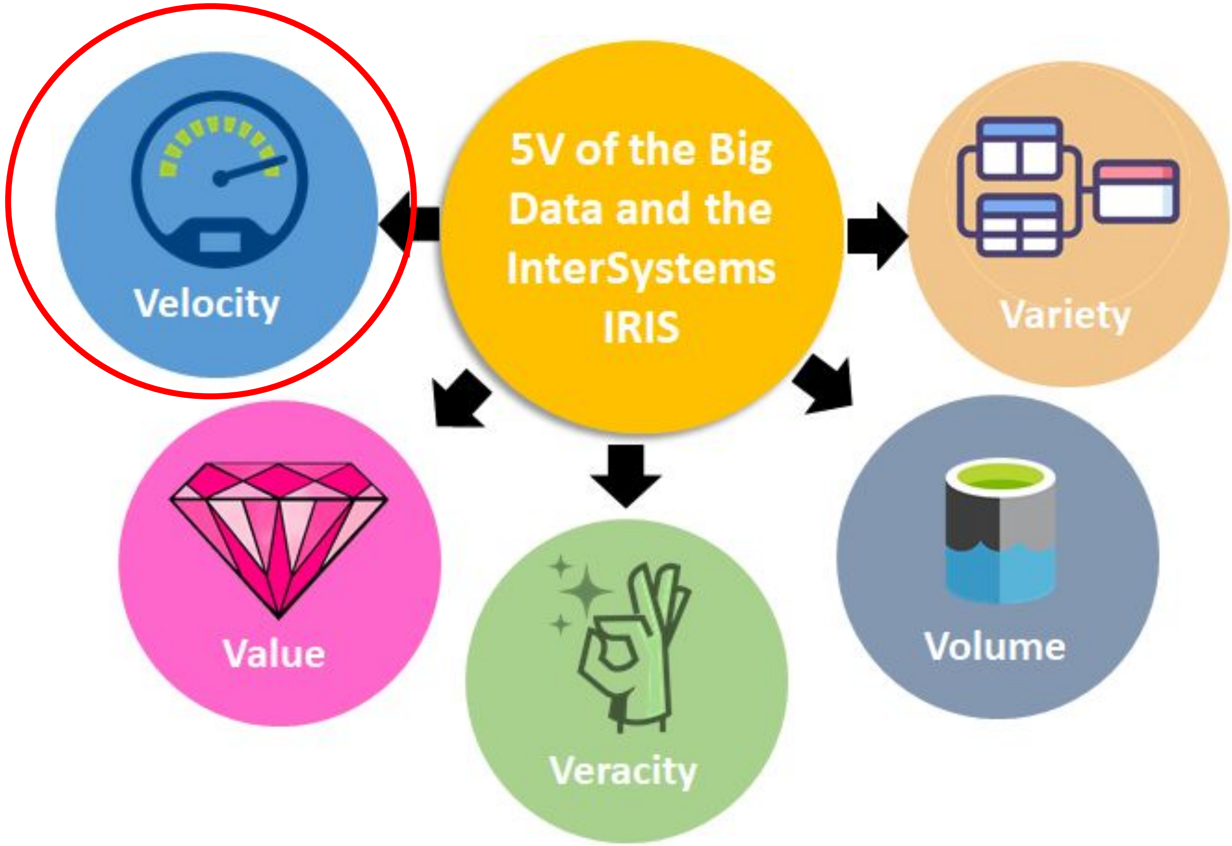
**Volume:** The amount of data

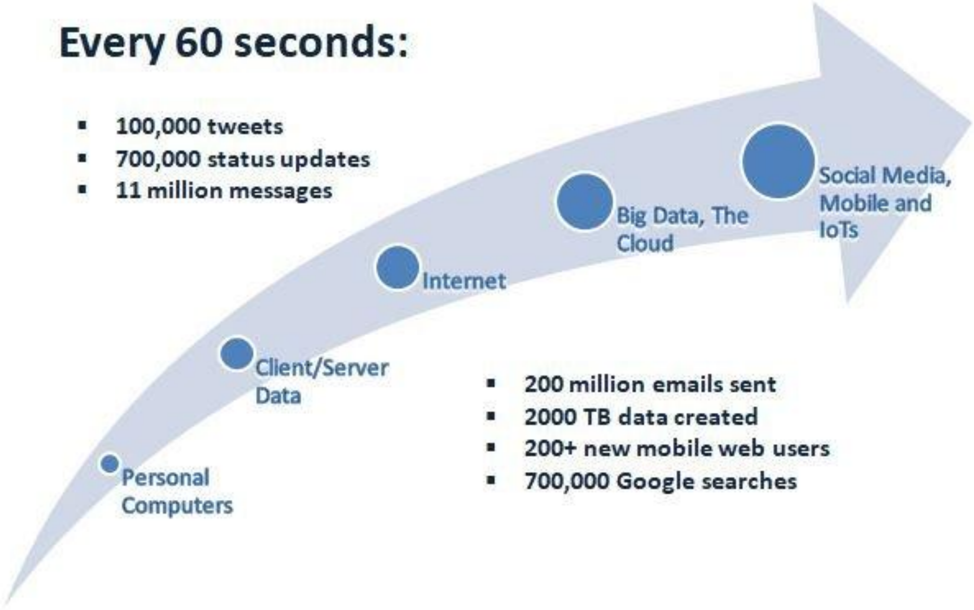


https://community.intersystems.com/post/big-data-5v-intersystems-iris

# Introduction

## ❖ Big Data Characteristics



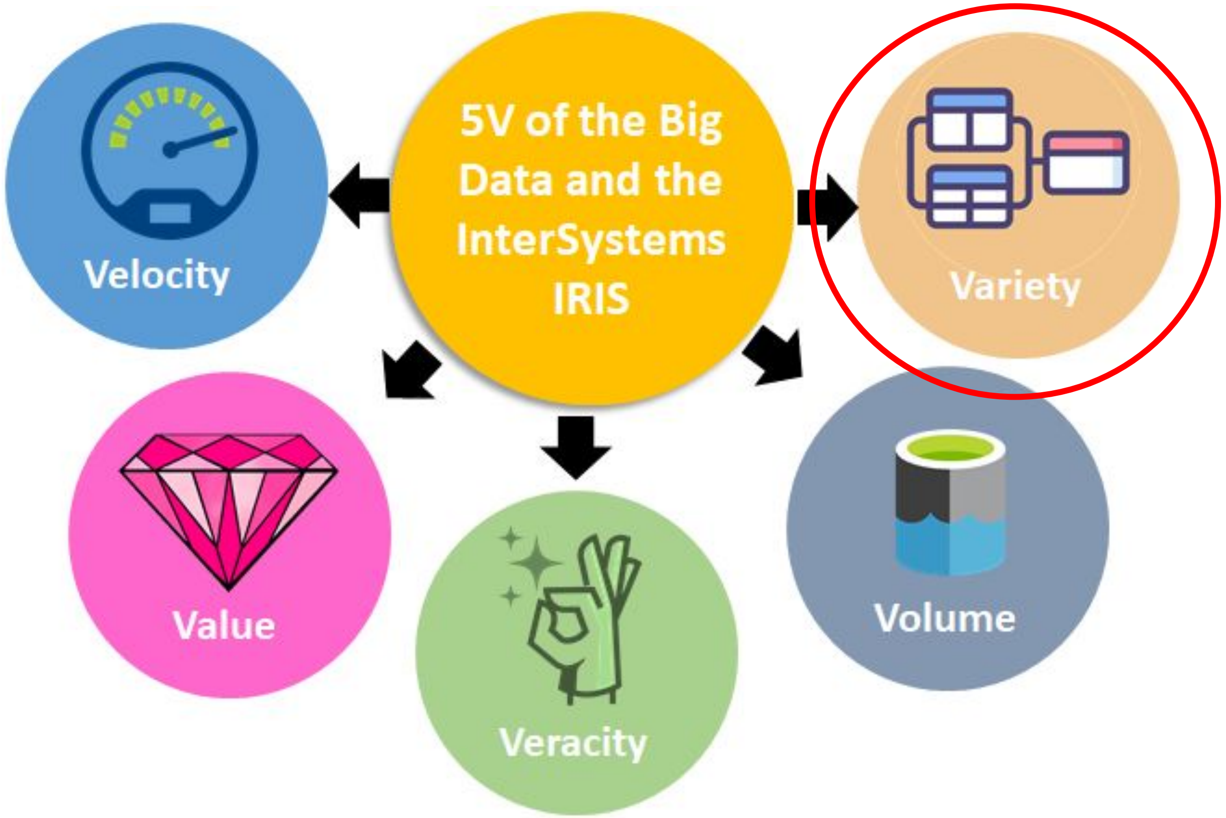

Data generated on Internet every 1 minute

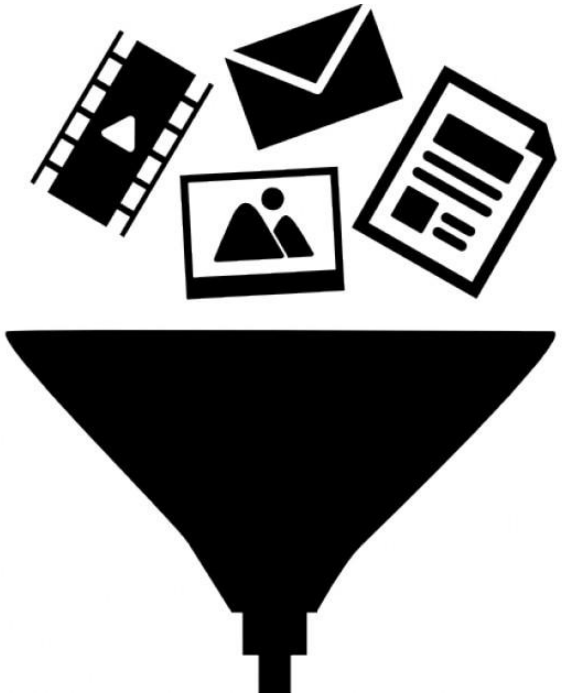
**Velocity:** The growth of data

https://community.intersystems.com/post/big-data-5v-intersystems-iris

# Introduction

## ❖ Big Data Characteristics



Lots of types of data to be processed

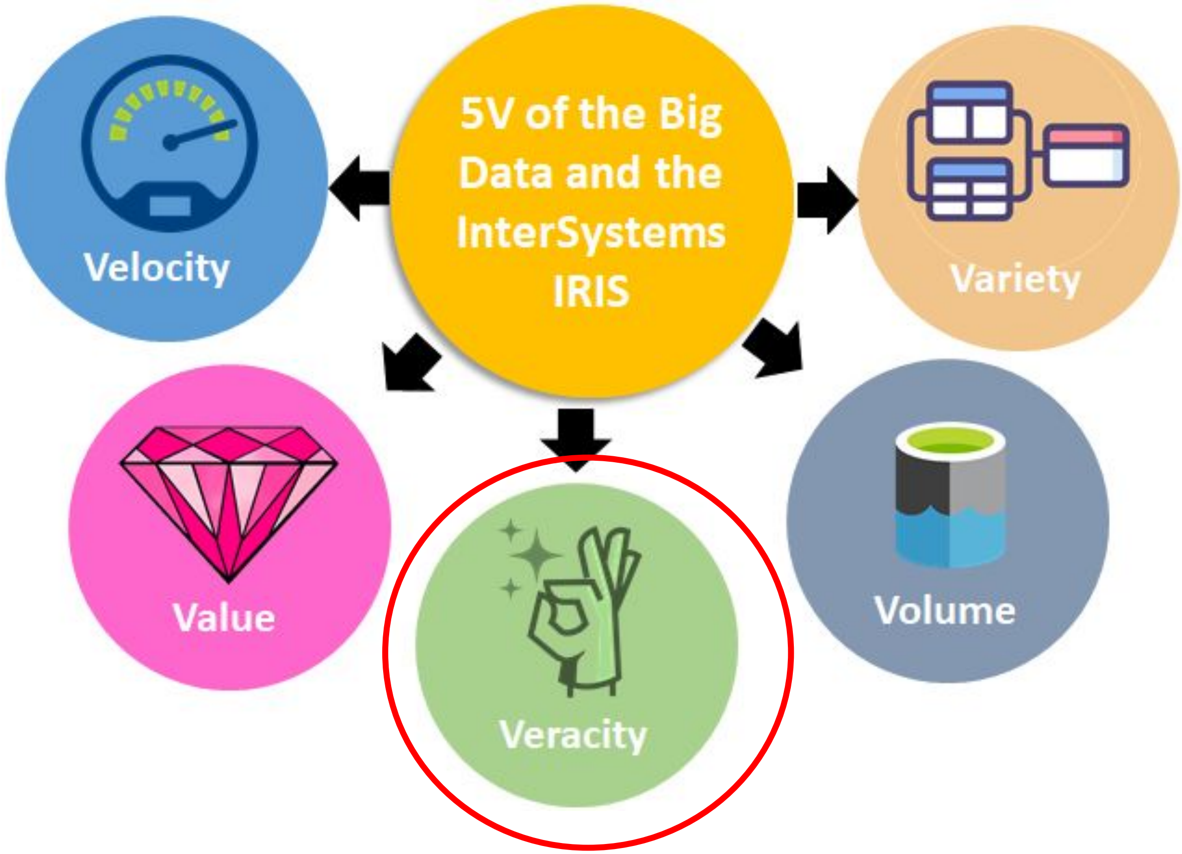
**Variety:** The diversity of data

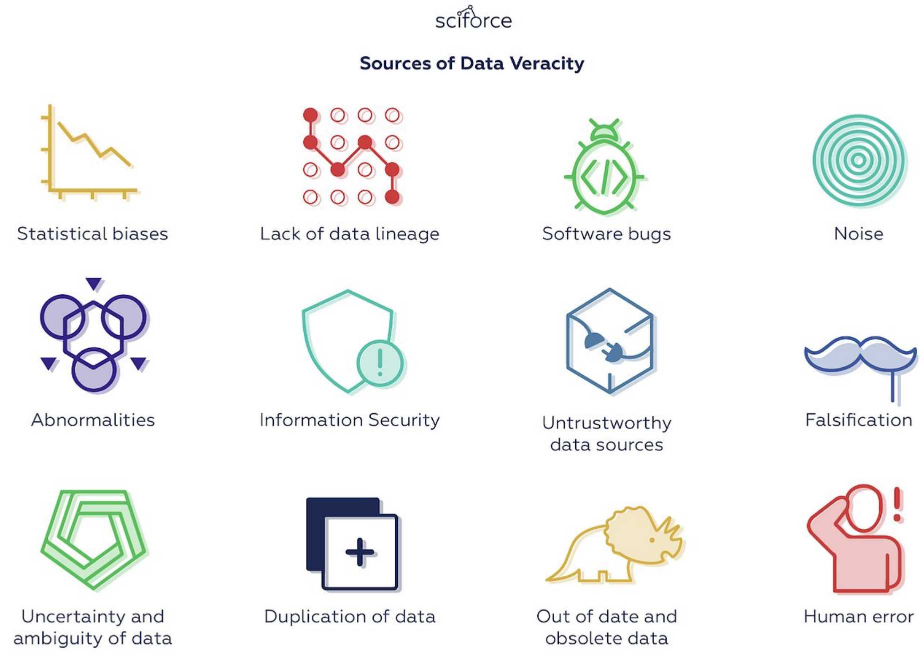


https://community.intersystems.com/post/big-data-5v-intersystems-iris

# Introduction

## ❖ Big Data Characteristics



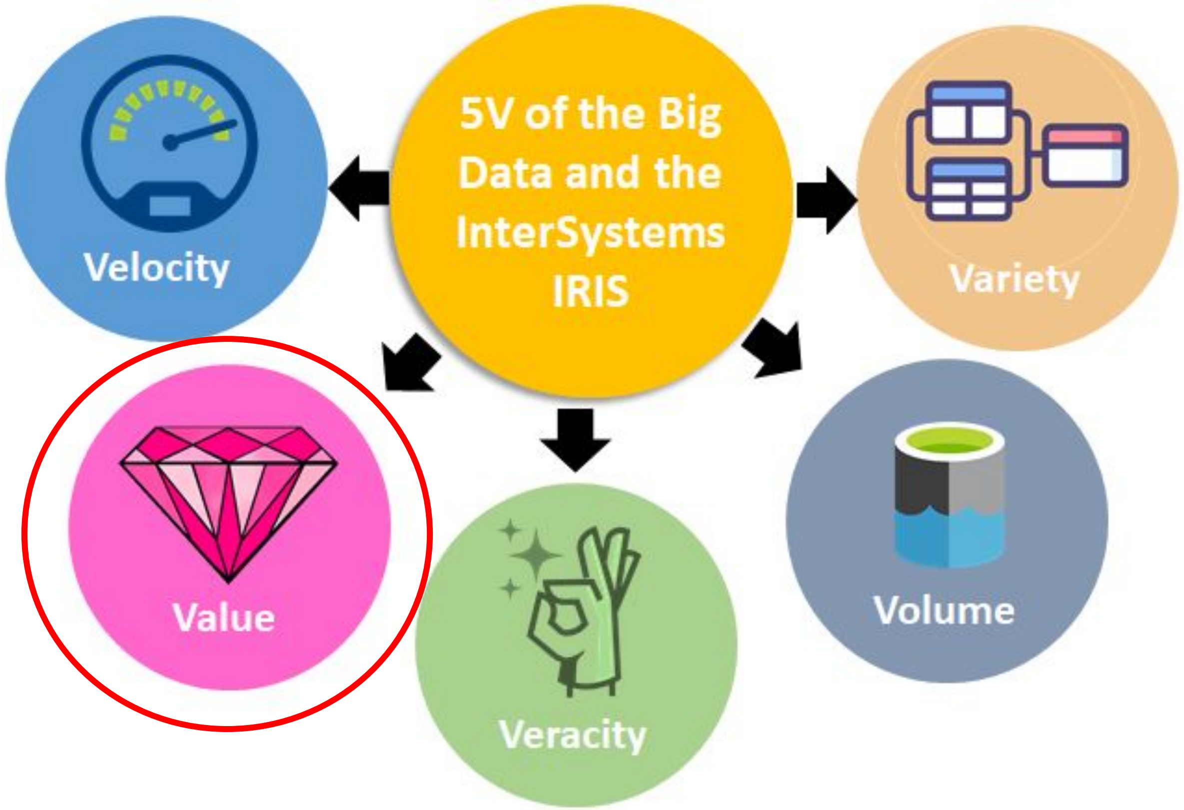

It's not just the quality of the data that is important, but how trustworthy the source, the type, and processing of the data are.

**Veracity:** The accuracy, reliability, and truthfulness of the data

https://medium.com/sciforce/data-veracity-a-new-key-to-big-data-38e110391c7d
https://community.intersystems.com/post/big-data-5v-intersystems-iris

# Introduction

## ❖ Big Data Characteristics





**Value:** The beneficial insights gained from data analysis.

Value in big data is essential for deriving meaningful insights and driving decisions.

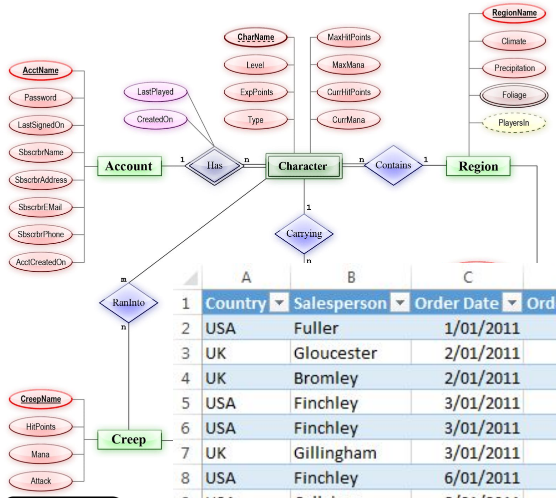


https://www.infilytics.in/solution/by-usecase/big-data-analysis
https://community.intersystems.com/post/big-data-5v-intersystems-iris

# Introduction

## ❖ Types of big data

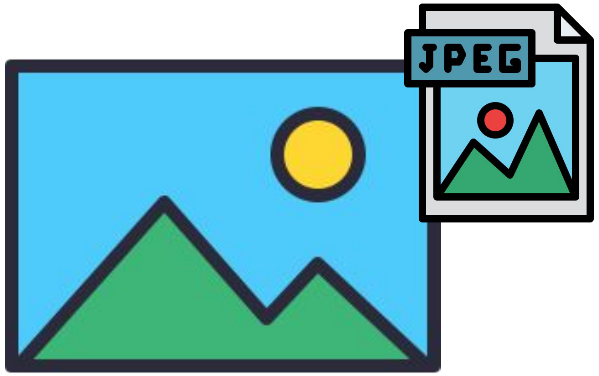
### Structured Data



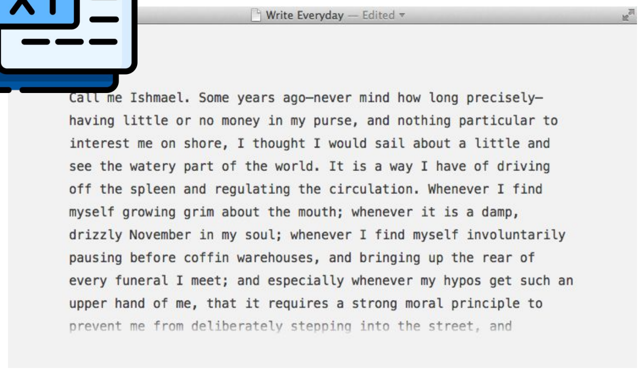


| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | Country | Salesperson | Order Date | OrderID | Units | Order Amount |
| 2 | USA | Fuller | 1/01/2011 | 10392 | 13 | 1,440.00 |
| 3 | UK | Gloucester | 2/01/2011 | 10397 | 17 | 716.72 |
| 4 | UK | Bromley | 2/01/2011 | 10771 | 18 | 344.00 |
| 5 | USA | Finchley | 3/01/2011 | 10393 | 16 | 2,556.95 |
| 6 | USA | Finchley | 3/01/2011 | 10394 | 10 | 442.00 |
| 7 | UK | Gillingham | 3/01/2011 | 10395 | 9 | 2,122.92 |
| 8 | USA | Finchley | 6/01/2011 | 10396 | 7 | 1,903.80 |
| 9 | USA | Callahan | 8/01/2011 | 10399 | 17 | 1,765.60 |
| 10 | USA | Fuller | 8/01/2011 | 10404 | 7 | 1,591.25 |
| 11 | USA | Fuller | 9/01/2011 | 10398 | 11 | 2,505.60 |
| 12 | USA | Coghill | 9/01/2011 | 10403 | 18 | 855.01 |
| 13 | USA | Finchley | 10/01/2011 | 10401 | 7 | 3,868.60 |
| 14 | USA | Callahan | 10/01/2011 | 10402 | 11 | 2,713.50 |
| 15 | UK | Rayleigh | 13/01/2011 | 10406 | 15 | 1,830.78 |
| 16 | USA | Callahan | 14/01/2011 | 10408 | 10 | 1,622.40 |
| 17 | USA | Farnham | 14/01/2011 | 10409 | 19 | 319.20 |
| 18 | USA | Farnham | 15/01/2011 | 10410 | 16 | 802.00 |

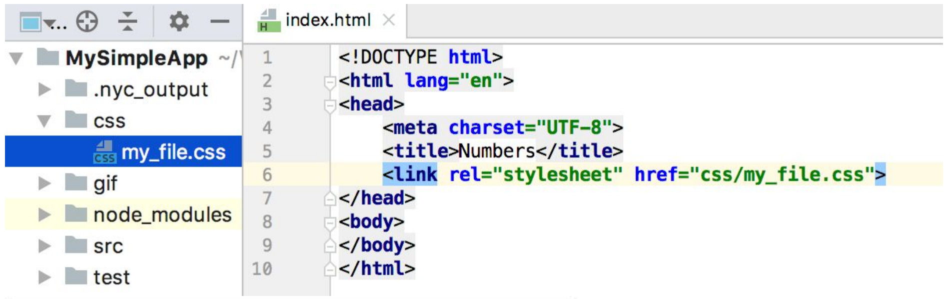
### Unstructured Data

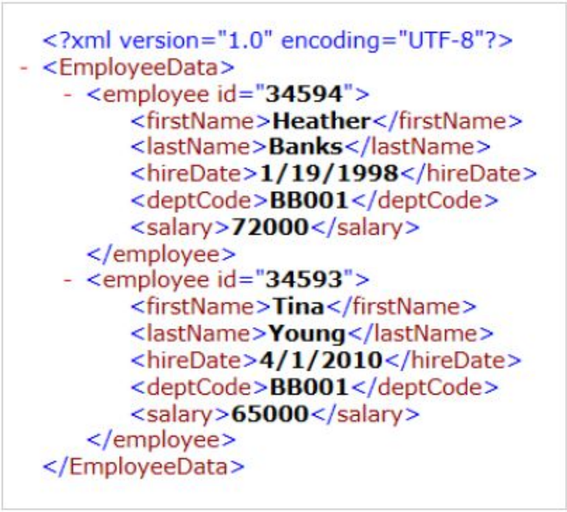


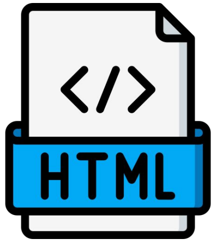





### Semi-structured Data

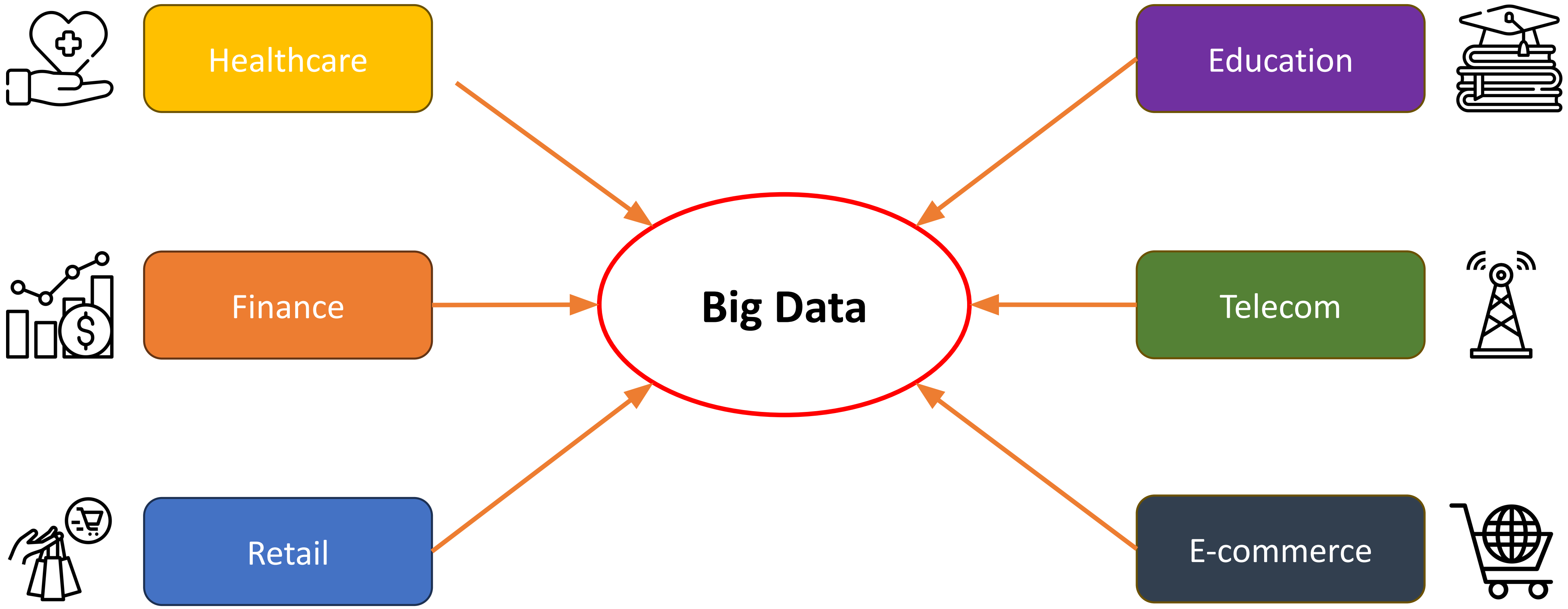


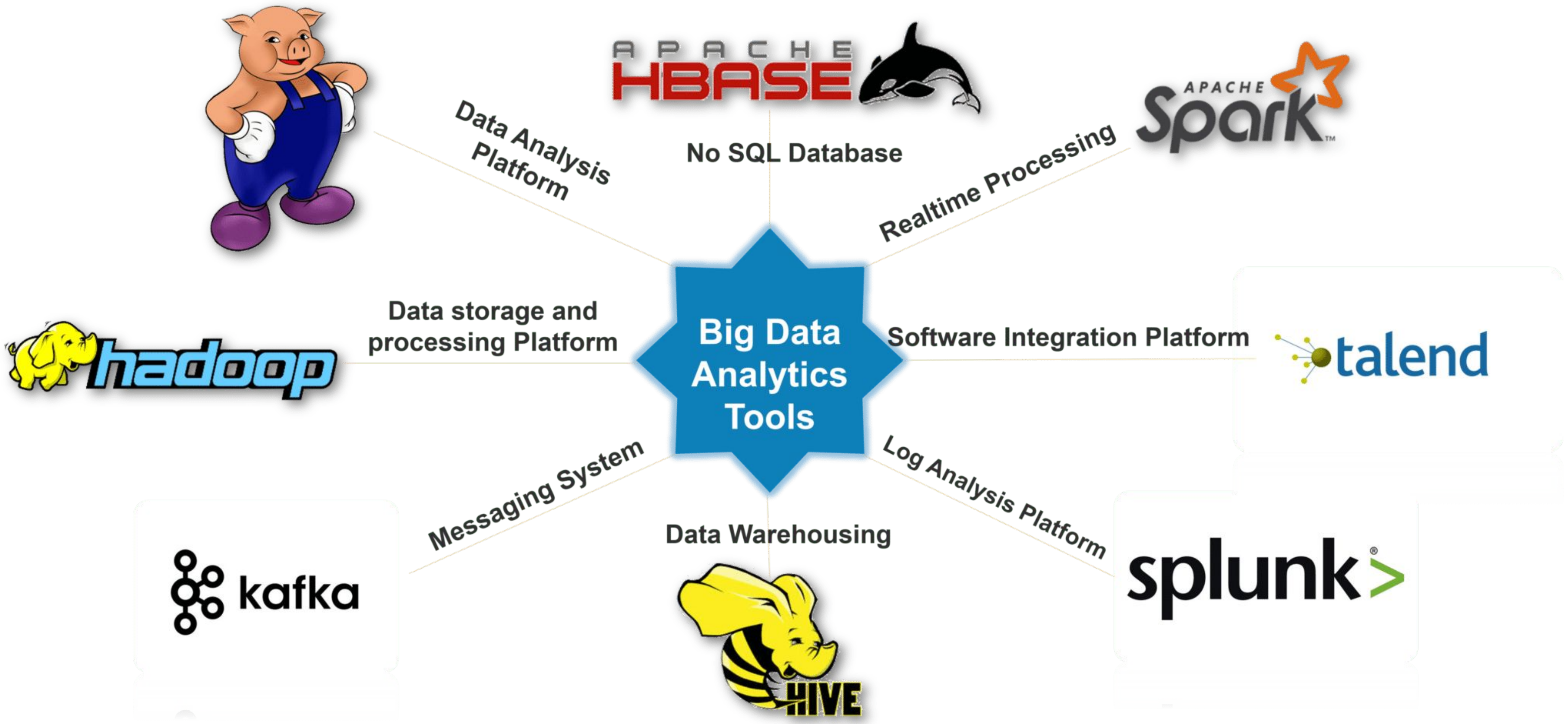

# Introduction

## ❖ Big Data Applications



# Introduction

## ❖ Tools for Big Data



# Hadoop

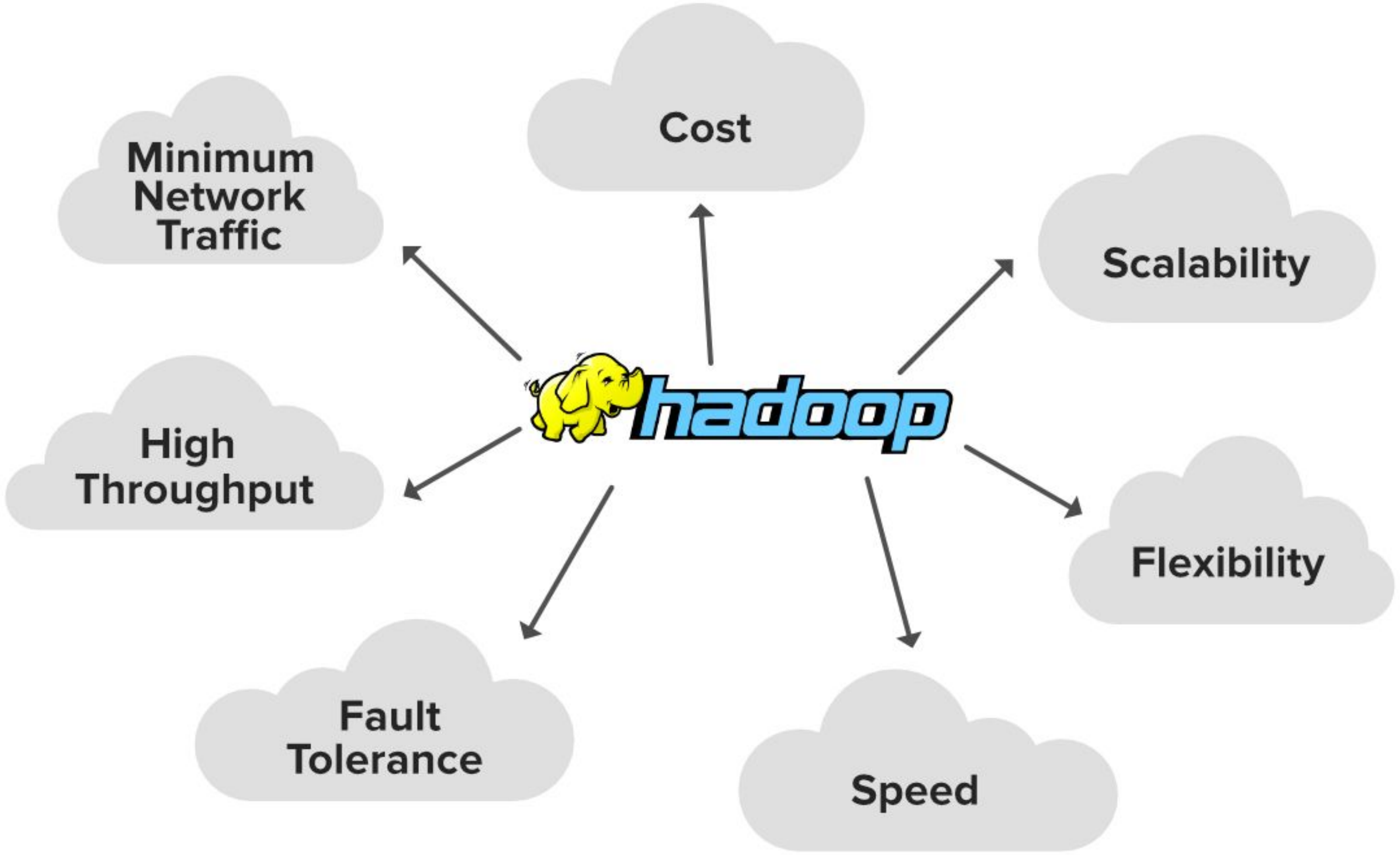
# Hadoop

## ❖ Introduction

**Apache Hadoop:** an open-source framework that is used to **efficiently store and process large datasets** ranging in size from gigabytes to petabytes of data.

# Hadoop

## ❖ Introduction




Hadoop Advantages

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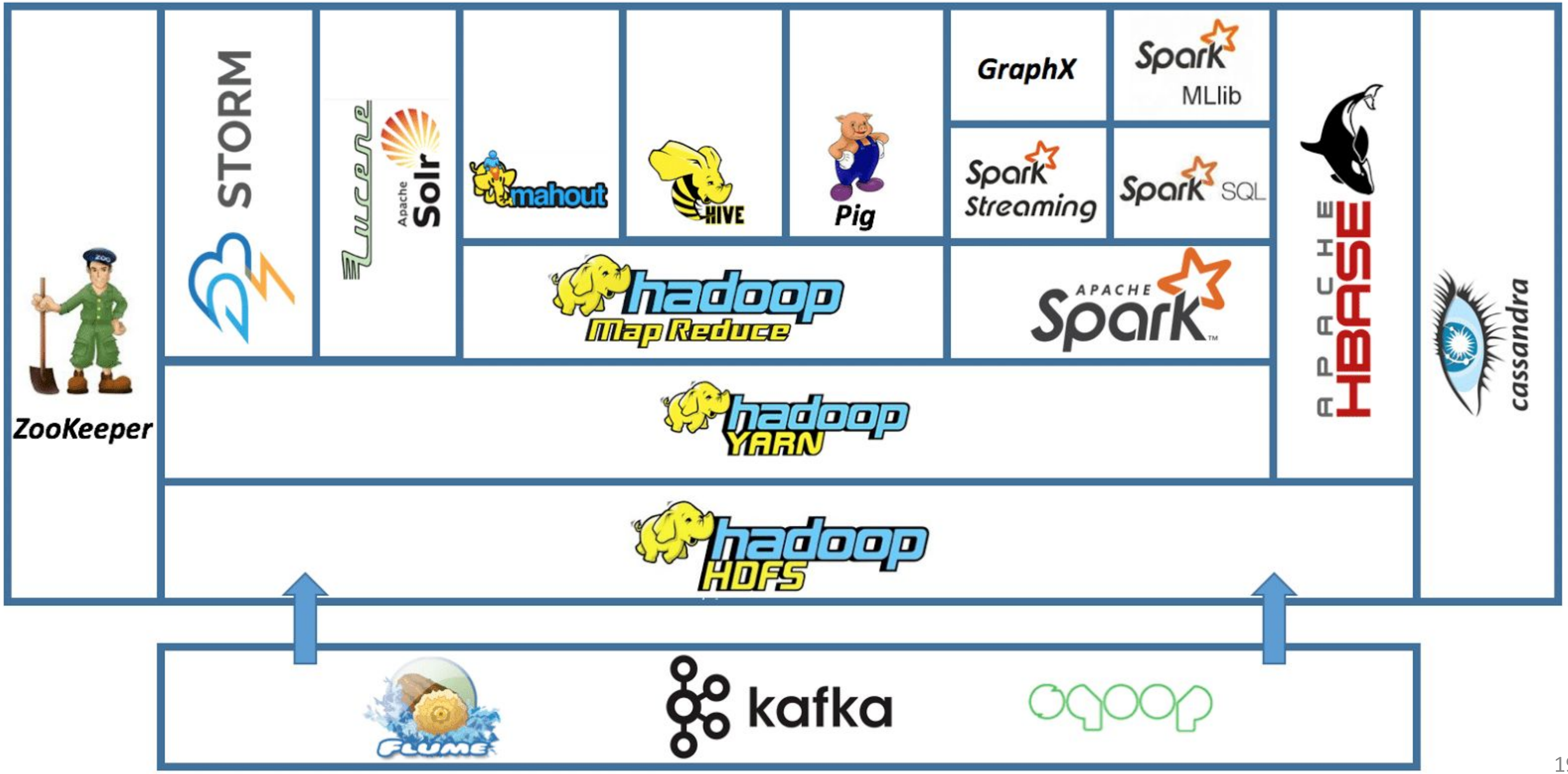
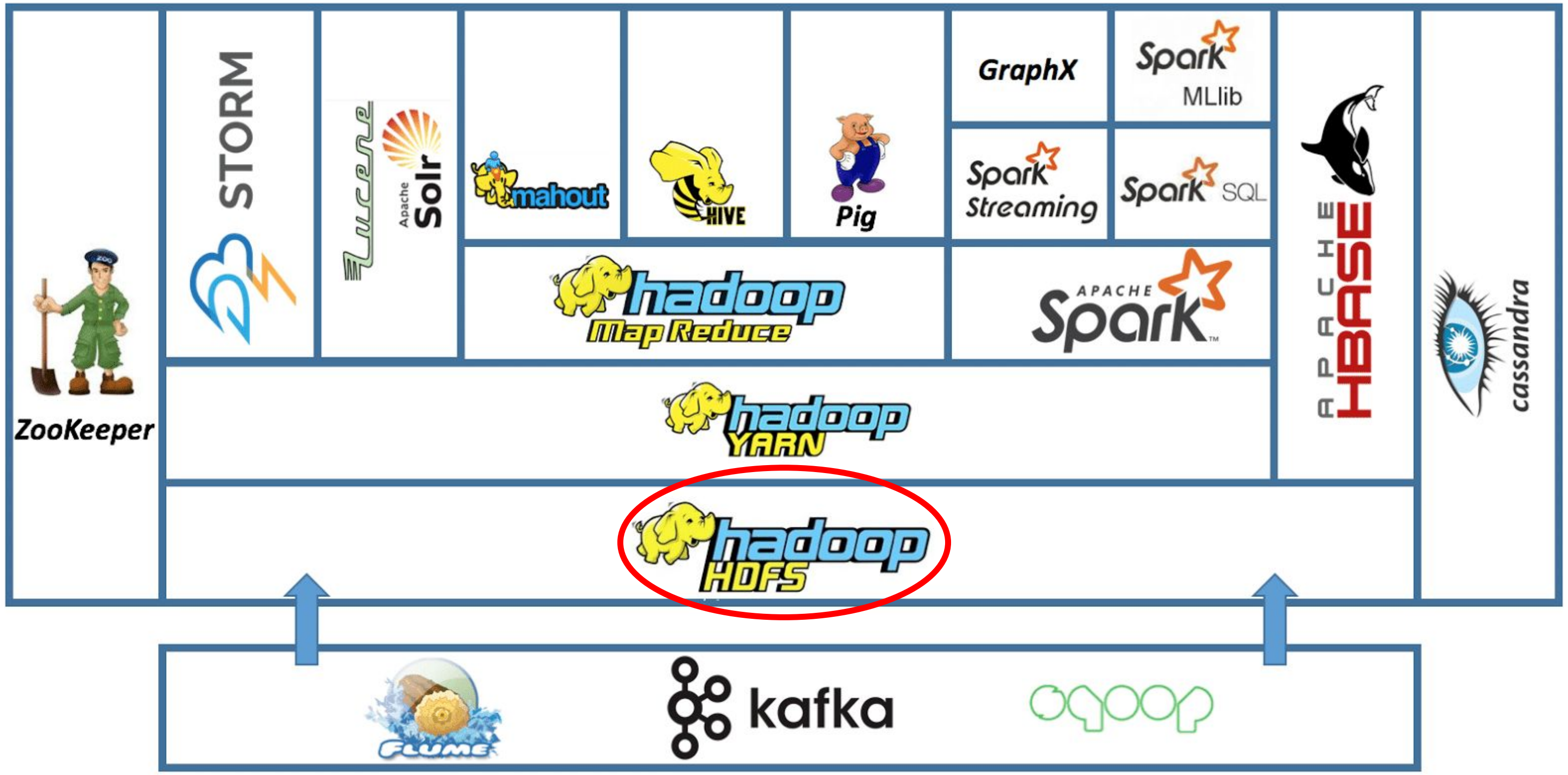
# Hadoop

## ❖ Hadoop Ecosystem



# Hadoop

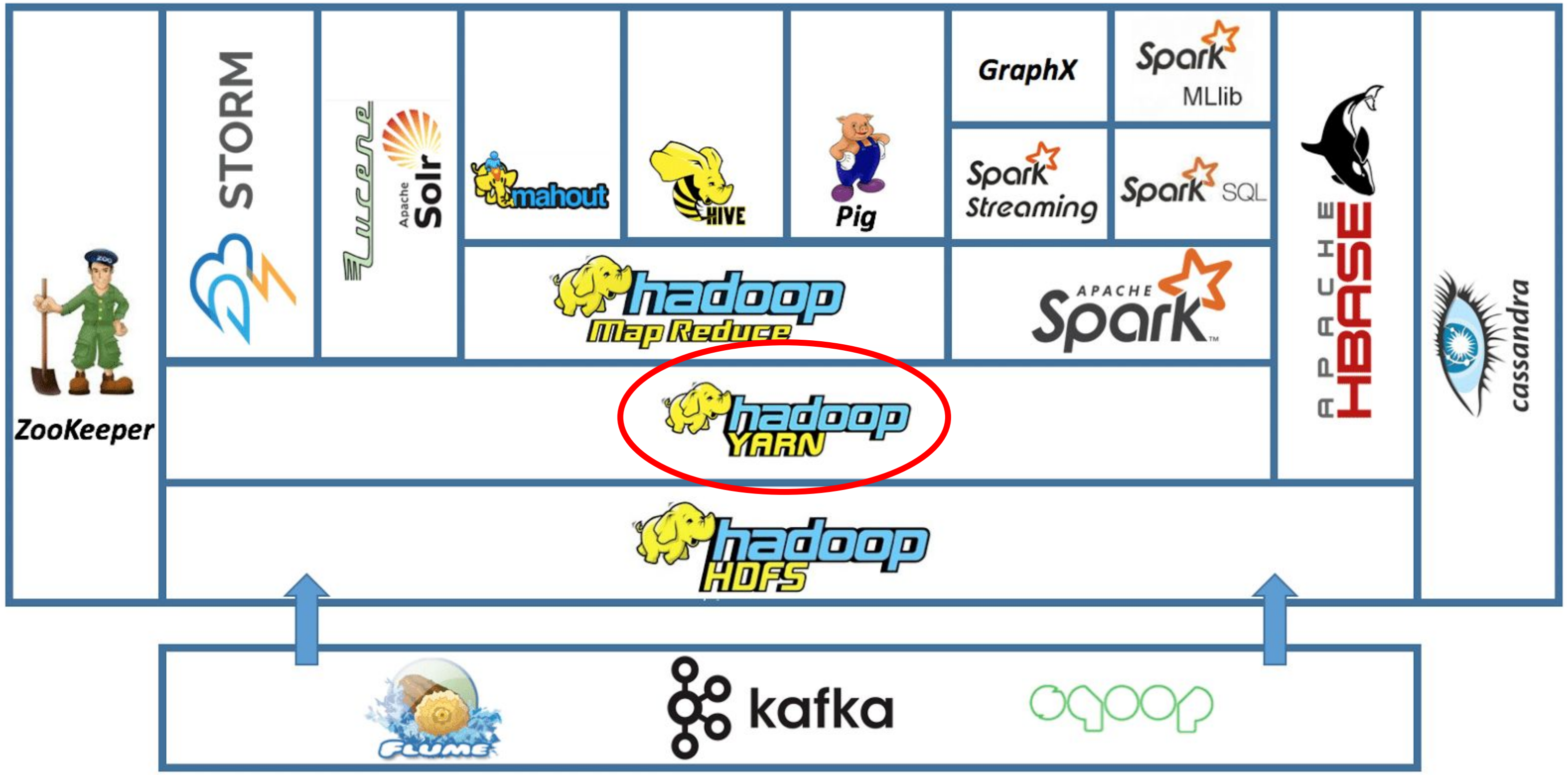
## ❖ Hadoop Ecosystem: Hadoop HDFS




**Hadoop HDFS (Hadoop Distributed File System):** A distributed file system that handles larget data sets running on commodity hardware.

# Hadoop

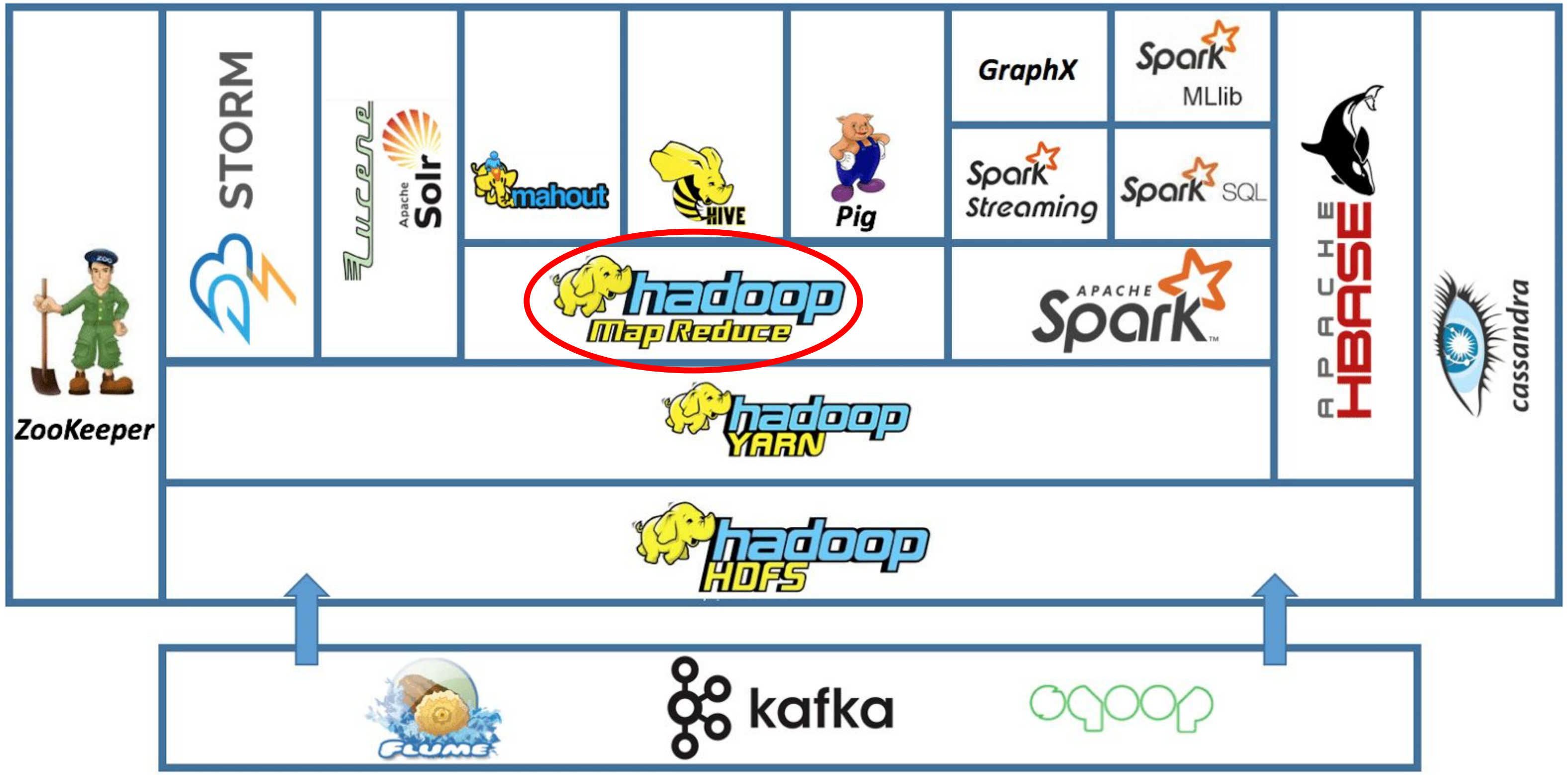
## ❖ Hadoop Ecosystem: Hadoop YARN




**Hadoop YARN (Yet Another Resource Negotiator):** the resources management and job scheduling technology.

# Hadoop

## ❖ Hadoop Ecosystem: Hadoop MapReduce

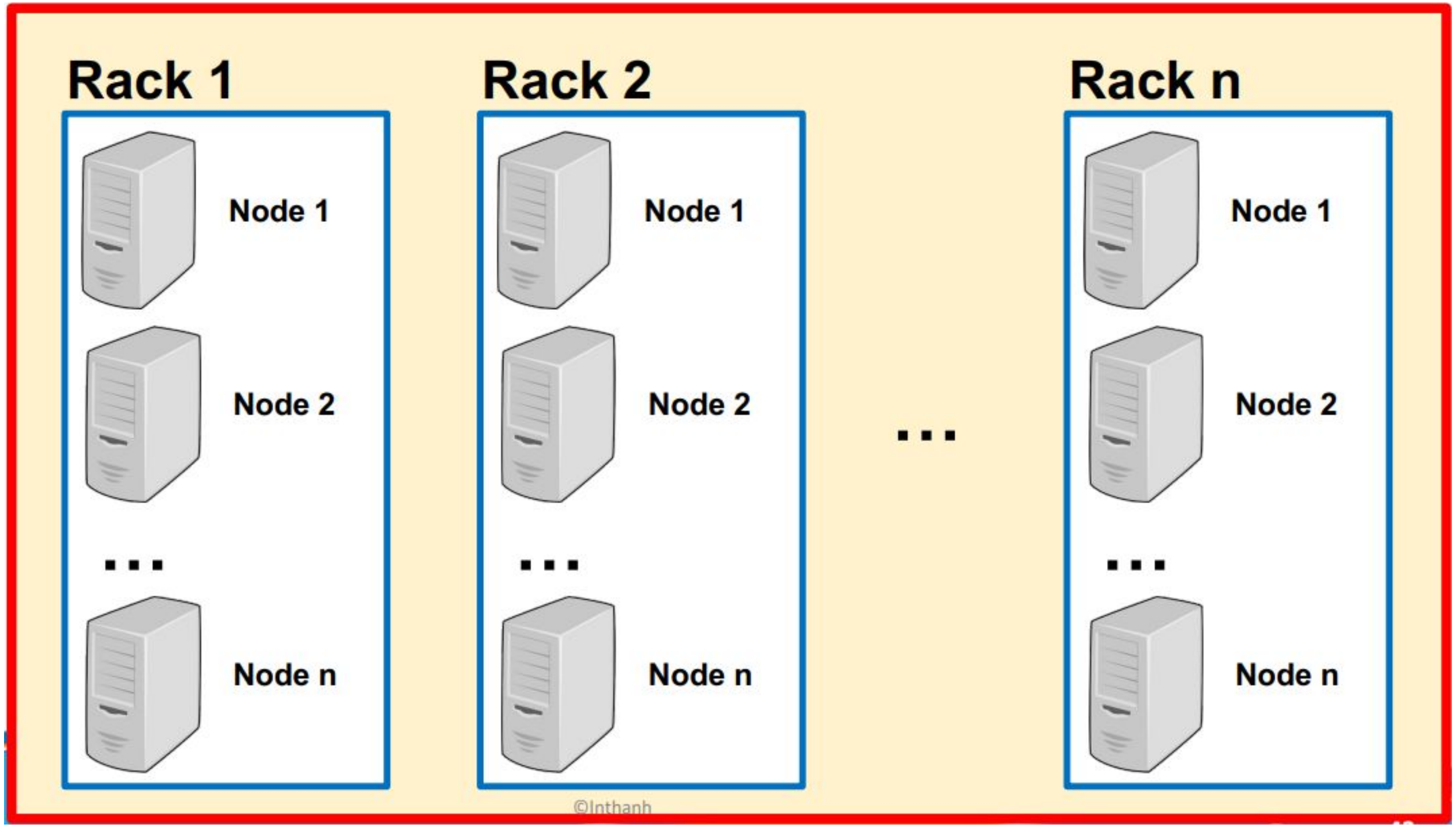



**Hadoop MapReduce:** A highly sufficient methodology for parallel processing of huge volumes of data.

# Hadoop

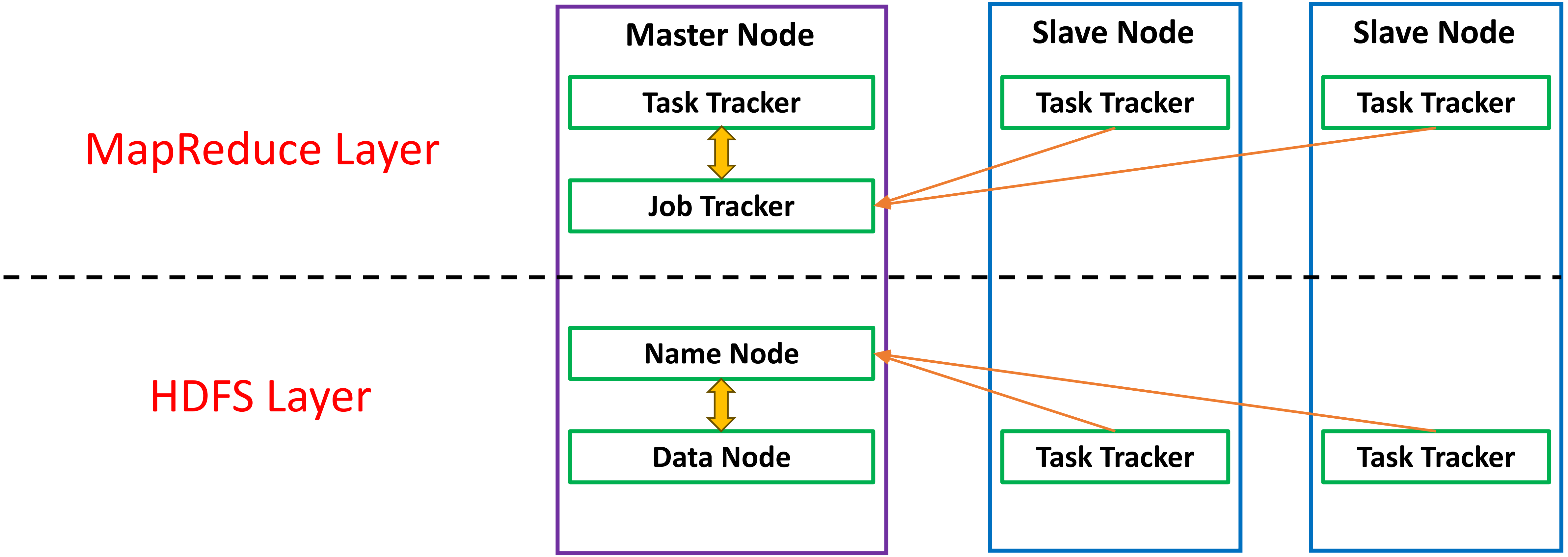
## ❖ Terminology

- Node
- Rack
- Cluster



# Hadoop

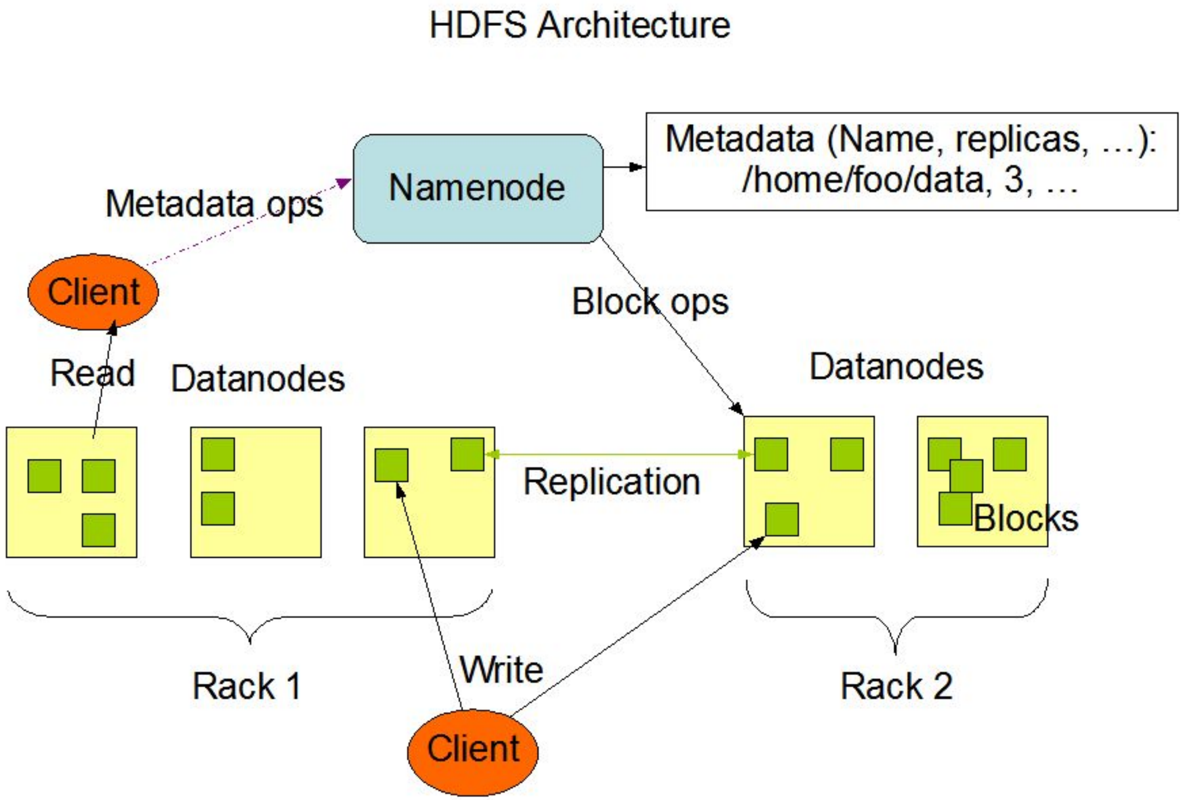
## ❖ Hadoop Architecture




High-level Architecture of Hadoop

# Hadoop

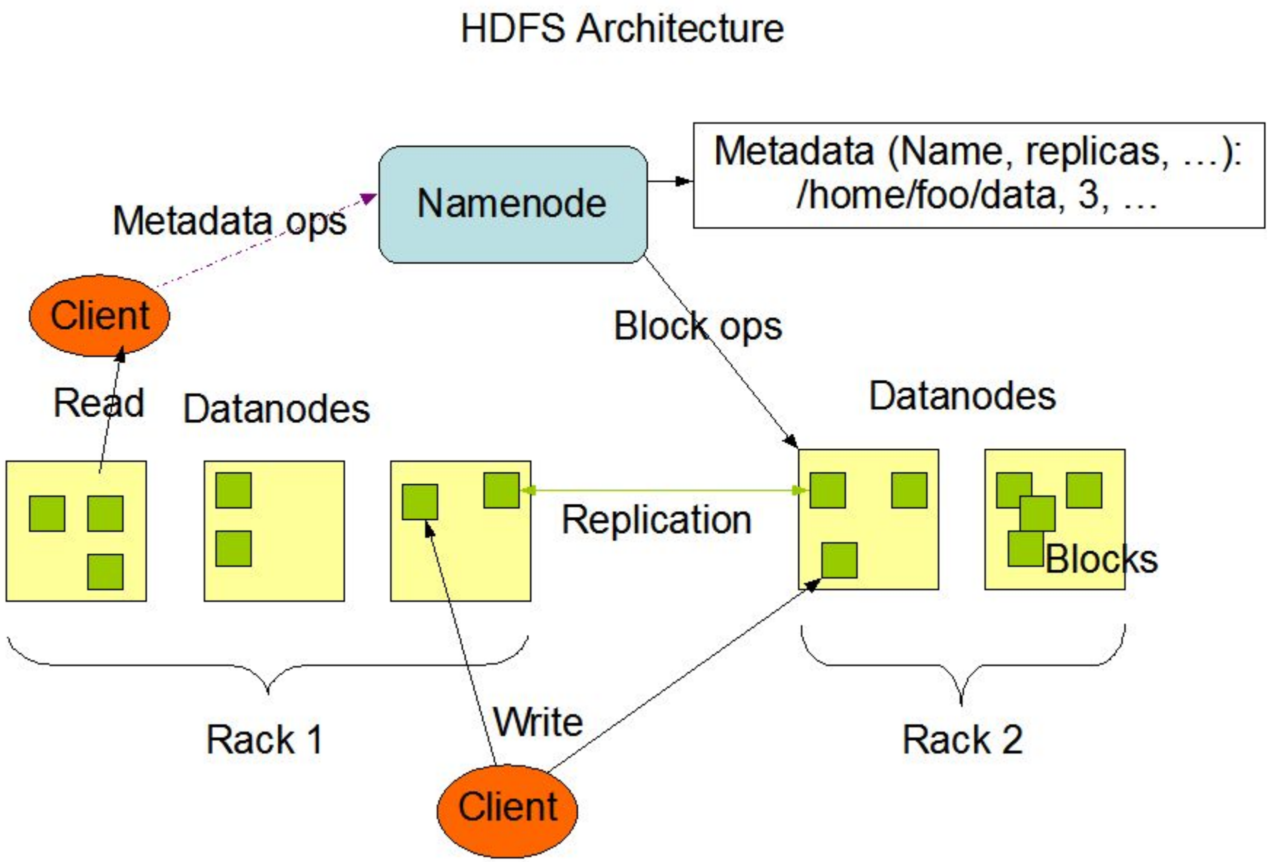
## ❖ Hadoop HDFS




**HDFS (Hadoop Distributed File System):** A distributed file system designed to store and process large amounts of data across multiple machines in a Hadoop cluster, providing high fault-tolerance and scalability.

# Hadoop

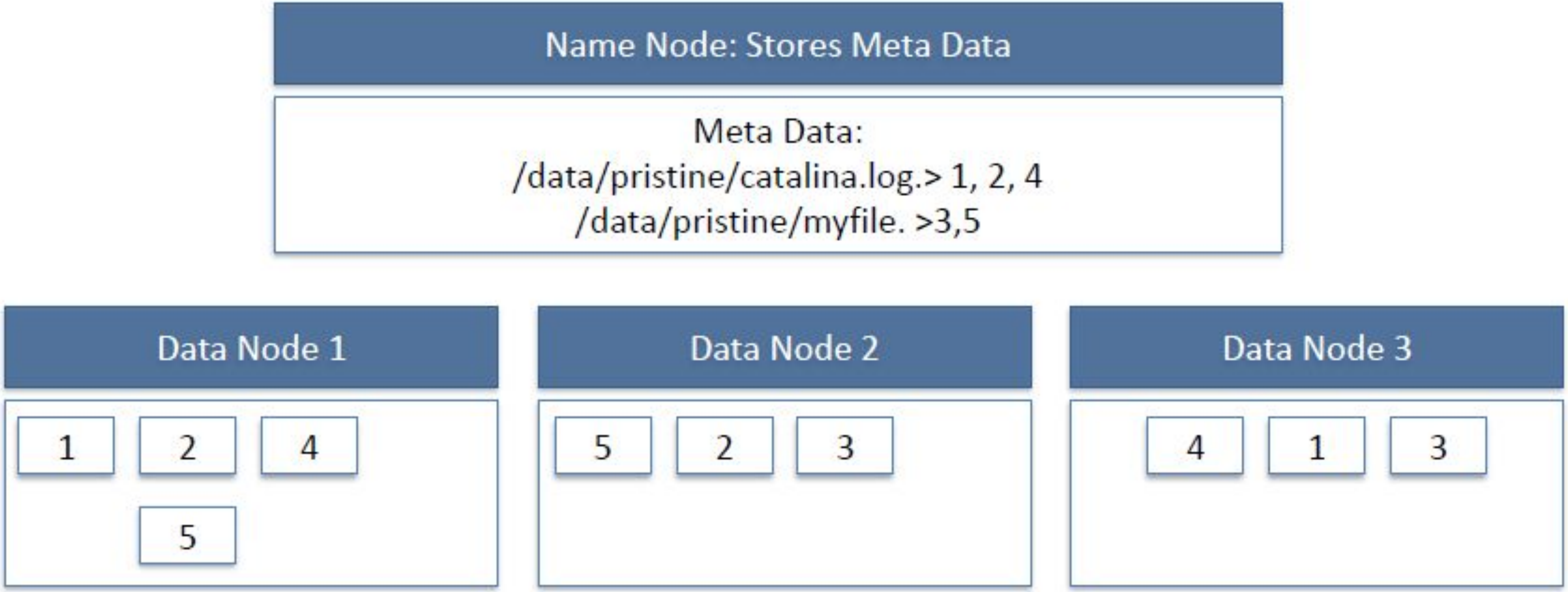
## ❖ Hadoop HDFS




- **Namenode:** Master node in a Hadoop cluster, managing file system's metadata.
- **Datanode:** Worker node in a Hadoop cluster, storing data blocks.
- **Blocks:** A fixed-size chunk of data. Default is 128MB (or 64MB) size.

# Hadoop

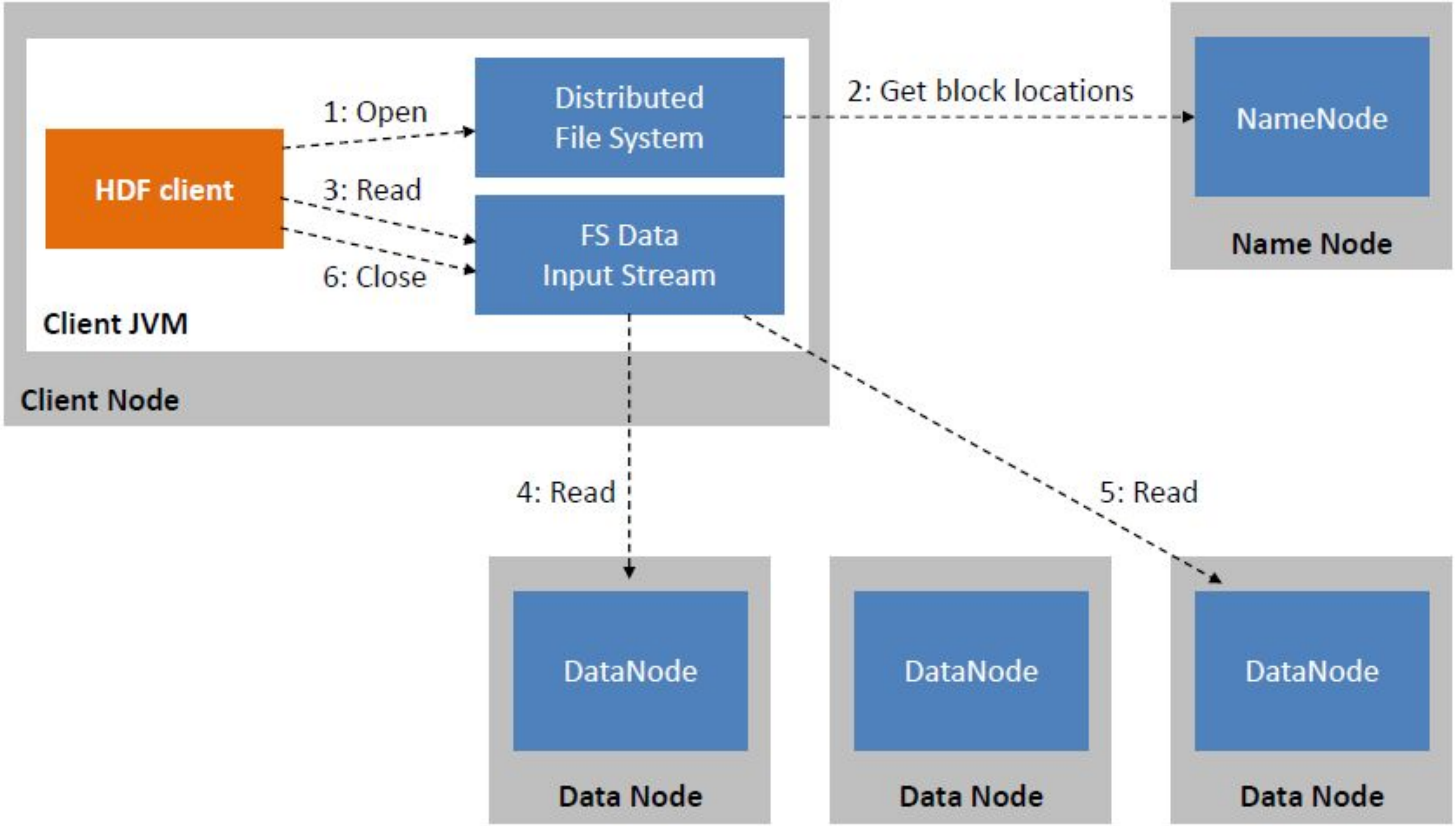
## ❖ Hadoop HDFS




https://www.javatpoint.com/hdfs

# Hadoop

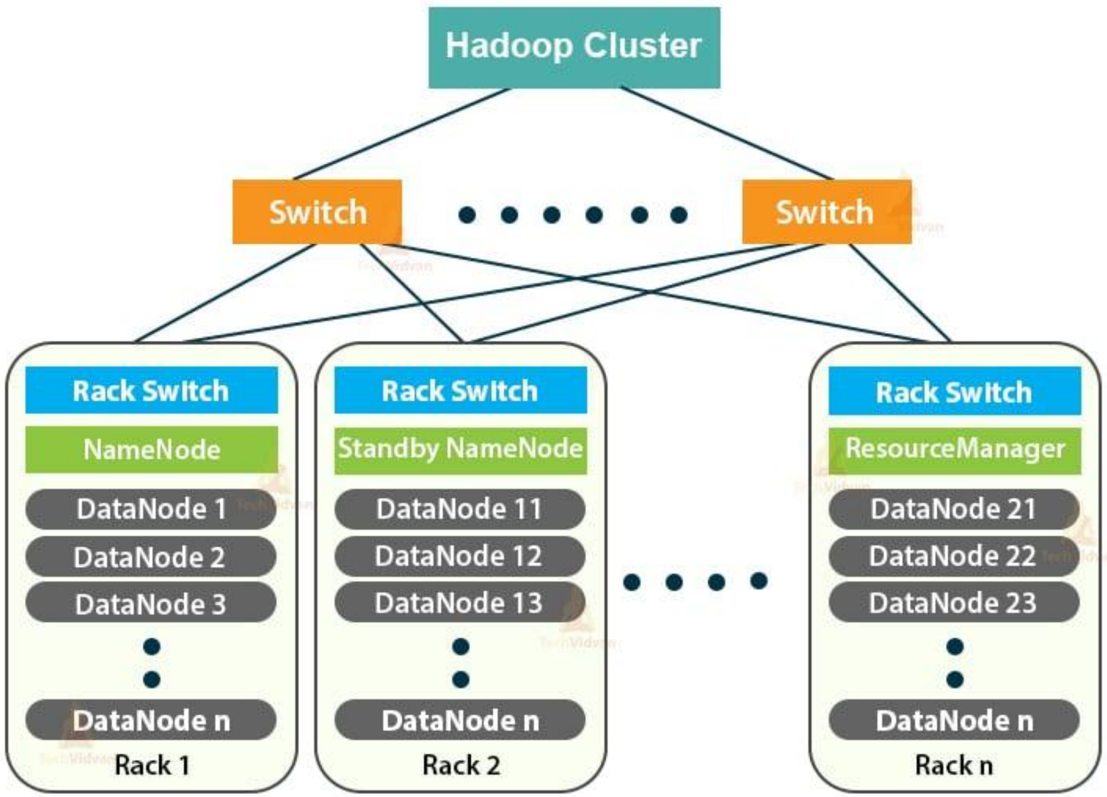
## ❖ Hadoop HDFS




https://www.javatpoint.com/hdfs

# Hadoop

## ❖ Hadoop HDFS





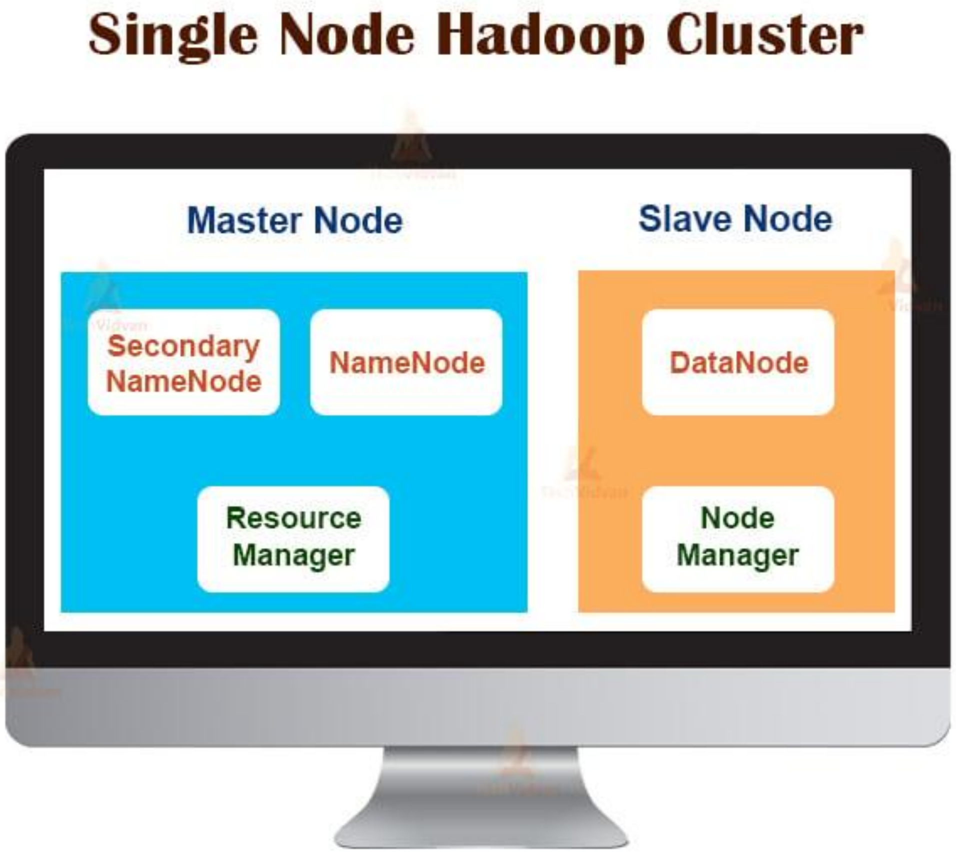
Hadoop Cluster Architecture



https://techvidvan.com/tutorials/hadoop-cluster/

# Hadoop

## ❖ Hadoop HDFS




Hadoop Single Node Cluster Architecture

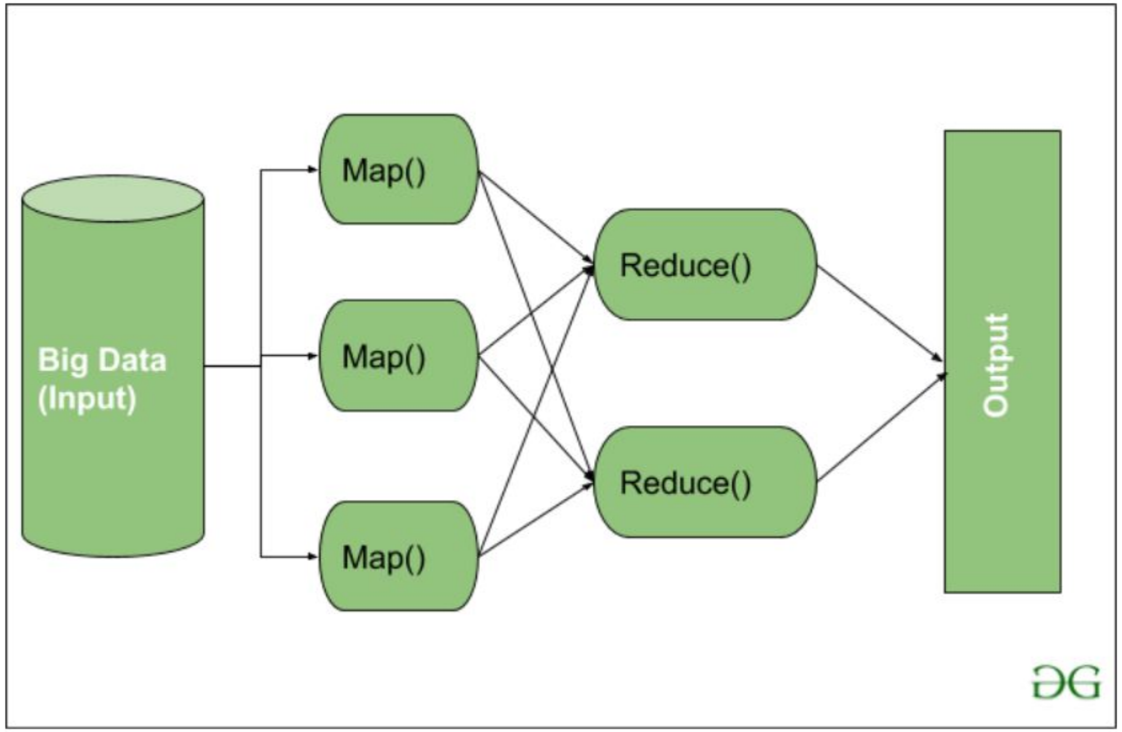


https://techvidvan.com/tutorials/hadoop-cluster/

# Hadoop

## ❖ Hadoop MapReduce

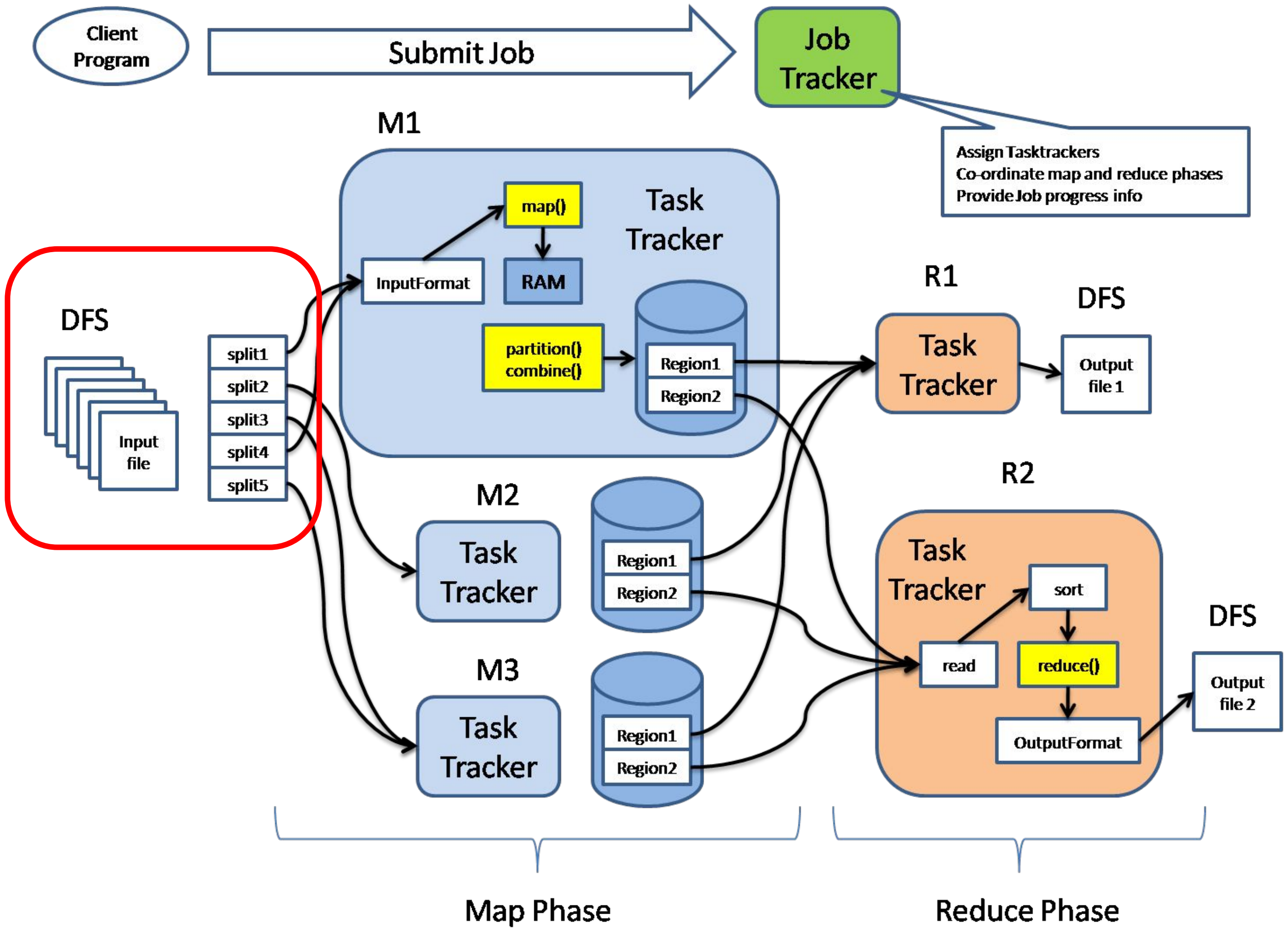
**Hadoop MapReduce:** A distributed programming model and framework that breaks down large-scale data processing tasks into smaller, parallelizable operations (map and reduce), allowing for efficient processing and analysis of big data across a cluster of machines.




https://www.geeksforgeeks.org/hadoop-architecture/

# Hadoop

## ❖ Hadoop MapReduce



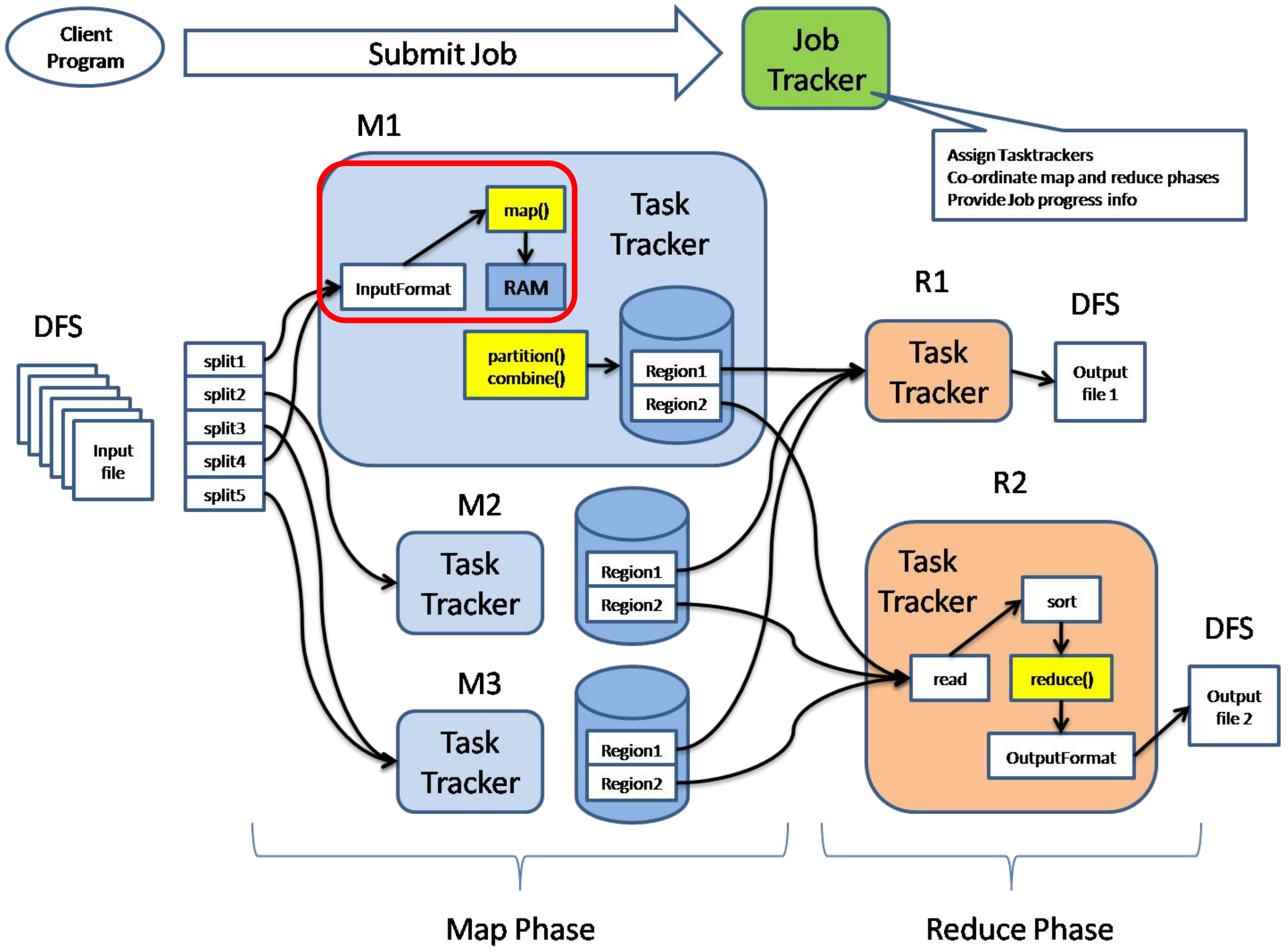

**InputReader:** Read the input data and split it into the data blocks.



https://www.analyticsvidhya.com/blog/2022/07/learn-everything-about-mapreduce-architecture-and-its-components/

# Hadoop

## ❖ Hadoop MapReduce

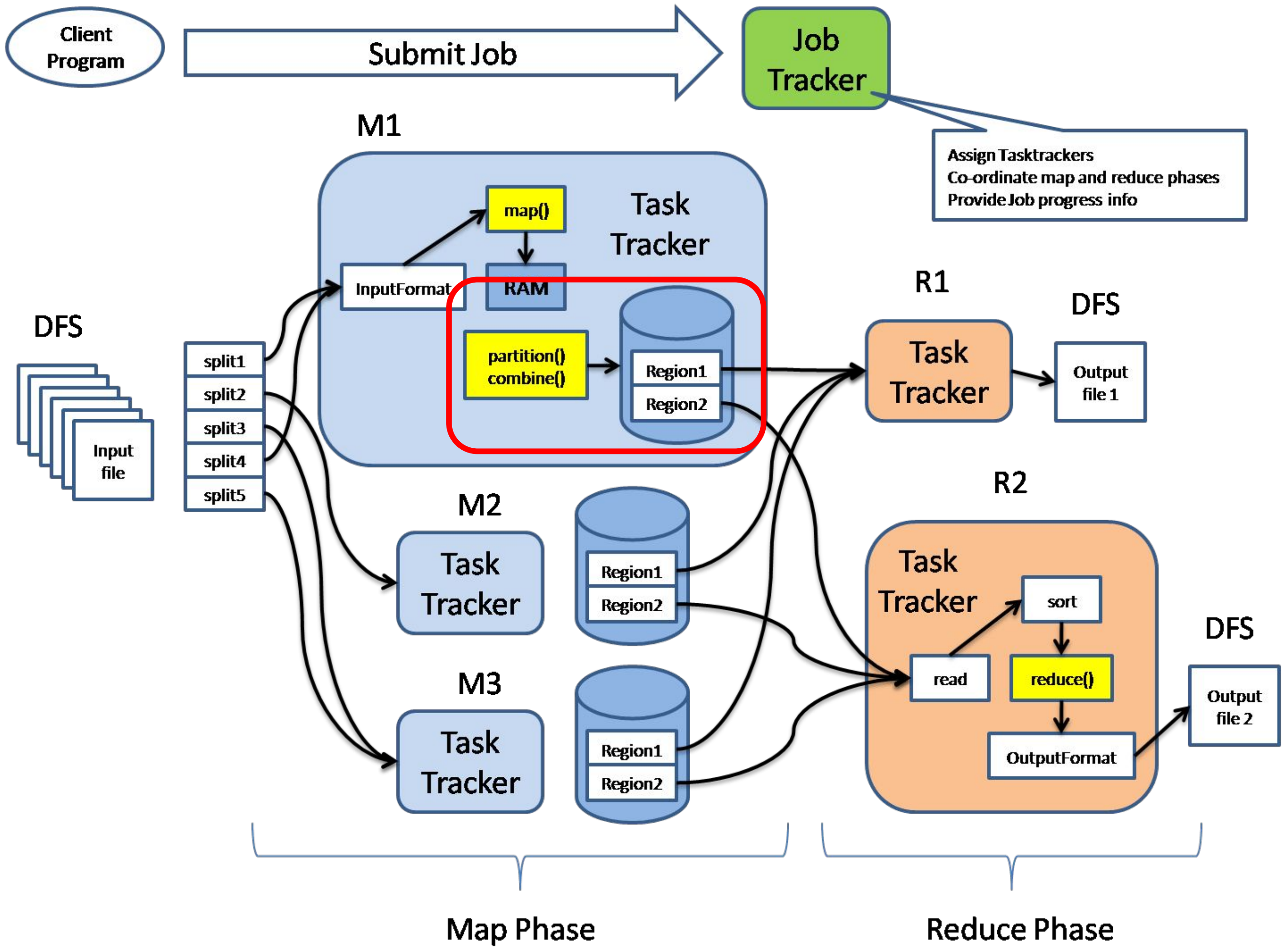



**MapFunction:** Process the key-value pairs and generate the corresponding output key-value pairs.

https://www.analyticsvidhya.com/blog/2022/07/learn-everything-about-mapreduce-architecture-and-its-components/

# Hadoop

## ❖ Hadoop MapReduce

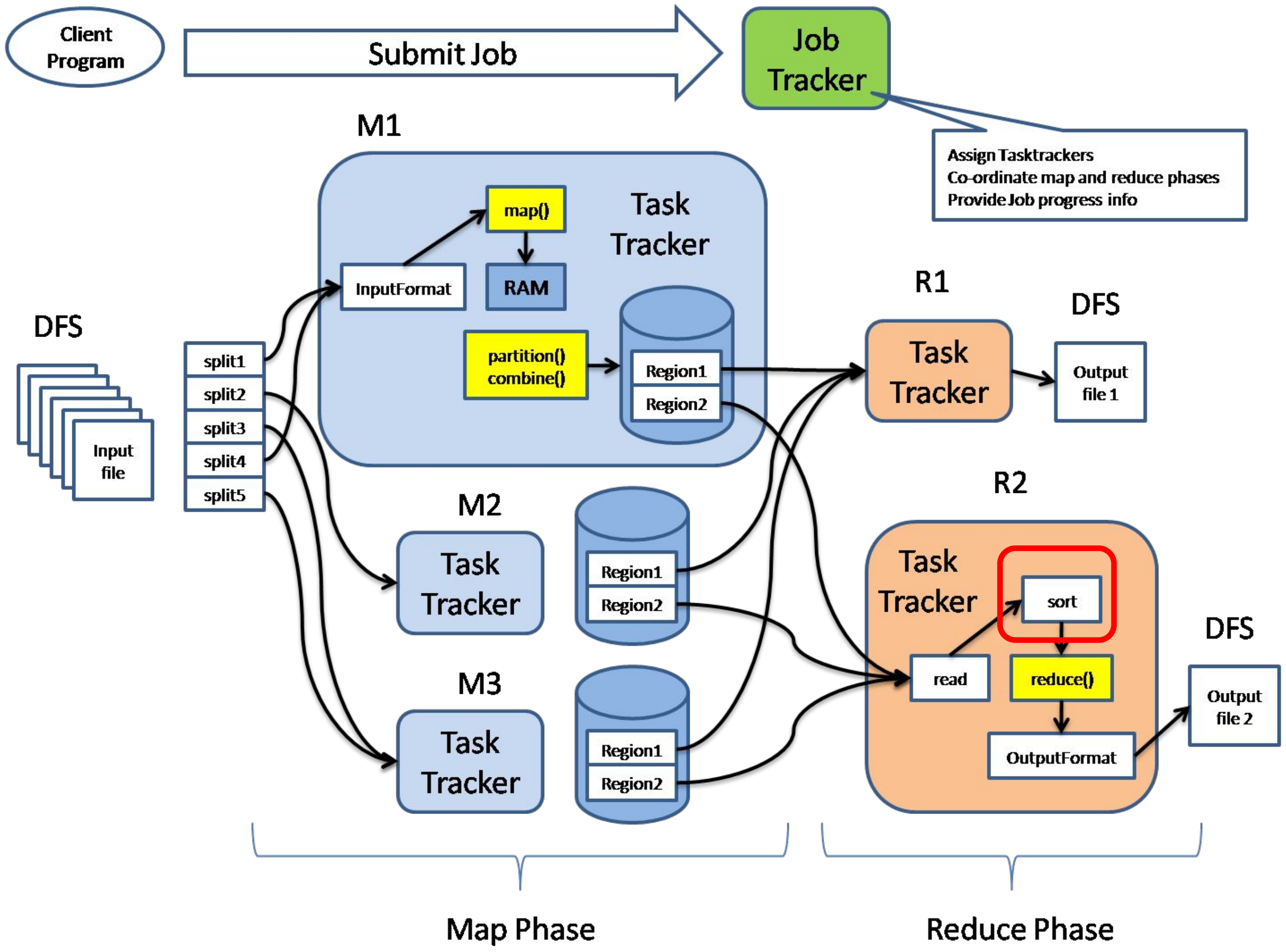



**PartitionFunction:** Assign the output of Mapping to the appropriate Reducer.

https://www.analyticsvidhya.com/blog/2022/07/learn-everything-about-mapreduce-architecture-and-its-components/

# Hadoop

## ❖ Hadoop MapReduce



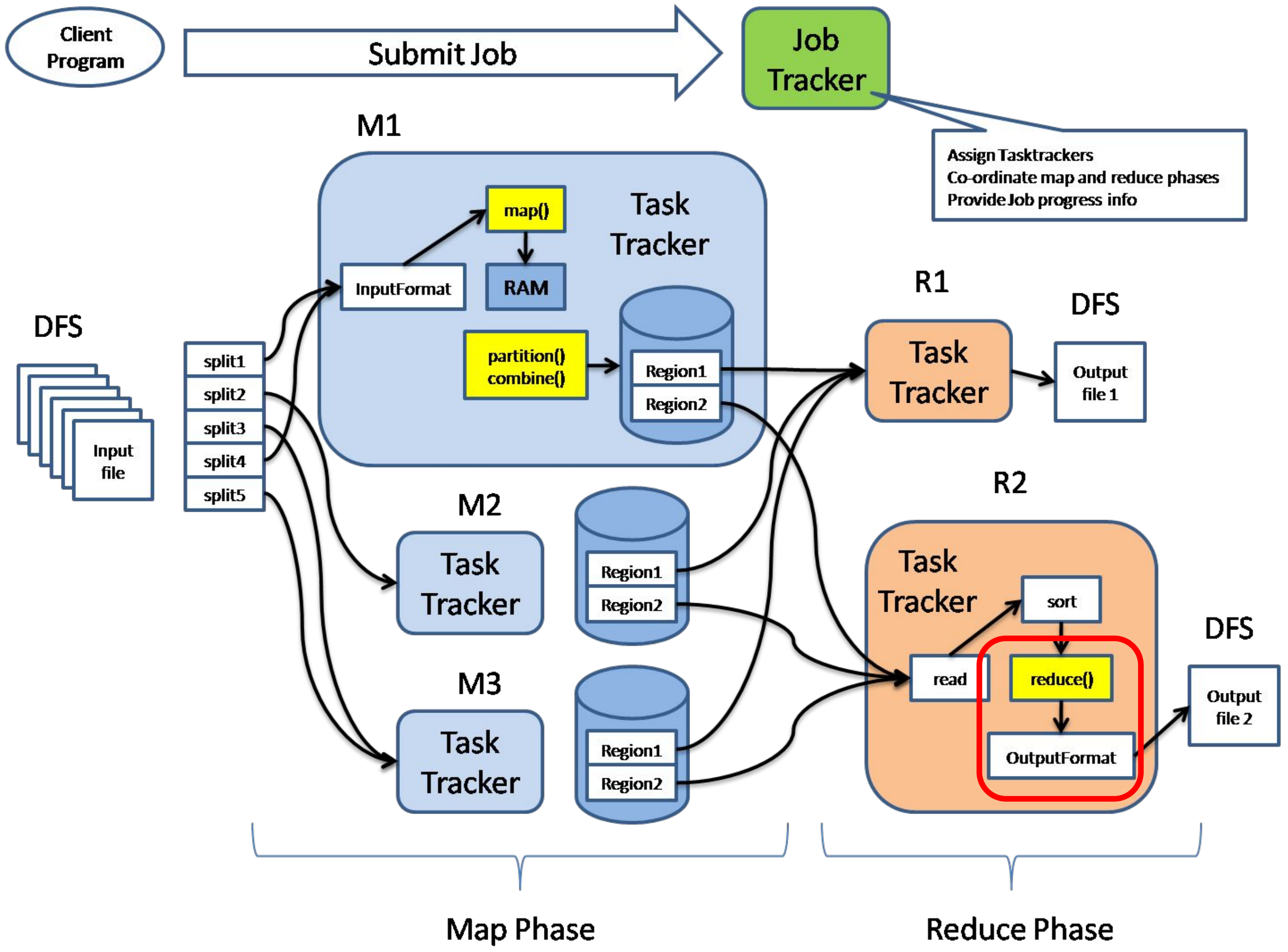

**ShufflingAndSorting:** Get the results of mapping and sorting



https://www.analyticsvidhya.com/blog/2022/07/learn-everything-about-mapreduce-architecture-and-its-components/

# Hadoop

## ❖ Hadoop MapReduce



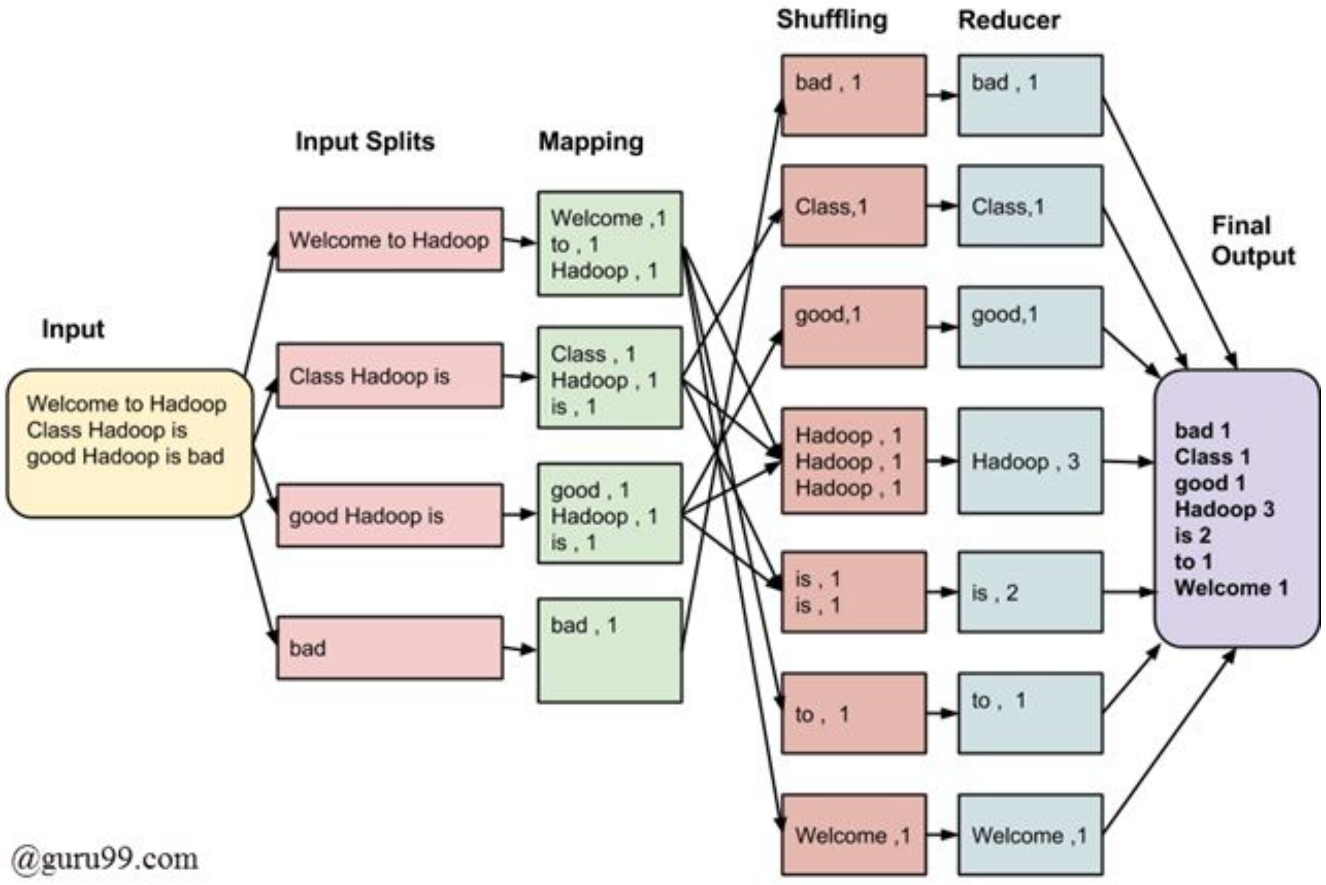

**ReduceFunction:** Grouping and processing the intermediate key-value pairs produced by the map phase.



https://www.analyticsvidhya.com/blog/2022/07/learn-everything-about-mapreduce-architecture-and-its-components/

# Hadoop

## ❖ Hadoop MapReduce: WordCount Example




https://www.guru99.com/introduction-to-mapreduce.html

# Hadoop

## ❖ Break-quiz

**1. What does HDFS stand for in Hadoop?**

A) Hadoop Distributed File System.

B) High-Definition File Storage.

C) Hadoop Data File System.

D) Hierarchical Distributed File Structure.

**2. Which programming model does Hadoop use for processing data?**

A) SQL.

B) MapReduce.

C) Machine Learning.

D) Graph Processing.

**3. What is the primary function of the NameNode in HDFS?**

A) Storing actual data.

B) Managing metadata and file system namespace.

C) Processing MapReduce jobs.

D) Coordinating data replication.

**4. How many copies of data does HDFS typically maintain by default?**

A) 1.

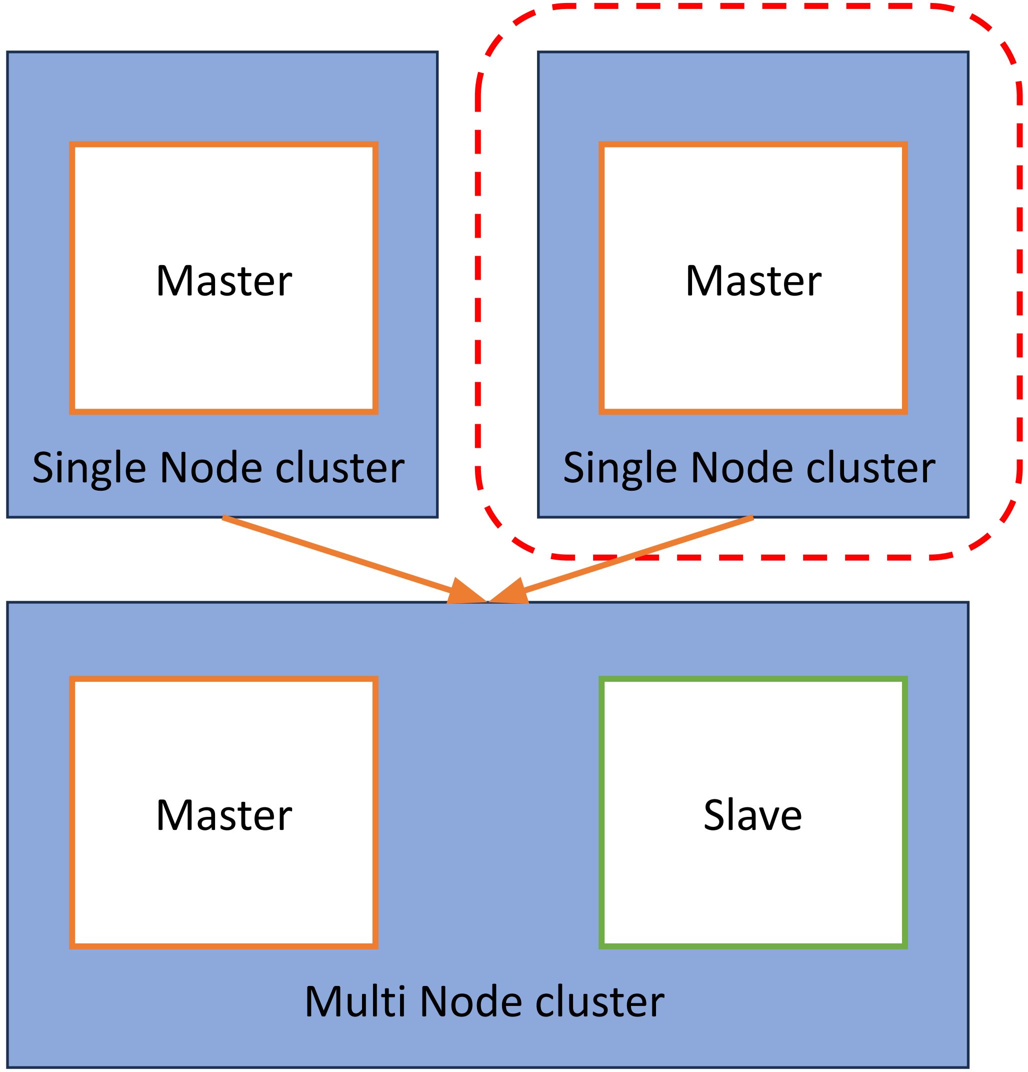
B) 2.

C) 3.

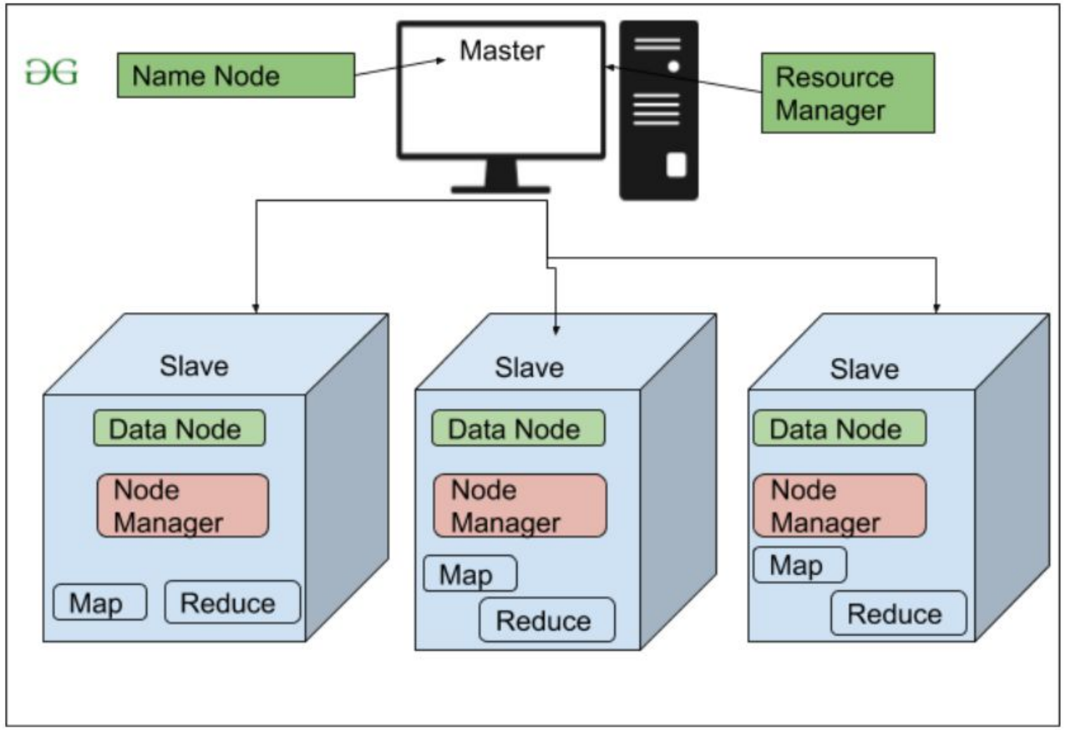
D) 4.

# Hadoop

## ❖ Installation




**Hadoop Single Node (Hadoop pseudo-distributed mode):** A configuration option in Hadoop that allows us to run Hadoop on a single machine.




https://www.geeksforgeeks.org/hadoop-architecture/

# Hadoop

## ❖ Installation: Requirements

1. Virtual Machine

VirtualBox

VMWare

Cloudera

2. CentOS

3. Hadoop

**Note:** The following installation demo is on Mac M1. You need to choose proper version for your computer.

# Hadoop

## ❖ Virtual Machine

| | | | |
|---|---|---|---|
| $VM_1$ Guest | $VM_2$ Guest | $VM_3$ Guest | $VM_4$ Guest |

Host Applications | Hypervisor

Host Operating System (Windows, Ubuntu, MacOS…)

Hardware Layer (CPU, GPU, RAM…)

**Virtual Machines (VMs):** An emulation of a computer system, where these machines use computer architectures to provide the functionality of a physical computer.


**Example:** A MacOS computer running two Windows VMs

# Hadoop

## ❖ Installation: VMware

**VMware:** A software providing machine virtualization

Download VMware Workstation Pro

VMware Workstation Pro is the industry standard desktop hypervisor for running virtual machines on Linux or Windows PCs. Start your free, fully functional 30-day trial today.

BUY ONLINE

**For Windows/Linux:** Use VMware Workstation

Run Windows and More on Mac

VMware Fusion

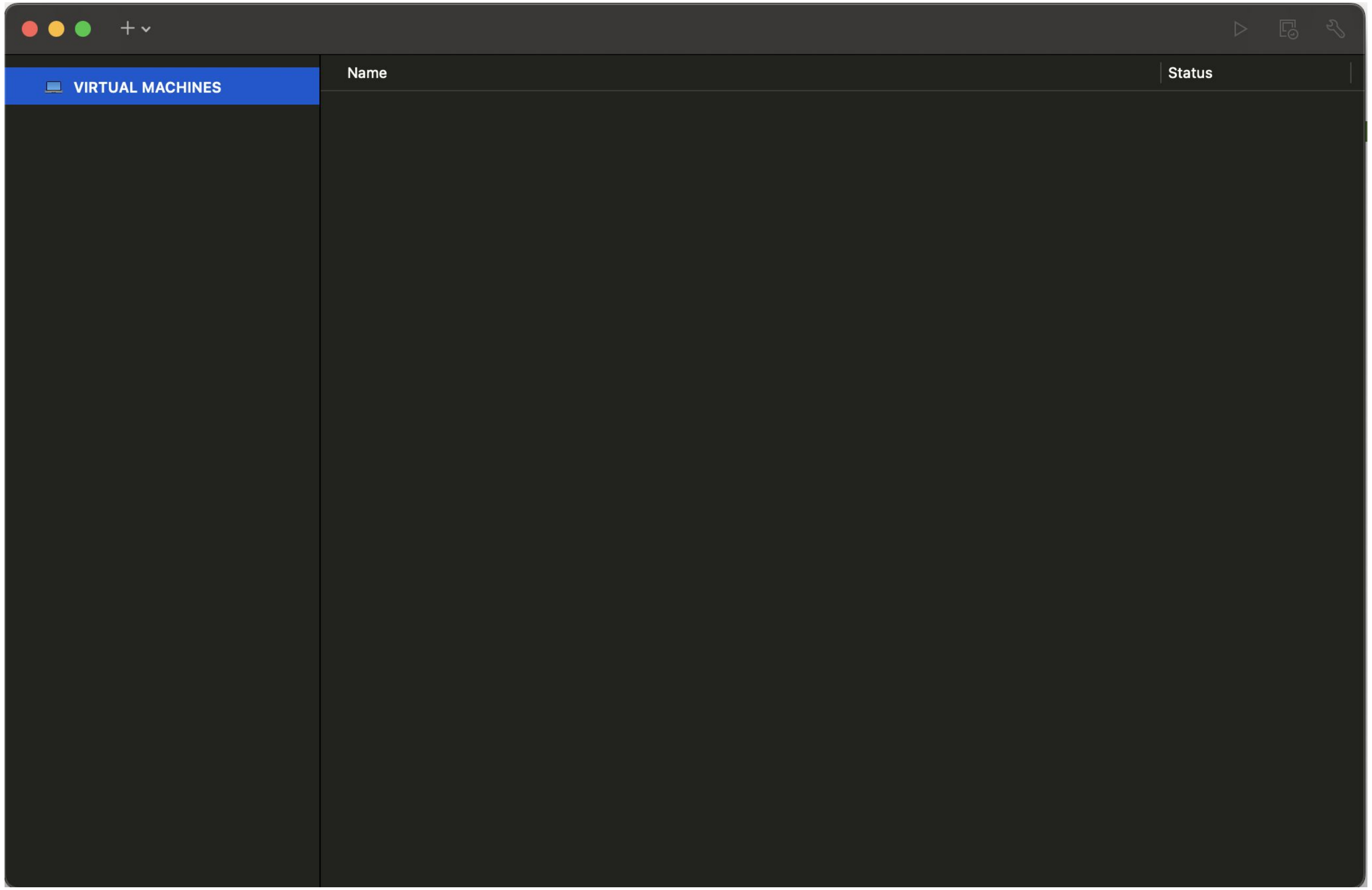
Harness the full power of your Mac when you use VMware Fusion to run Windows, Linux, containers, Kubernetes and more in virtual machines (VMs) without rebooting.

BUY ONLINE TRY FOR FREE

**For MacOS:** Use Vmware Fusion

# Hadoop

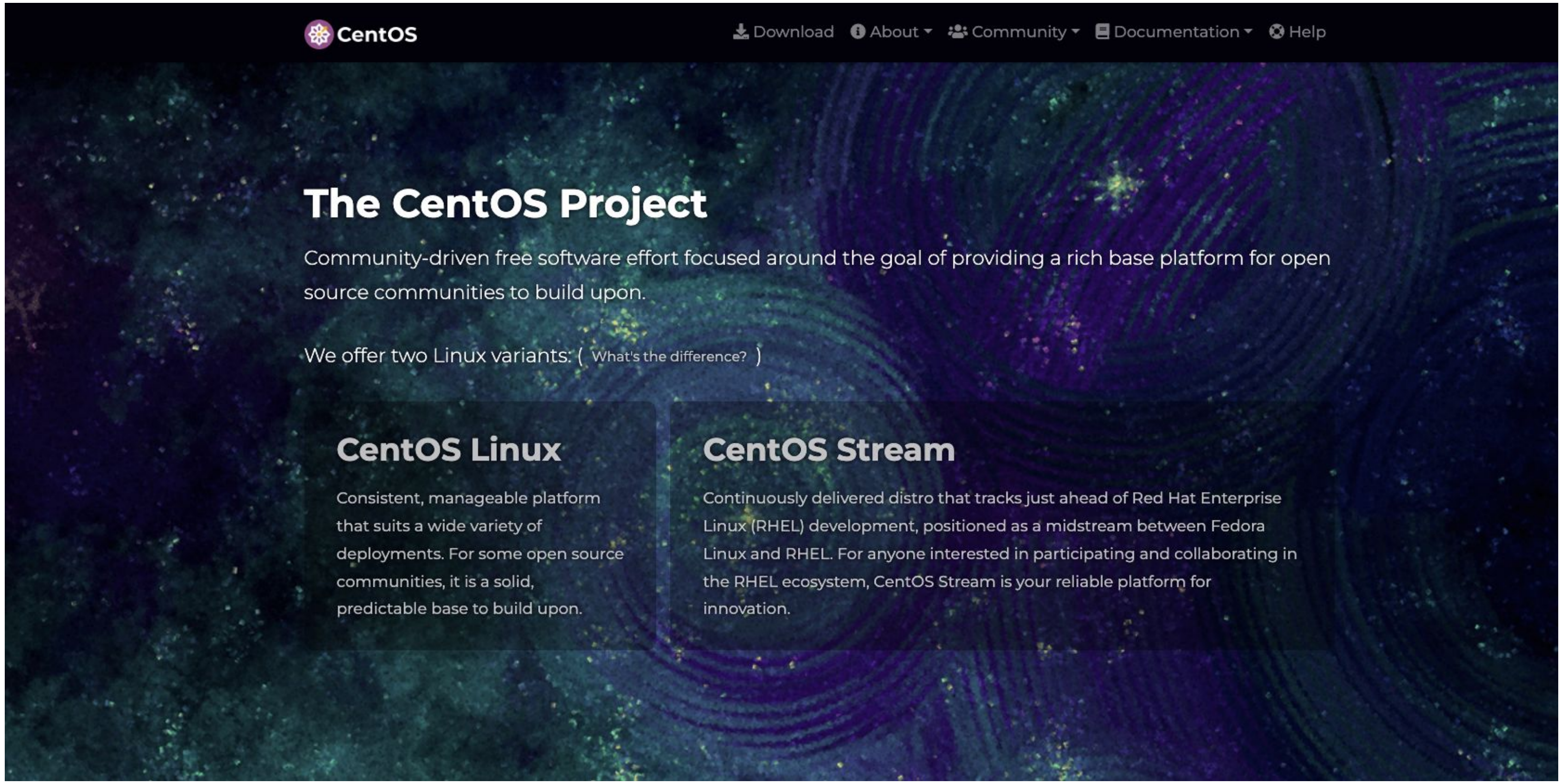
## ❖ Installation: VMware




VM Fusion Interface

# Hadoop

## ❖ Installation: CentOS




**CentOS:** Community-driven free software effort focused around the goal of providing a rich base platform for open source communities to build upon.

## ❖ Installation: CentOS

### Download

CentOS Stream

9 | 8

| Architectures | Packages | Others |
|---|---|---|
| x86_64 | RPMs | Cloud \| Containers \| Vagrant |
| ARM64 (aarch64) | RPMs | Cloud \| Containers \| Vagrant |
| IBM Power (ppc64le) | RPMs | Cloud \| Containers \| Vagrant |
| IBM Z (s390x) | RPMs | Cloud \| Containers \| Vagrant |

**Documentation**

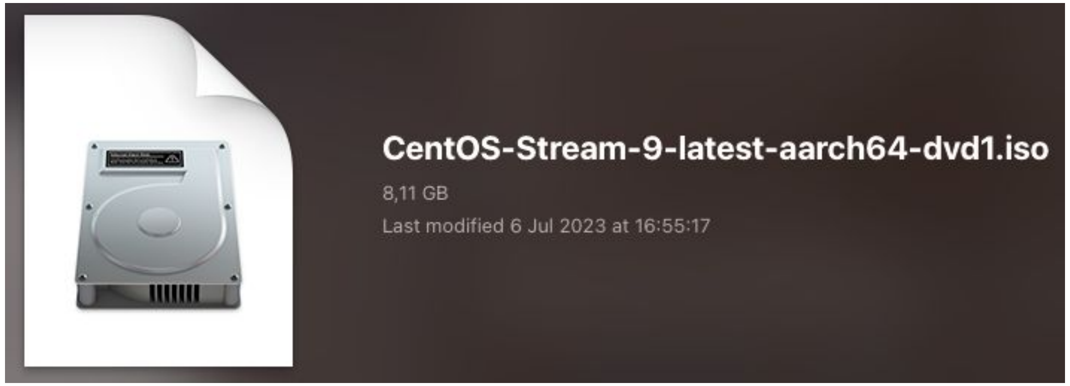
Release Notes | Release Email | Website

**End-of-life**

End of RHEL9 full support phase.

**Choose version:**

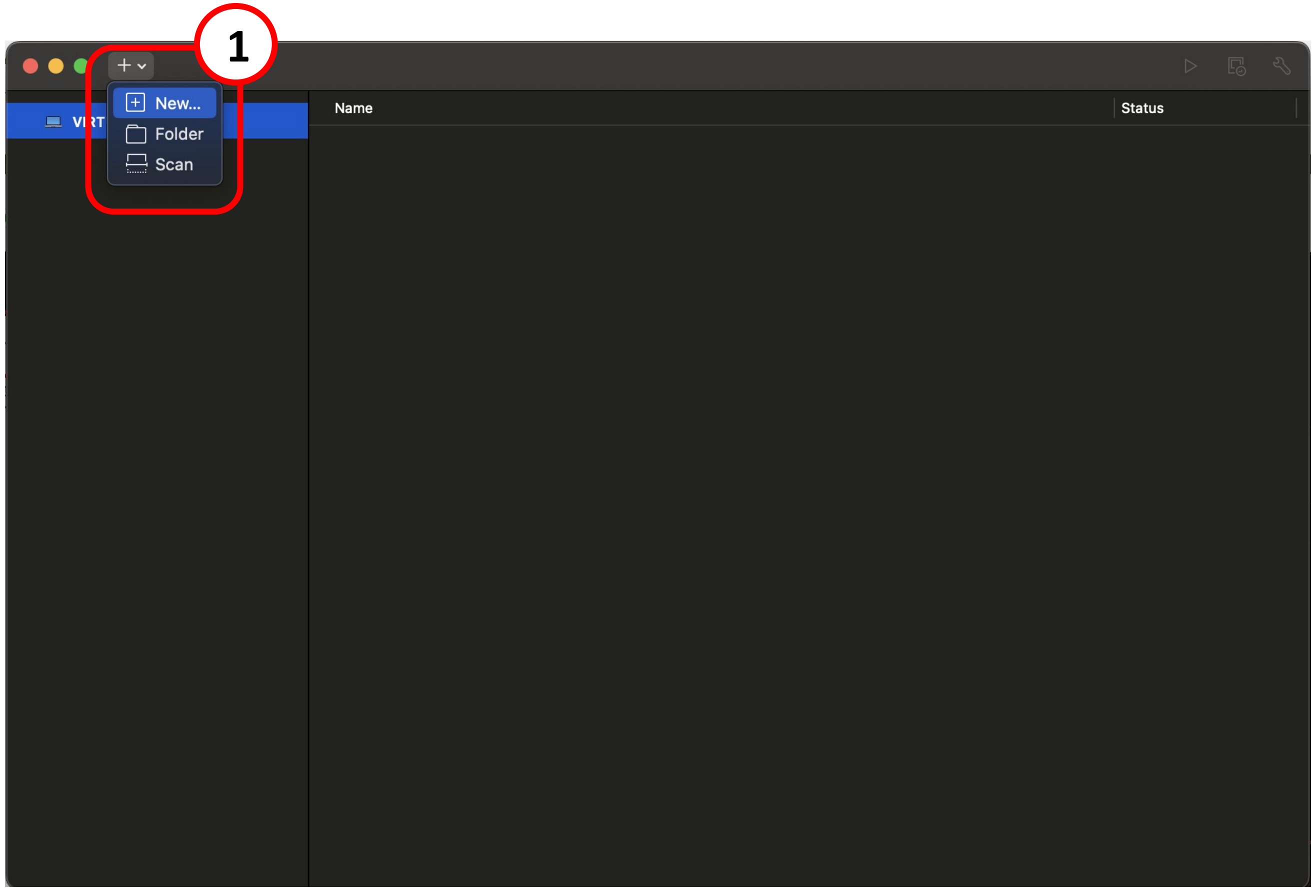
- **x86_64:** For Windows/Ubuntu/Mac Intel
- **ARM64 (aarch64):** For Mac M1/M2

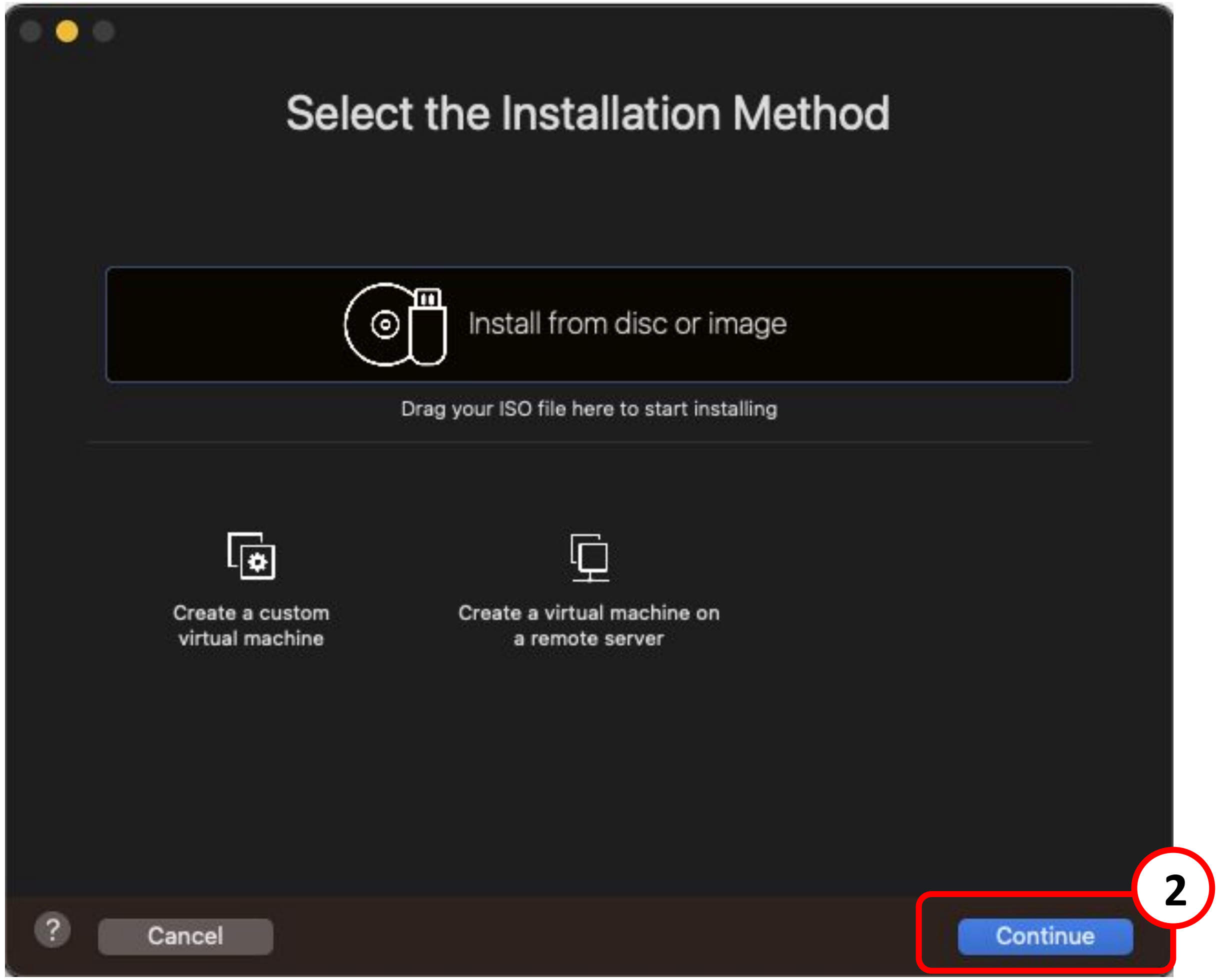




Download file .iso of CentOS: Link

# Hadoop

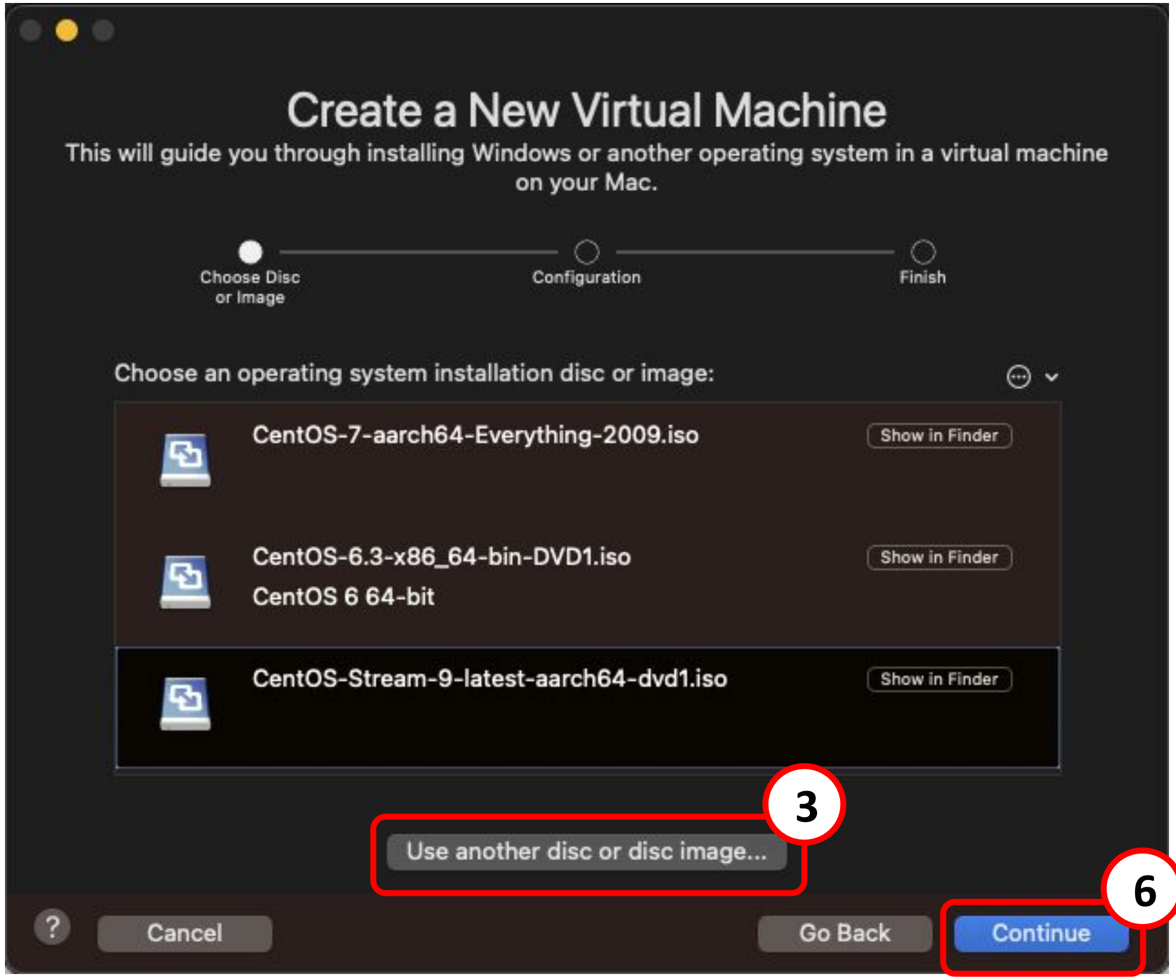
## ❖ Installation: Run CentOS in VMware

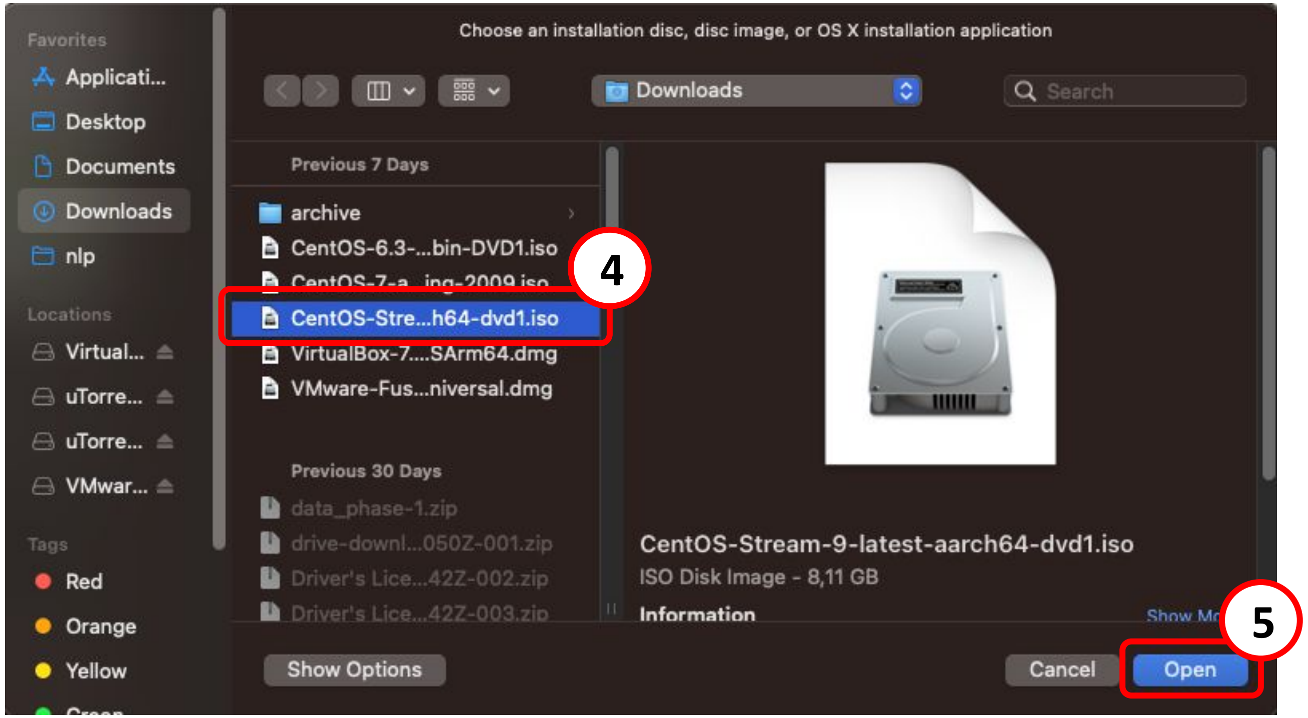


# Hadoop

## ❖ Installation: Run CentOS in VMware

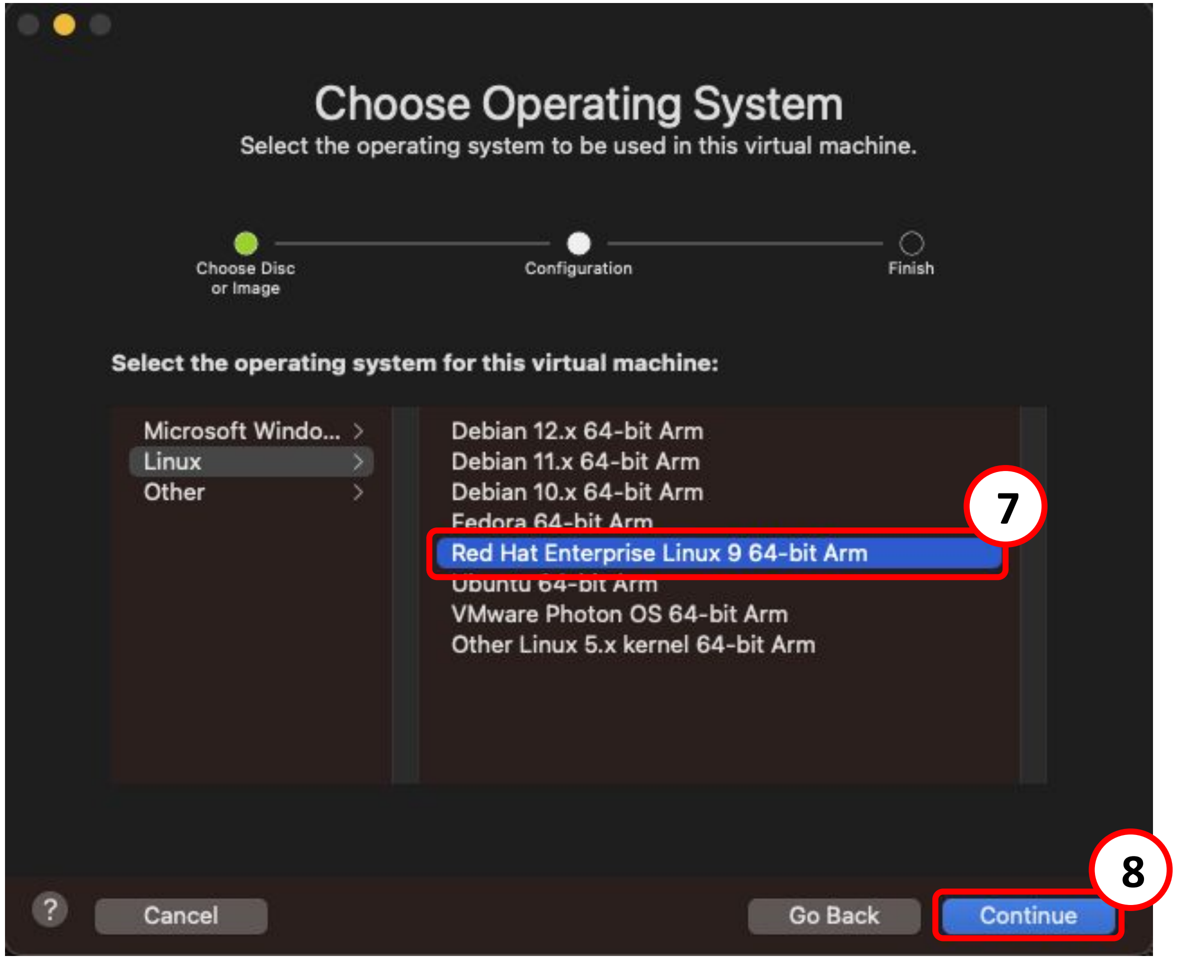


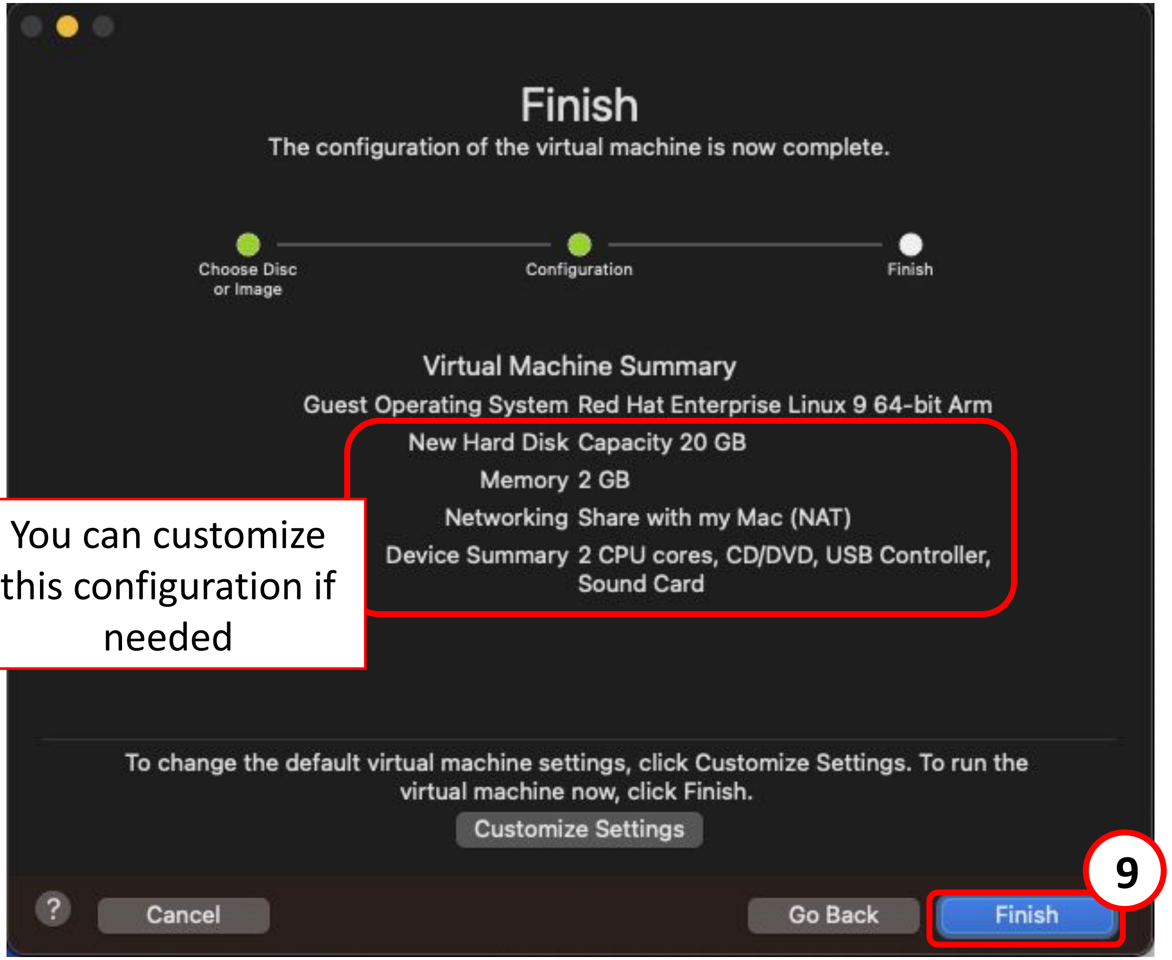

Choose CentOS .iso file

# Hadoop

## ❖ Installation: Run CentOS in VMware

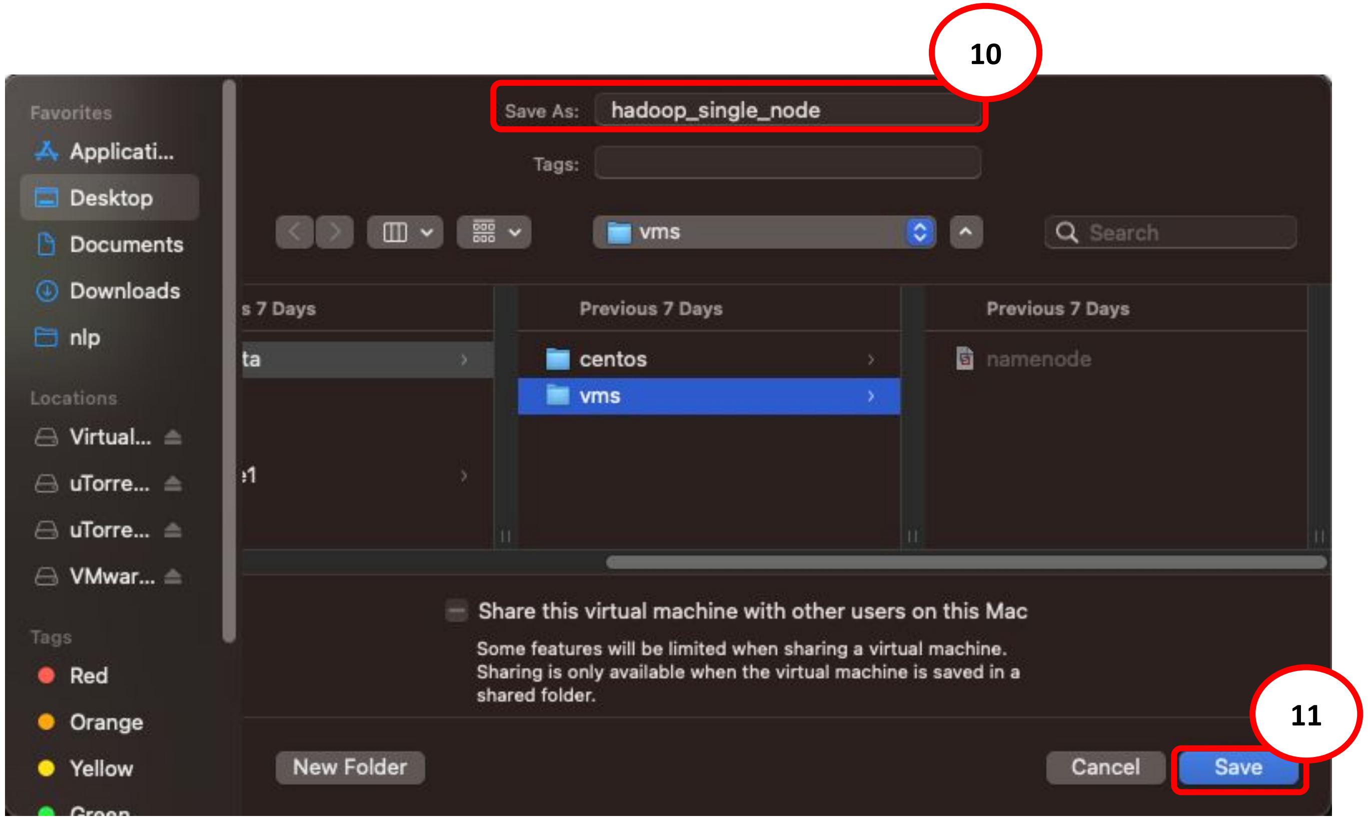


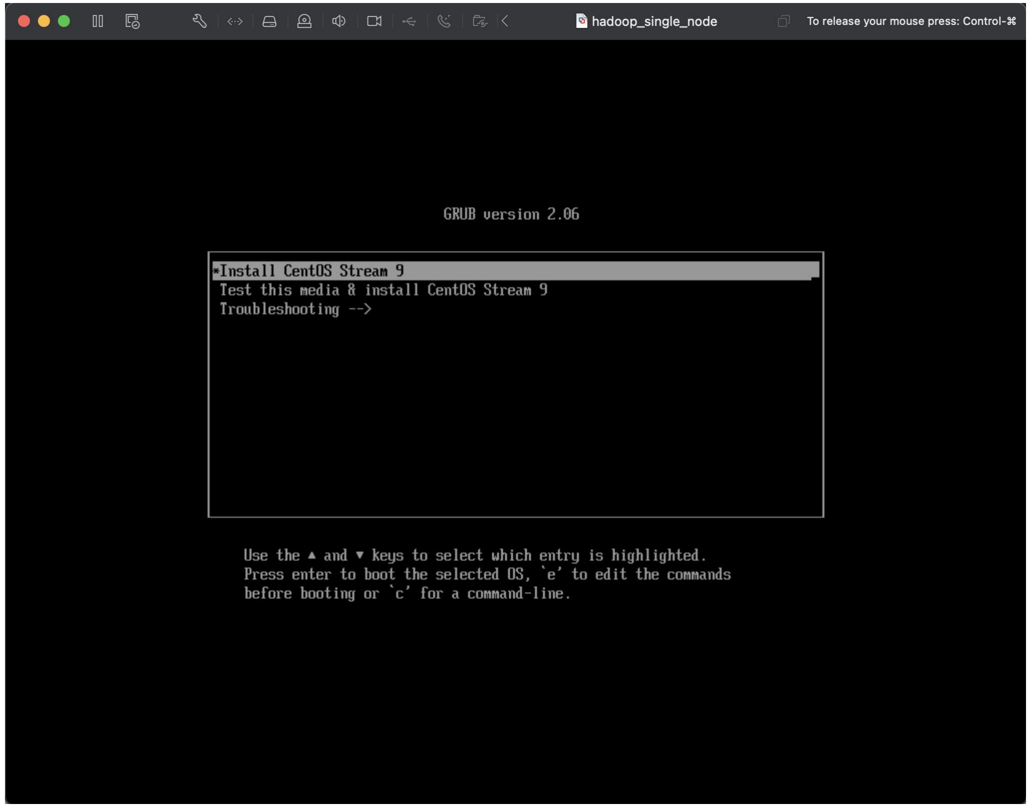

You can customize this configuration if needed



# Hadoop

## ❖ Installation: Run CentOS in VMware

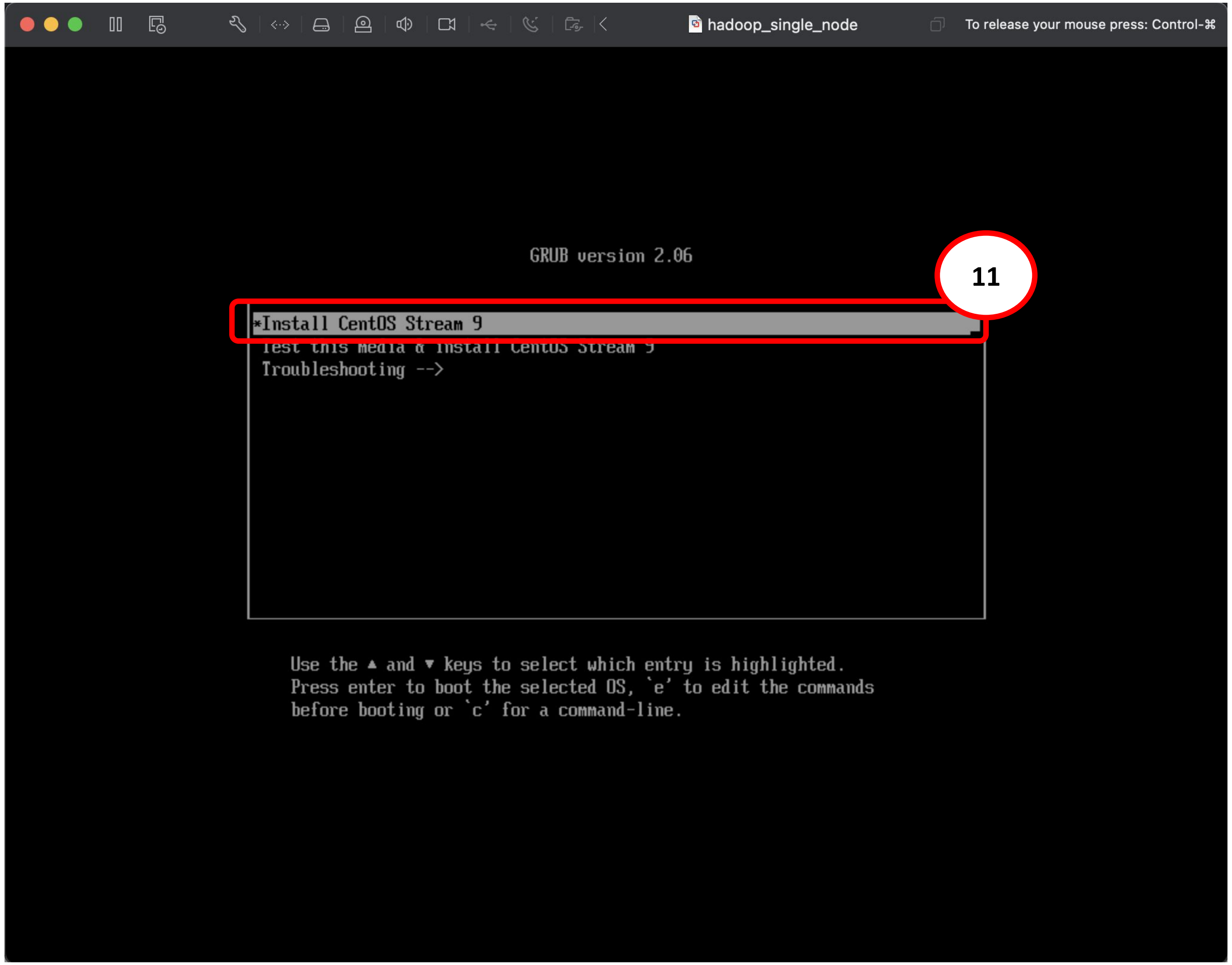



Choose where to save the new virtual machine

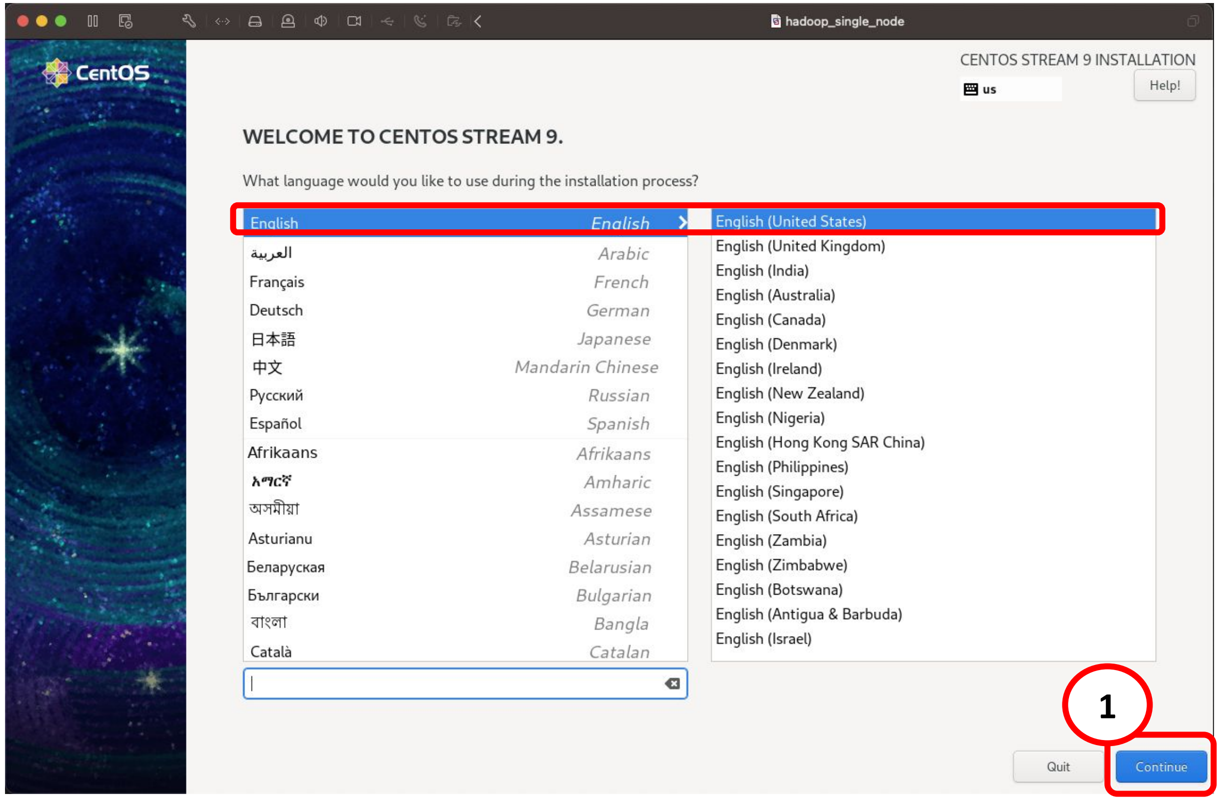
# Hadoop

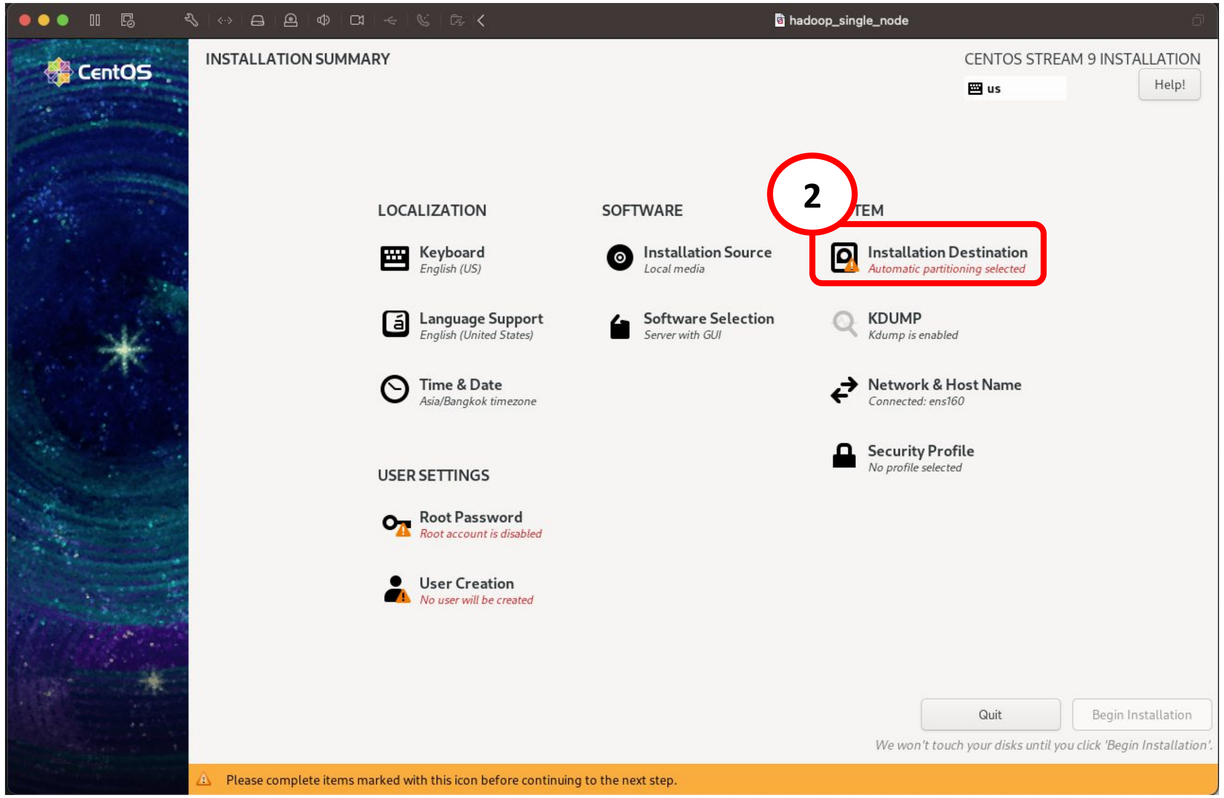
## ❖ Installation: Run CentOS in VMware



# Hadoop

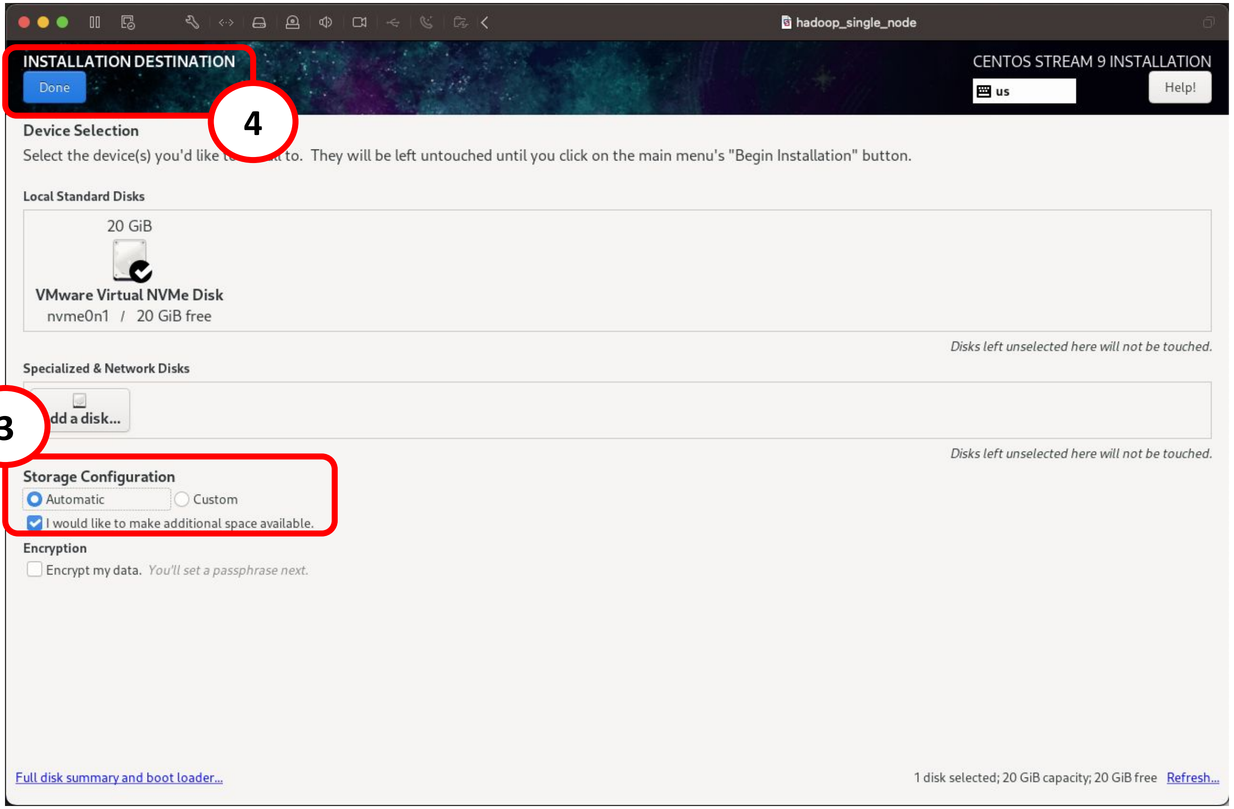
## ❖ Installation: Setup CentOS

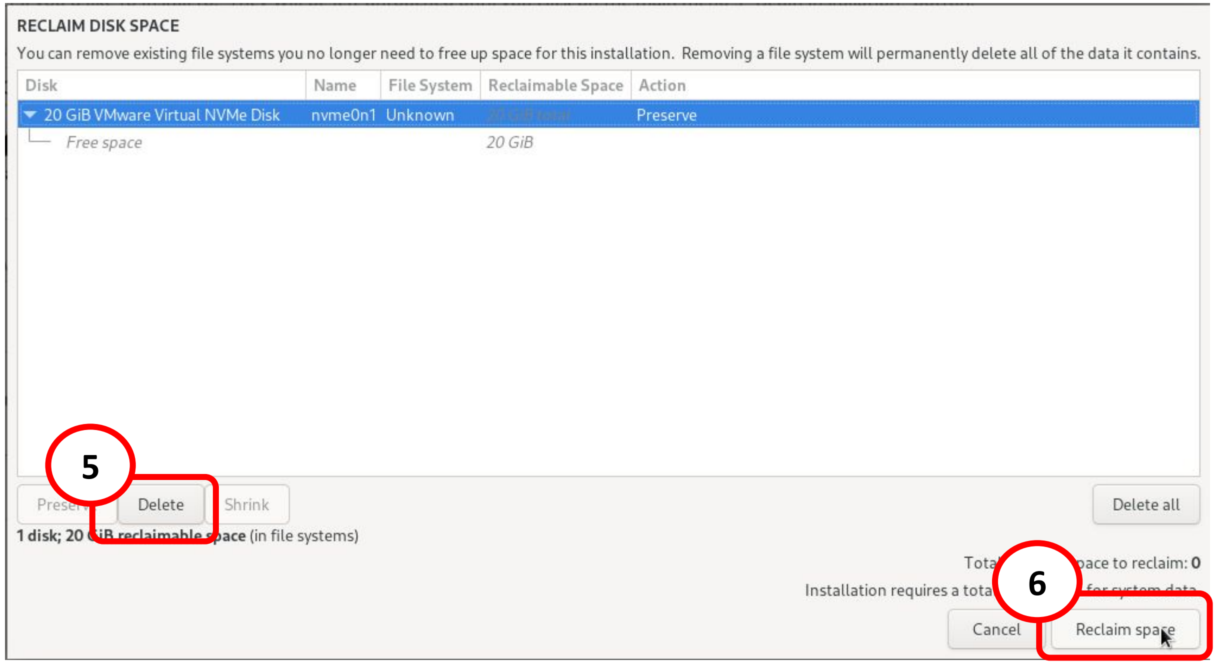


# Hadoop

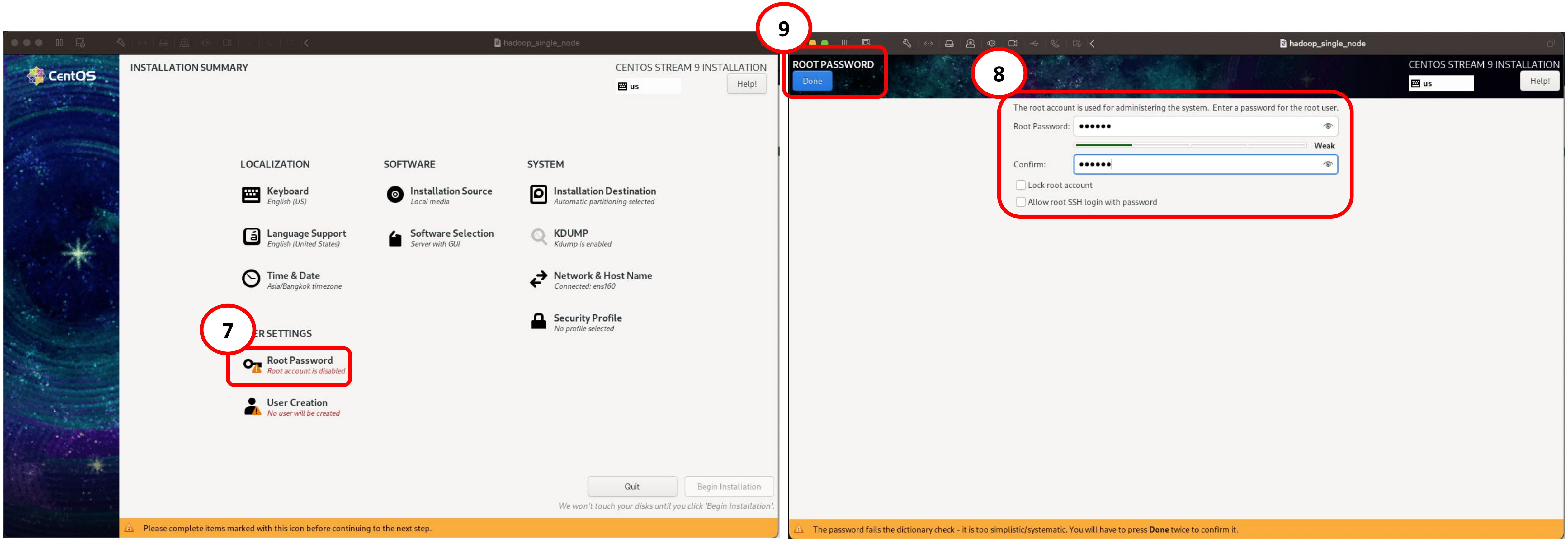
## ❖ Installation: Setup CentOS



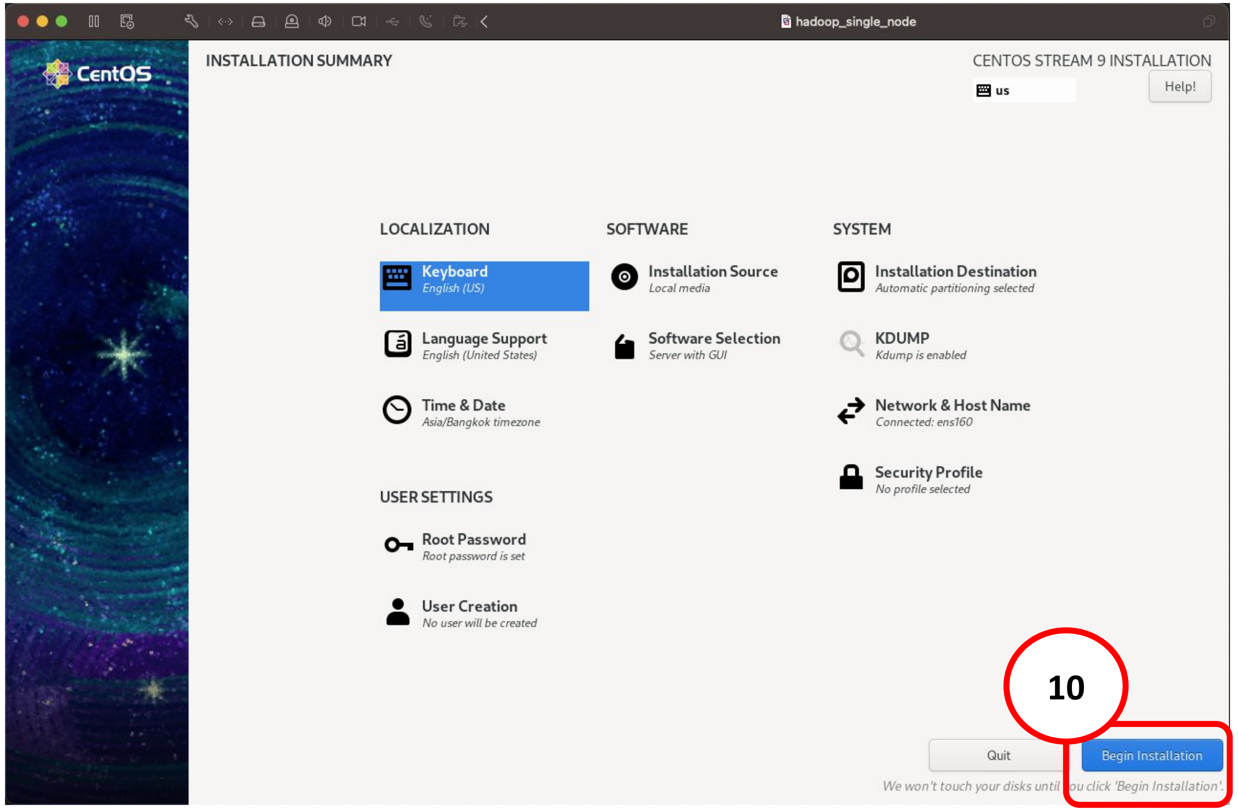
# Hadoop

## ❖ Installation: Setup CentOS



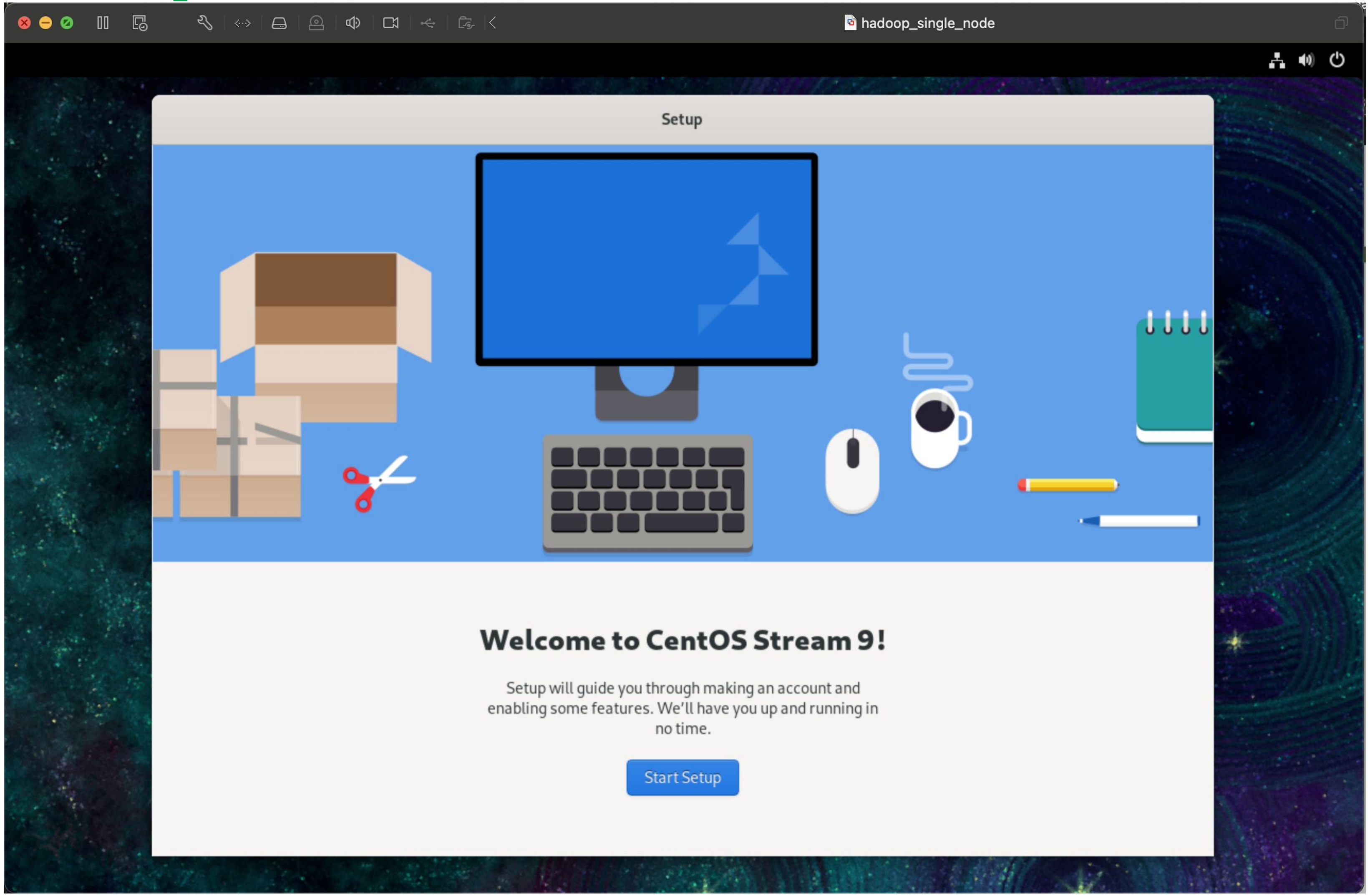
# Hadoop

## ❖ Installation: Setup CentOS



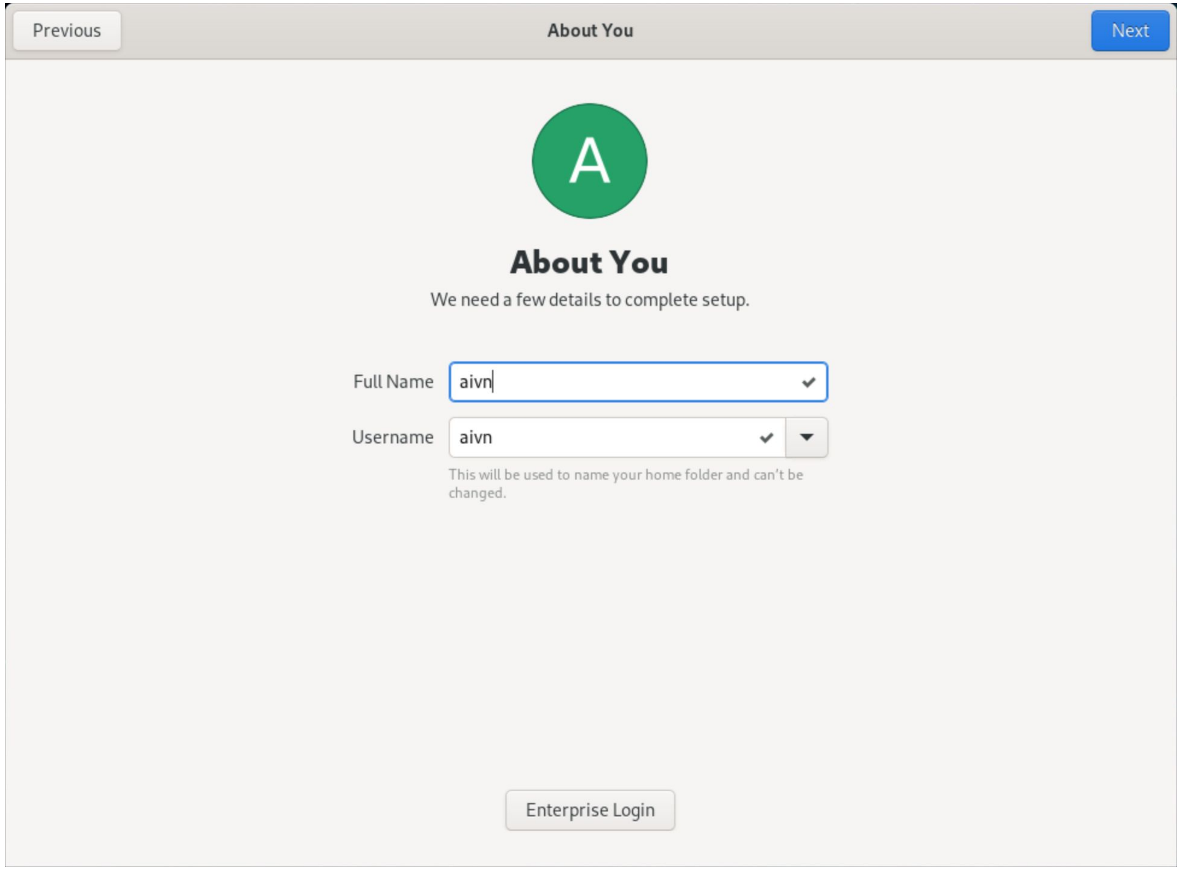

# Hadoop

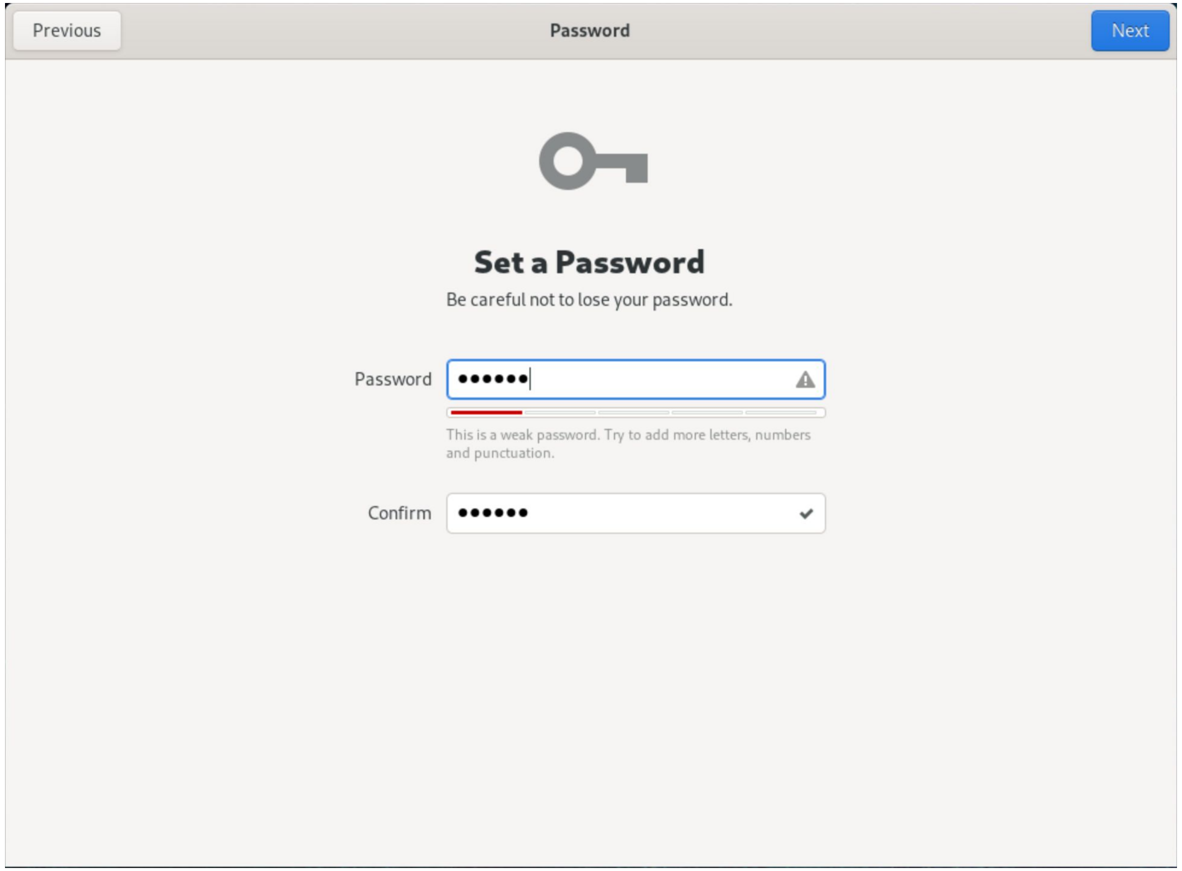
## ❖ Installation: Setup CentOS



# Hadoop

## ❖ Installation: Setup CentOS



# Hadoop

## ❖ Installation: Hadoop Requirements

1. Install Java

2. Install Hadoop

3. Set up Hadoop

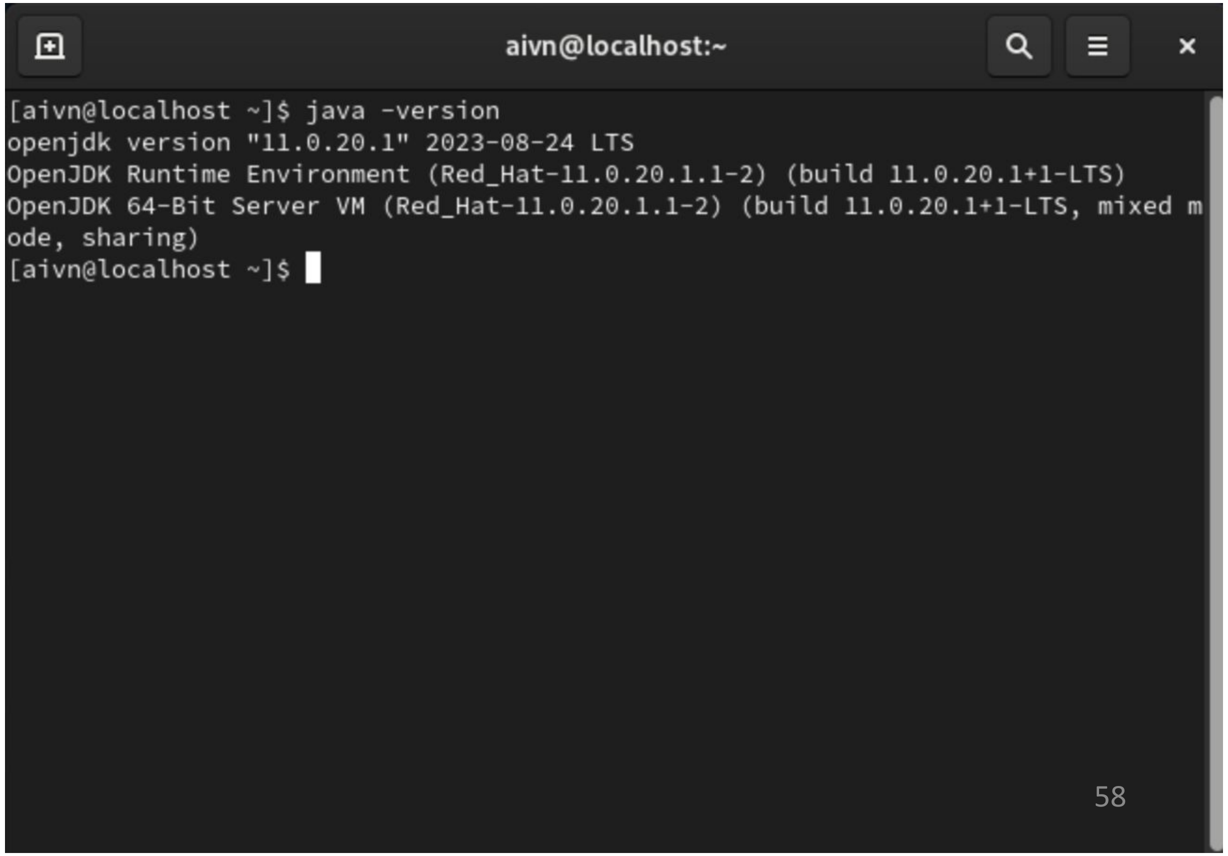
# Hadoop

## ❖ Installation: Java

**Java:** A high-level, class-based, object-oriented programming language that is designed to have as few implementation dependencies as possible.

**Install Java 8 on CentOS:**

1. `sudo yum update -y`
2. `sudo yum install java-11-openjdk-devel -y`



# Hadoop

## ❖ Installation: Config ssh

aivn@localhost:~

```
[aivn@localhost ~]$ ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
Generating public/private rsa key pair.
Created directory '/home/aivn/.ssh'.
Your identification has been saved in /home/aivn/.ssh/id_rsa
Your public key has been saved in /home/aivn/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:uPJLiVZDYYIbdlRATYHfG+2lPIl7XGUfKdlhqfJ+onM aivn@localhost.localdomain
The key's randomart image is:
+---[RSA 3072]----+
|    +=**.       .|
|   + oo..       + |
| . + .... .    = o|
|  .  ...o ...+o+ |
|       + S* +oo...|
|      o +o * ..  .|
|     + +  o o.    |
|    . +  . o. E . |
|       o. . .+ o  |
+----[SHA256]-----+
[aivn@localhost ~]$
```

aivn@localhost:~ — ssh localhost

```
[aivn@localhost ~]$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
[aivn@localhost ~]$ chmod 0600 ~/.ssh/authorized_keys
[aivn@localhost ~]$ ssh localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is SHA256:hLsA7L850MS4eyT+i+dsSnw9iDrE7ErypPzwTorOXig.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Fri Aug  2 23:40:23 2024
[aivn@localhost ~]$
```

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(TA Session)

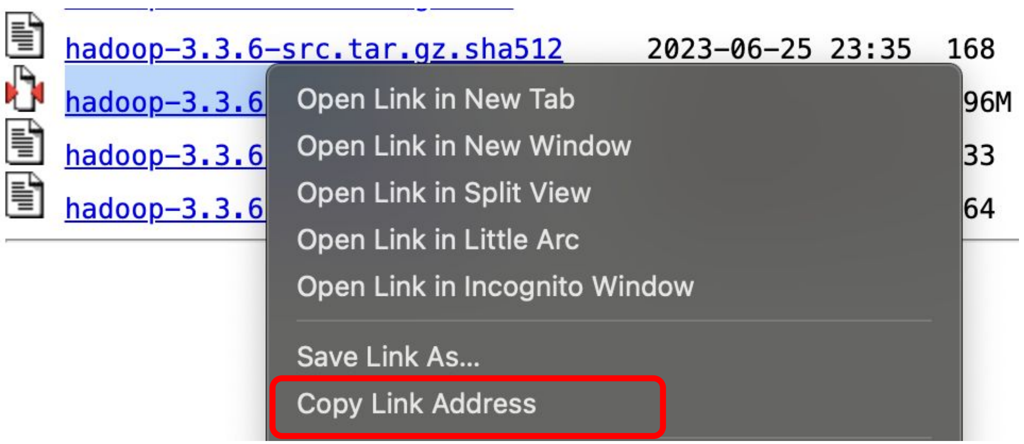
# Hadoop

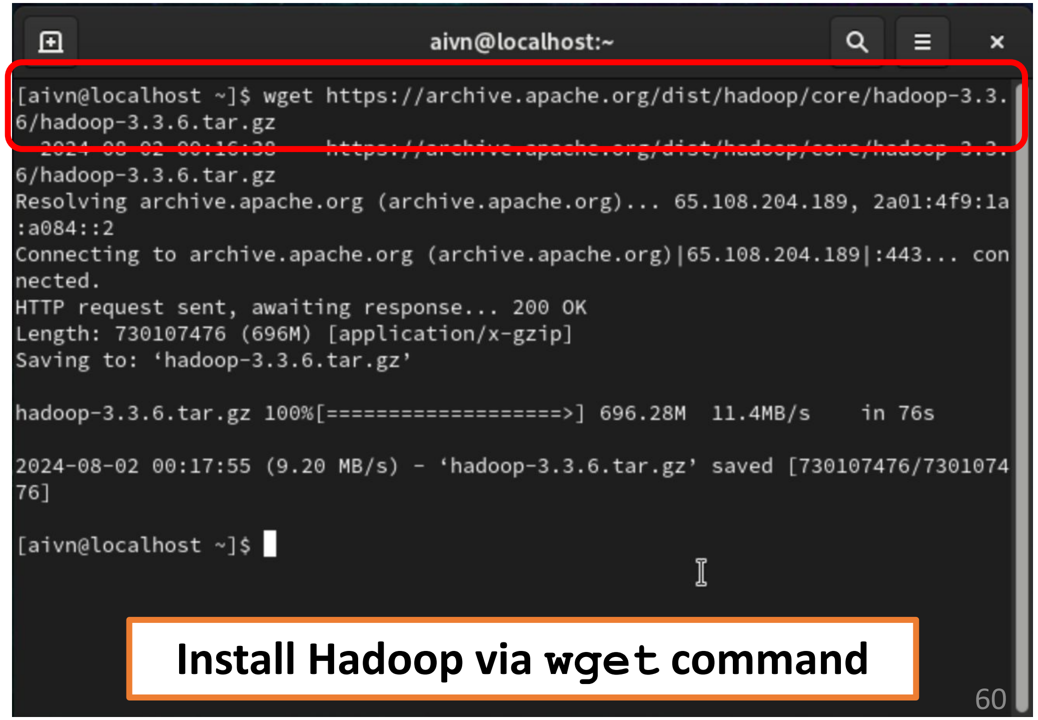
## ❖ Installation: Hadoop

### Index of /dist/hadoop/core/hadoop-3.3.6

| Name | Last modified | Size | Description |
|---|---|---|---|
| Parent Directory | | - | |
| CHANGELOG.md | 2023-06-25 23:35 | 22K | |
| CHANGELOG.md.asc | 2023-06-25 23:35 | 833 | |
| CHANGELOG.md.sha512 | 2023-06-25 23:35 | 153 | |
| RELEASENOTES.md | 2023-06-25 23:35 | 1.4K | |
| RELEASENOTES.md.asc | 2023-06-25 23:35 | 833 | |
| RELEASENOTES.md.sha512 | 2023-06-25 23:35 | 156 | |
| hadoop-3.3.6-aarch64.tar.gz | 2023-06-25 23:35 | 710M | |
| hadoop-3.3.6-aarch64.tar.gz.asc | 2023-06-25 23:35 | 833 | |
| hadoop-3.3.6-aarch64.tar.gz.sha512 | 2023-06-25 23:35 | 164 | |
| hadoop-3.3.6-rat.txt | 2023-06-25 23:35 | 2.0M | |
| hadoop-3.3.6-rat.txt.asc | 2023-06-25 23:35 | 833 | |
| hadoop-3.3.6-rat.txt.sha512 | 2023-06-25 23:35 | 165 | |
| hadoop-3.3.6-site.tar.gz | 2023-06-25 23:35 | 41M | |
| hadoop-3.3.6-site.tar.gz.asc | 2023-06-25 23:35 | 833 | |
| hadoop-3.3.6-site.tar.gz.sha512 | 2023-06-25 23:35 | 169 | |
| hadoop-3.3.6-src.tar.gz | 2023-06-25 23:35 | 36M | |
| hadoop-3.3.6-src.tar.gz.asc | 2023-06-25 23:35 | 833 | |
| hadoop-3.3.6-src.tar.gz.sha512 | 2023-06-25 23:35 | 168 | |
| hadoop-3.3.6.tar.gz | 2023-06-25 23:35 | 696M | |
| hadoop-3.3.6.tar.gz.asc | 2023-06-25 23:35 | 833 | |
| hadoop-3.3.6.tar.gz.sha512 | 2023-06-25 23:35 | 164 | |

Hadoop 3.3.6 download page

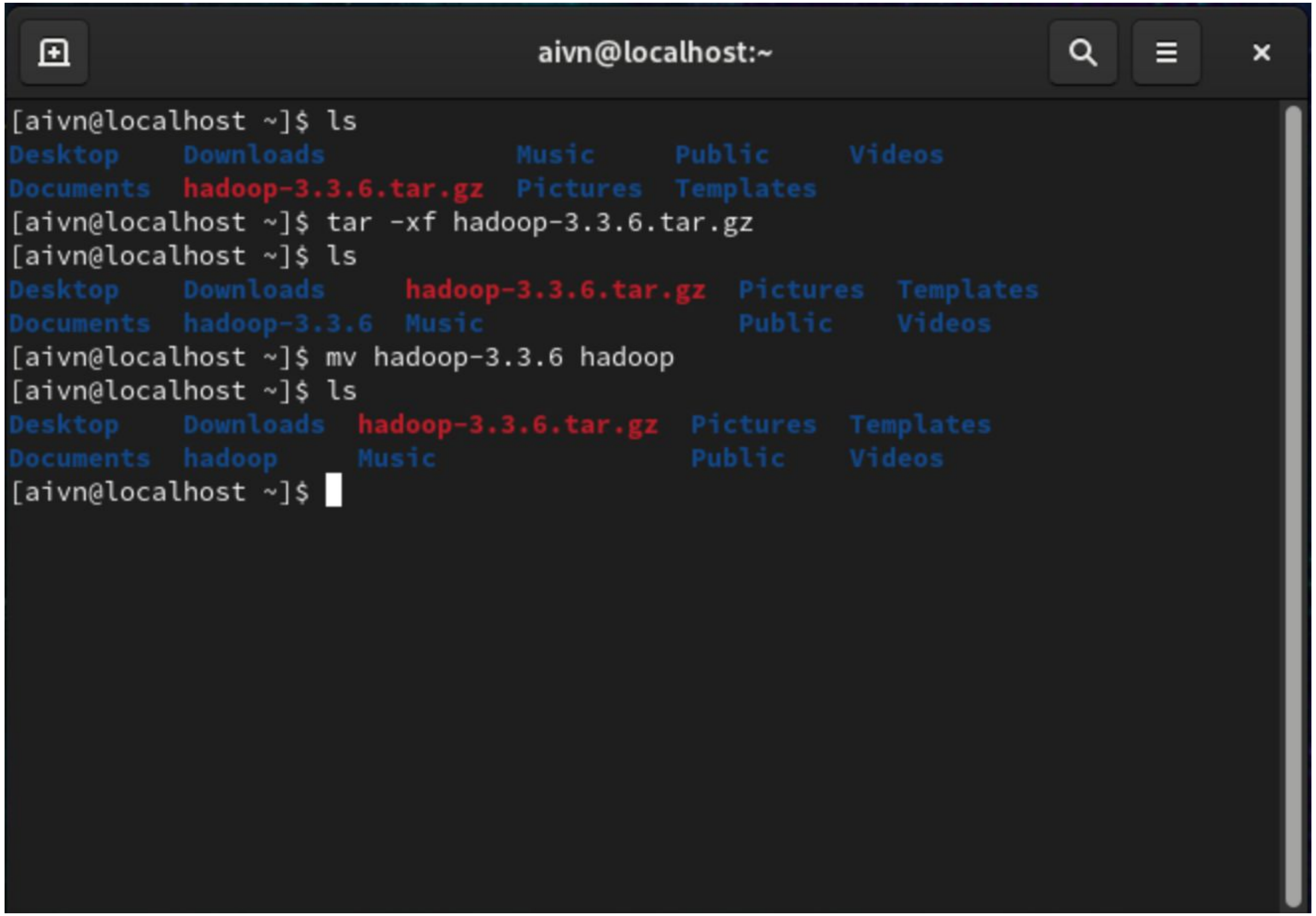



Install Hadoop via wget command

# Hadoop

## ❖ Installation: Hadoop




- **Extract downloaded file via tar command:**

```
tar -xf hadoop-3.3.6.tar.gz
```

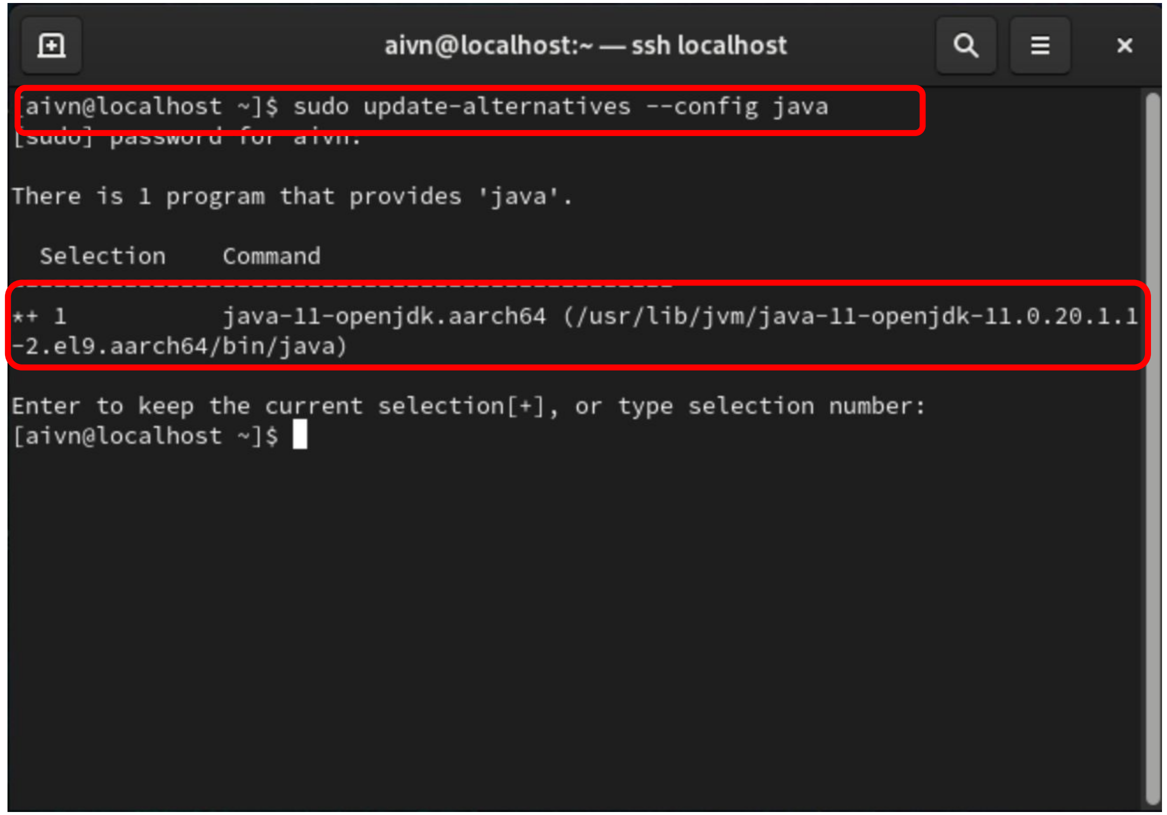
- **Rename Hadoop folder:**

```
mv hadoop-3.3.6 hadoop
```

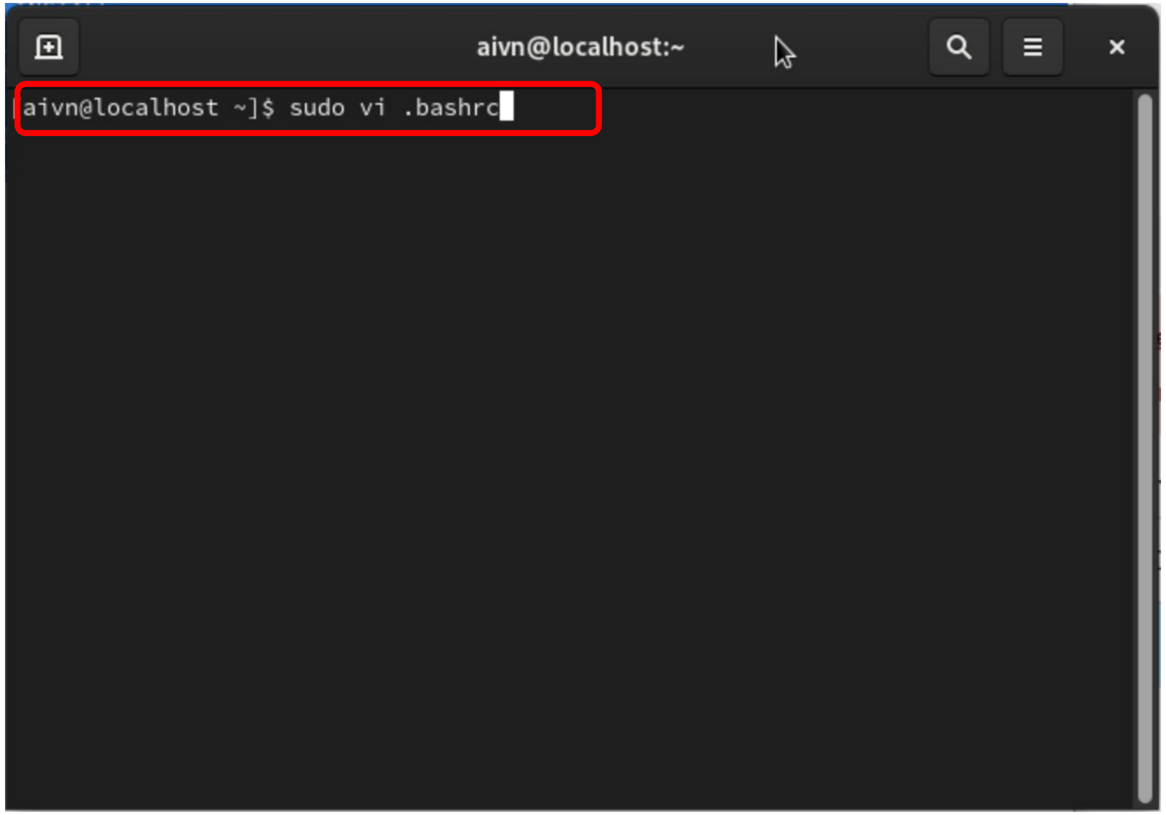
# Hadoop

## ❖ Installation: Config Hadoop

**1. Check Java configuration**



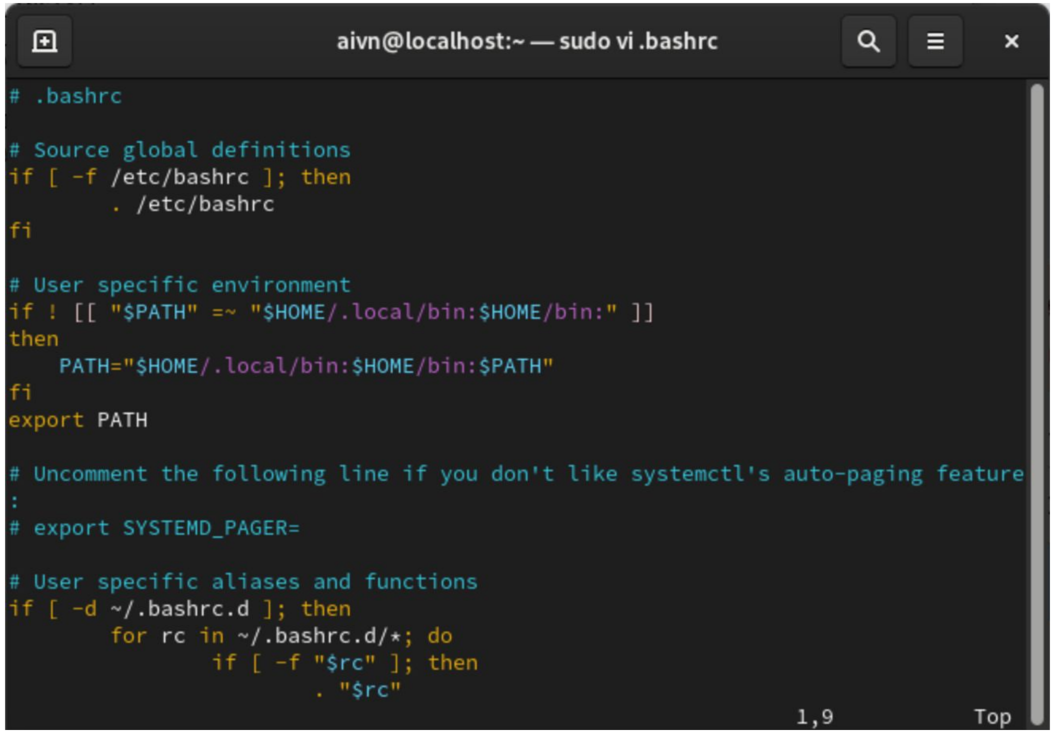

**2. Enter ~/.bashrc script**



# Hadoop

## ❖ Installation: Hadoop

**3. Set up .bashrc script**




**Paste the code below:**

```
export HDFS_NAMENODE_USER="aivn"
export HDFS_DATANODE_USER="aivn"
export HDFS_SECONDARYNAMENODE_USER="aivn"
export YARN_RESOURCEMANAGER_USER="aivn"
export YARN_NODEMANAGER_USER="aivn"
export
JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.20.1.1-2.el9.a
arch64
export HADOOP_HOME=/home/aivn/hadoop
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export
HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"
```

# Hadoop

## ❖ Installation: Hadoop

```
aivn@localhost:~

# .bashrc

# Source global definitions
if [ -f /etc/bashrc ]; then
        . /etc/bashrc
fi

export HDFS_NAMENODE_USER="aivn"
export HDFS_DATANODE_USER="aivn"
export HDFS_SECONDARYNAMENODE_USER="aivn"
export YARN_RESOURCEMANAGER_USER="aivn"
export YARN_NODEMANAGER_USER="aivn"
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.20.1.1-2.el9.aarch64
export HADOOP_HOME=/home/aivn/hadoop
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"

export SPARK_HOME=/home/aivn/spark
                                                      1,1           Top
```

```
aivn@localhost:~

[aivn@localhost ~]$ source .bashrc
[aivn@localhost ~]$
```

**3. Set up .bashrc script then run it**

# Hadoop

## ❖ Installation: Hadoop

**4. Config hadoop-env.sh**

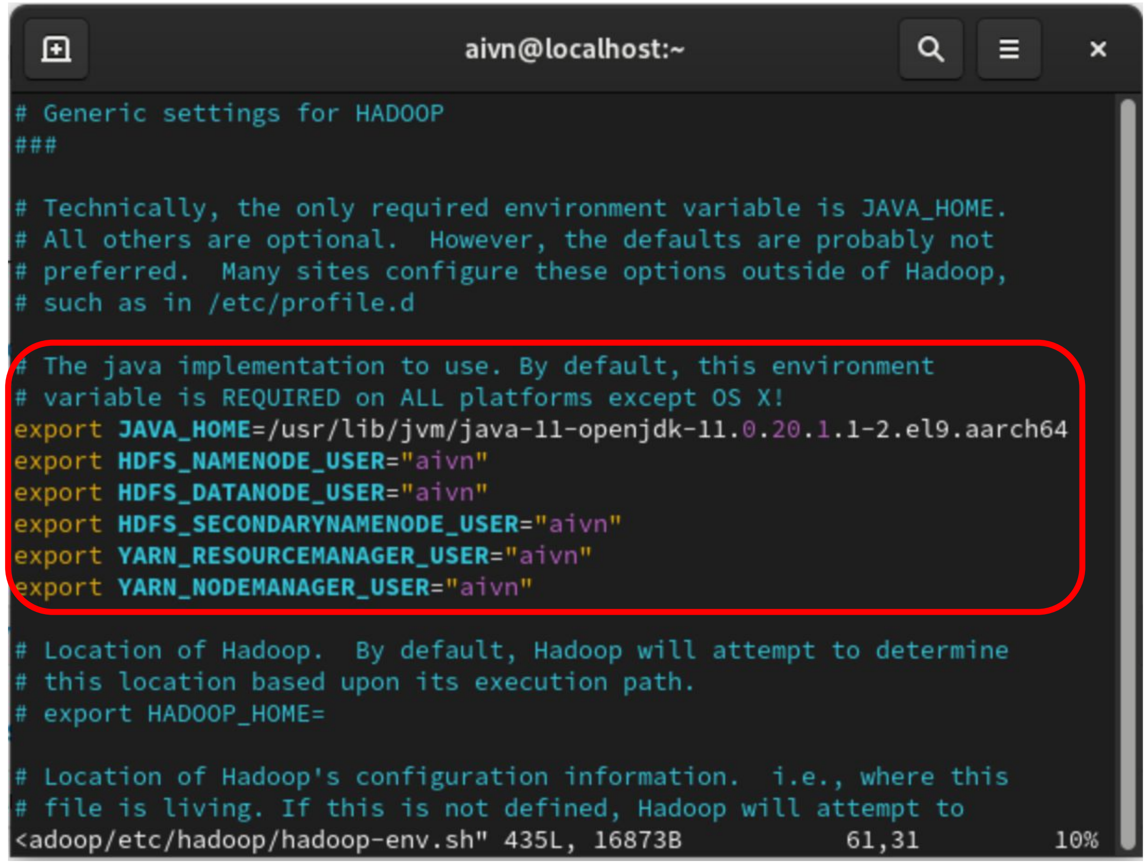
```
aivn@localhost:~
[aivn@localhost ~]$ sudo vi hadoop/etc/hadoop/hadoop-env.sh
```

```
aivn@localhost:~
# Generic settings for HADOOP
###

# Technically, the only required environment variable is JAVA_HOME.
# All others are optional.  However, the defaults are probably not
# preferred.  Many sites configure these options outside of Hadoop,
# such as in /etc/profile.d

# The java implementation to use. By default, this environment
# variable is REQUIRED on ALL platforms except OS X!
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.20.1.1-2.el9.aarch64
export HDFS_NAMENODE_USER="aivn"
export HDFS_DATANODE_USER="aivn"
export HDFS_SECONDARYNAMENODE_USER="aivn"
export YARN_RESOURCEMANAGER_USER="aivn"
export YARN_NODEMANAGER_USER="aivn"

# Location of Hadoop.  By default, Hadoop will attempt to determine
# this location based upon its execution path.
# export HADOOP_HOME=

# Location of Hadoop's configuration information.  i.e., where this
# file is living. If this is not defined, Hadoop will attempt to
<adoop/etc/hadoop/hadoop-env.sh" 435L, 16873B            61,31          10%
```

# Hadoop

## ❖ Installation: Hadoop

**4. Config hadoop-env.sh**



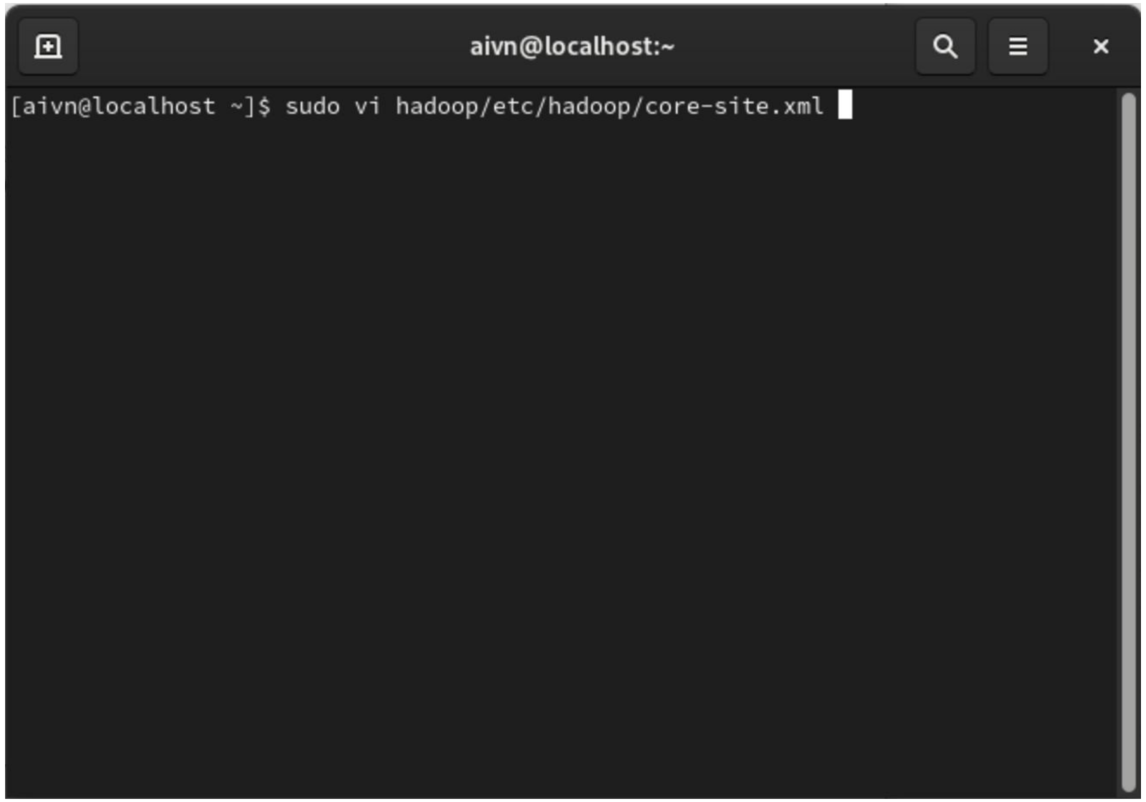
```
aivn@localhost:~
# Generic settings for HADOOP
###

# Technically, the only required environment variable is JAVA_HOME.
# All others are optional.  However, the defaults are probably not
# preferred.  Many sites configure these options outside of Hadoop,
# such as in /etc/profile.d

# The java implementation to use. By default, this environment
# variable is REQUIRED on ALL platforms except OS X!
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.20.1.1-2.el9.aarch64
export HDFS_NAMENODE_USER="aivn"
export HDFS_DATANODE_USER="aivn"
export HDFS_SECONDARYNAMENODE_USER="aivn"
export YARN_RESOURCEMANAGER_USER="aivn"
export YARN_NODEMANAGER_USER="aivn"

# Location of Hadoop.  By default, Hadoop will attempt to determine
# this location based upon its execution path.
# export HADOOP_HOME=

# Location of Hadoop's configuration information.  i.e., where this
# file is living. If this is not defined, Hadoop will attempt to
<adoop/etc/hadoop/hadoop-env.sh" 435L, 16873B          61,31          10%
```

- export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-11.0.20.1.1-2.el9.aarch64
- export HDFS_NAMENODE_USER="aivn"
- export HDFS_DATANODE_USER="aivn"
- export HDFS_SECONDARYNAMENODE_USER="aivn"
- export YARN_RESOURCEMANAGER_USER="aivn"
- export YARN_NODEMANAGER_USER="aivn"

# Hadoop

## ❖ Installation: Hadoop

**5. Config core-site.xml**




**Paste the code below:**

```
<configuration>
<property>
    <name>fs.default.name</name>
    <value>hdfs://localhost:9000</value>
</property>
</configuration>
```

# Hadoop

## ❖ Installation: Hadoop

aivn@localhost:~

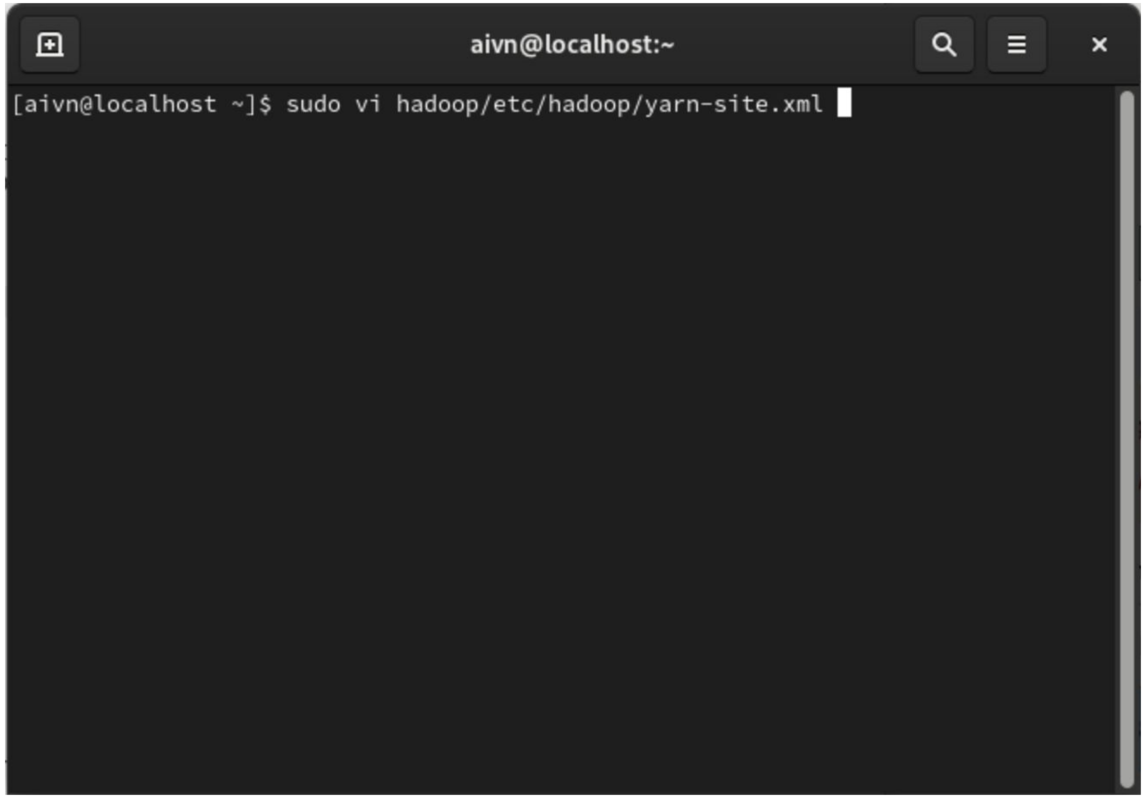
```
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
        <name>fs.default.name</name>
        <value>hdfs://localhost:9000</value>
</property>
</configuration>
~
                                                    23,11         Bot
```

**5. Config core-site.xml**

# Hadoop

## ❖ Installation: Hadoop

**6. Config yarn-site.xml**




**Paste the code below:**

```
<configuration>
<property>
      <name>yarn.nodemanager.aux-services</name>
      <value>mapreduce_shuffle</value>
</property>
<property>
      <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
      <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
<property>
      <name>yarn.resourcemanager.hostname</name>
      <value>127.0.0.1</value>
</property>
<property>
      <name>yarn.acl.enable</name>
      <value>0</value>
</property>
<property>
      <name>yarn.nodemanager.env-whitelist</name>
      <value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_PERPEND_DISTCACHE,HADOOP_YARN_HOME,HADOOP_MAPRED_HOME</value>
</property>
</configuration>
```

# Hadoop

## ❖ Installation: Hadoop

aivn@localhost:~ — ssh localhost

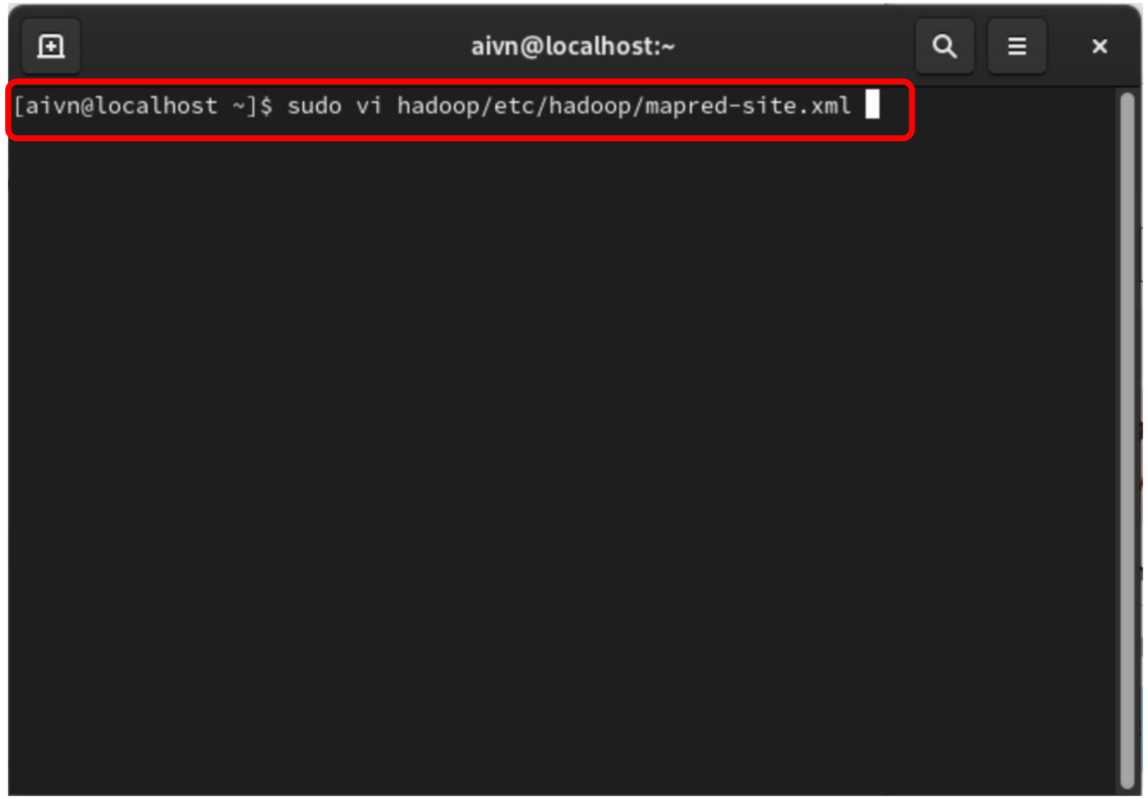
```
<configuration>
<property>
        <name>yarn.nodemanager.aux-services</name>
        <value>mapreduce_shuffle</value>
</property>
<property>
        <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
        <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
<property>
        <name>yarn.resourcemanager.hostname</name>
        <value>127.0.0.1</value>
</property>
<property>
        <name>yarn.acl.enable</name>
        <value>0</value>
</property>
<property>
        <name>yarn.nodemanager.env-whitelist</name>
        <value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLA
SSPATH_PERPEND_DISTCACHE,HADOOP_YARN_HOME,HADOOP_MAPRED_HOME</value>
</property>
</configuration>
:wq
```

**6. Config yarn-site.xml**

# Hadoop

## ❖ Installation: Hadoop

**7. Config mapred-site.xml**

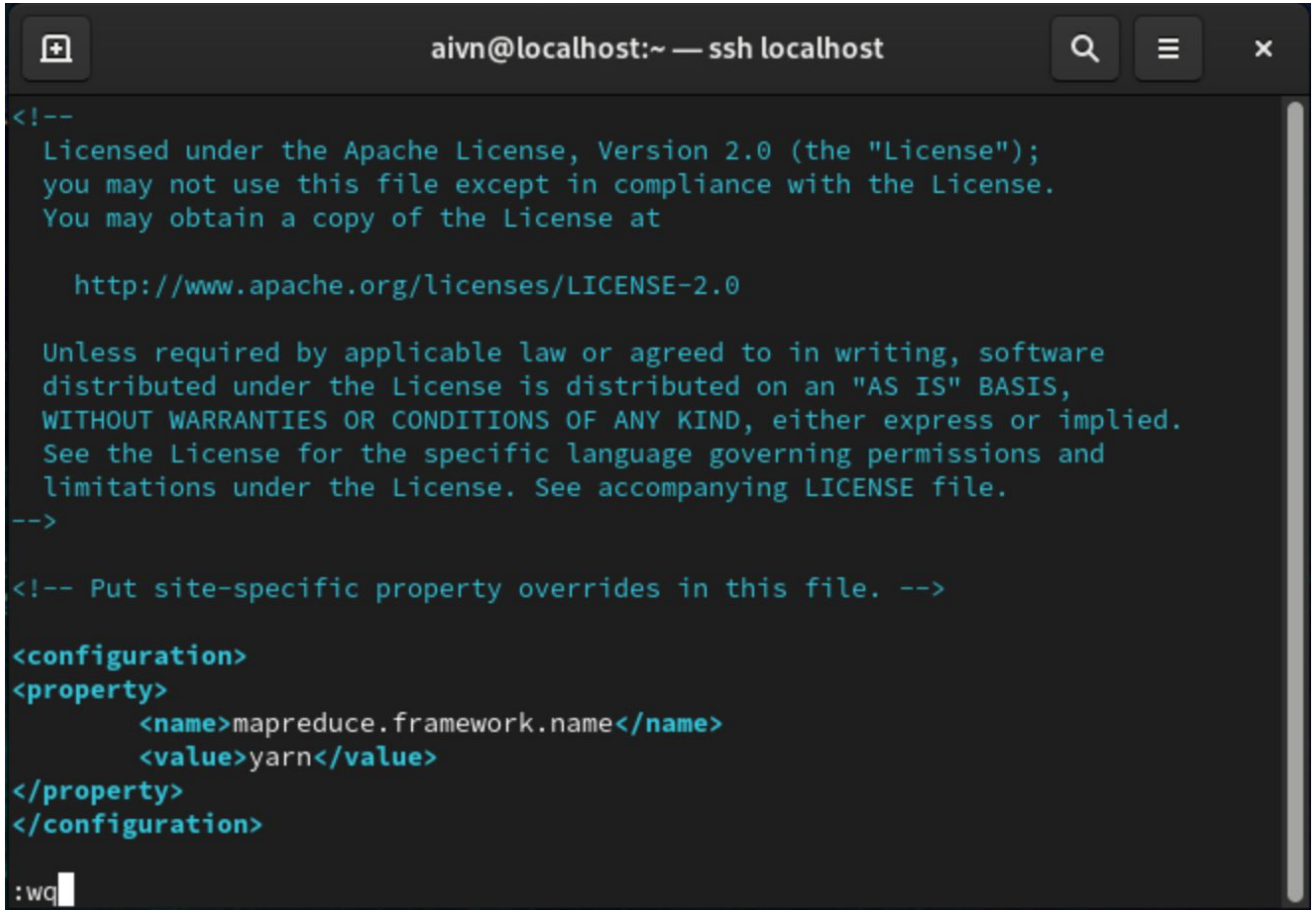



**Paste the code below:**

```
<configuration>
<property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
</property>
</configuration>
```

# Hadoop

## ❖ Installation: Hadoop




```
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
        <name>mapreduce.framework.name</name>
        <value>yarn</value>
</property>
</configuration>

:wq
```

**7. Config mapred-site.xml**

# Hadoop

## ❖ Installation: Hadoop

### 8.1. Create namenode and datanode folder

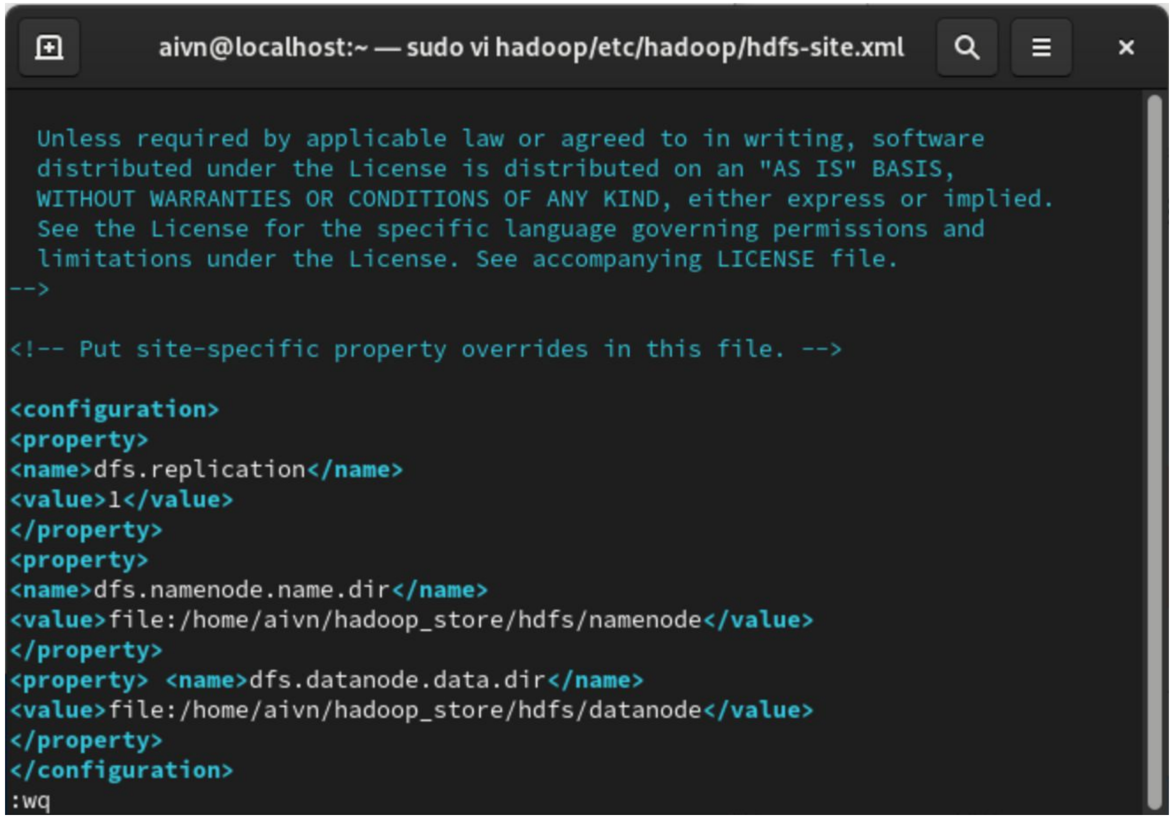
```
aivn@localhost:~
[aivn@localhost ~]$ mkdir -p hadoop_store/hdfs/namenode
[aivn@localhost ~]$ mkdir -p hadoop_store/hdfs/datanode
[aivn@localhost ~]$ ls hadoop_store/hdfs/
datanode  namenode
[aivn@localhost ~]$
```

### 8.2. Config hdfs-site.xml

```
aivn@localhost:~
[aivn@localhost ~]$ sudo vi hadoop/etc/hadoop/hdfs-site.xml
```

# Hadoop

## ❖ Installation: Hadoop




**Paste the code below:**

```
<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
<name>dfs.namenode.name.dir</name>
<value>file:/home/aivn/hadoop_store/hdfs/namenode</value>
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>file:/home/aivn/hadoop_store/hdfs/datanode</value>
</property>
</configuration>
```

# Hadoop

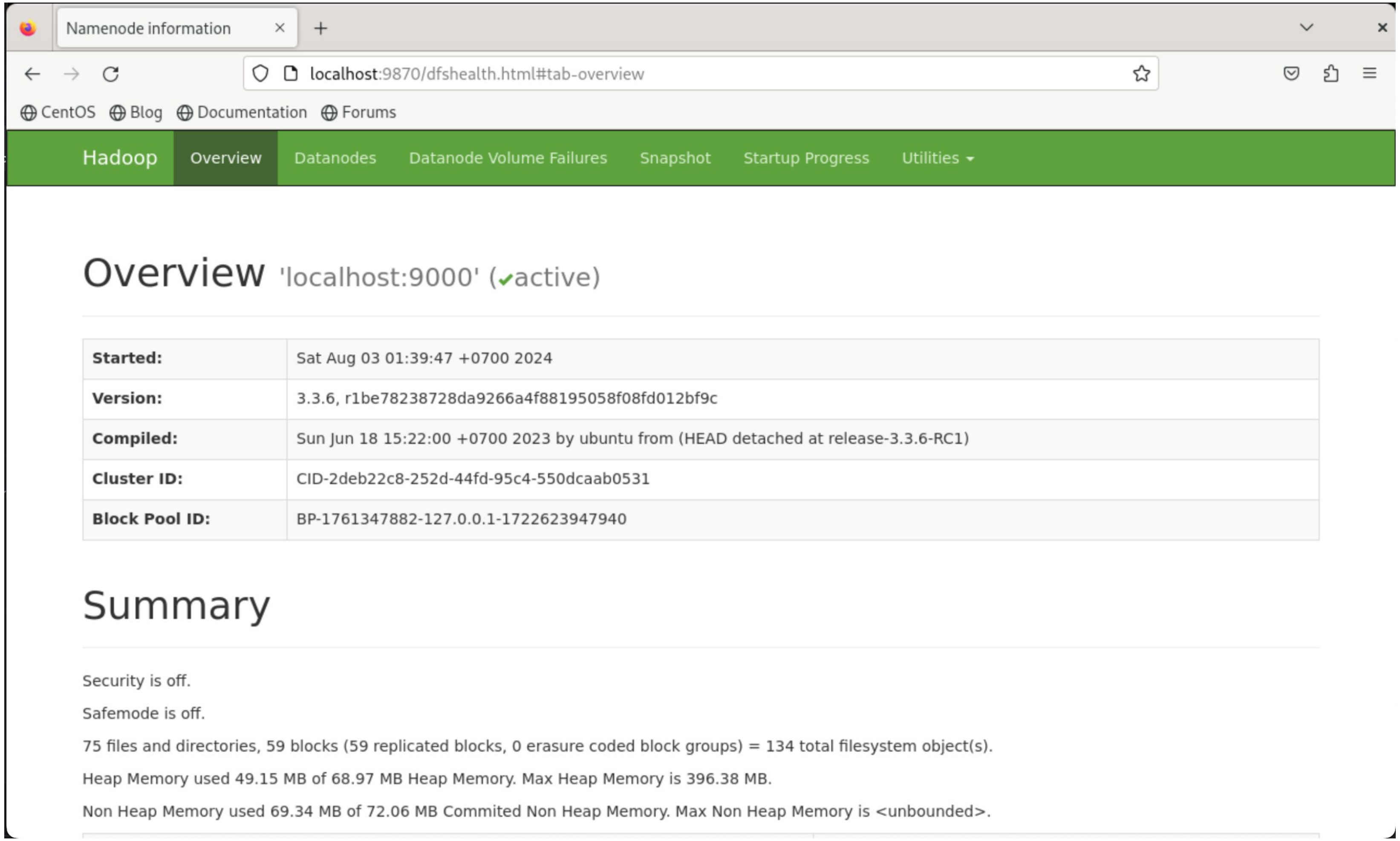
## ❖ Check Hadoop

```
[aivn@localhost ~]$ cd hadoop
[aivn@localhost hadoop]$ ./bin/hadoop
Usage: hadoop [OPTIONS] SUBCOMMAND [SUBCOMMAND OPTIONS]
 or    hadoop [OPTIONS] CLASSNAME [CLASSNAME OPTIONS]
  where CLASSNAME is a user-provided Java class

  OPTIONS is none or any of:

buildpaths                       attempt to add class files from build tree
--config dir                     Hadoop config directory
--debug                          turn on shell script debug mode
--help                           usage information
hostnames list[,of,host,names]   hosts to use in worker mode
hosts filename                   list of hosts to use in worker mode
loglevel level                   set the log4j level for this command
workers                          turn on worker mode

  SUBCOMMAND is one of:


    Admin Commands:

daemonlog     get/set the log level for each daemon
```

**Check Hadoop Installation**

# Hadoop

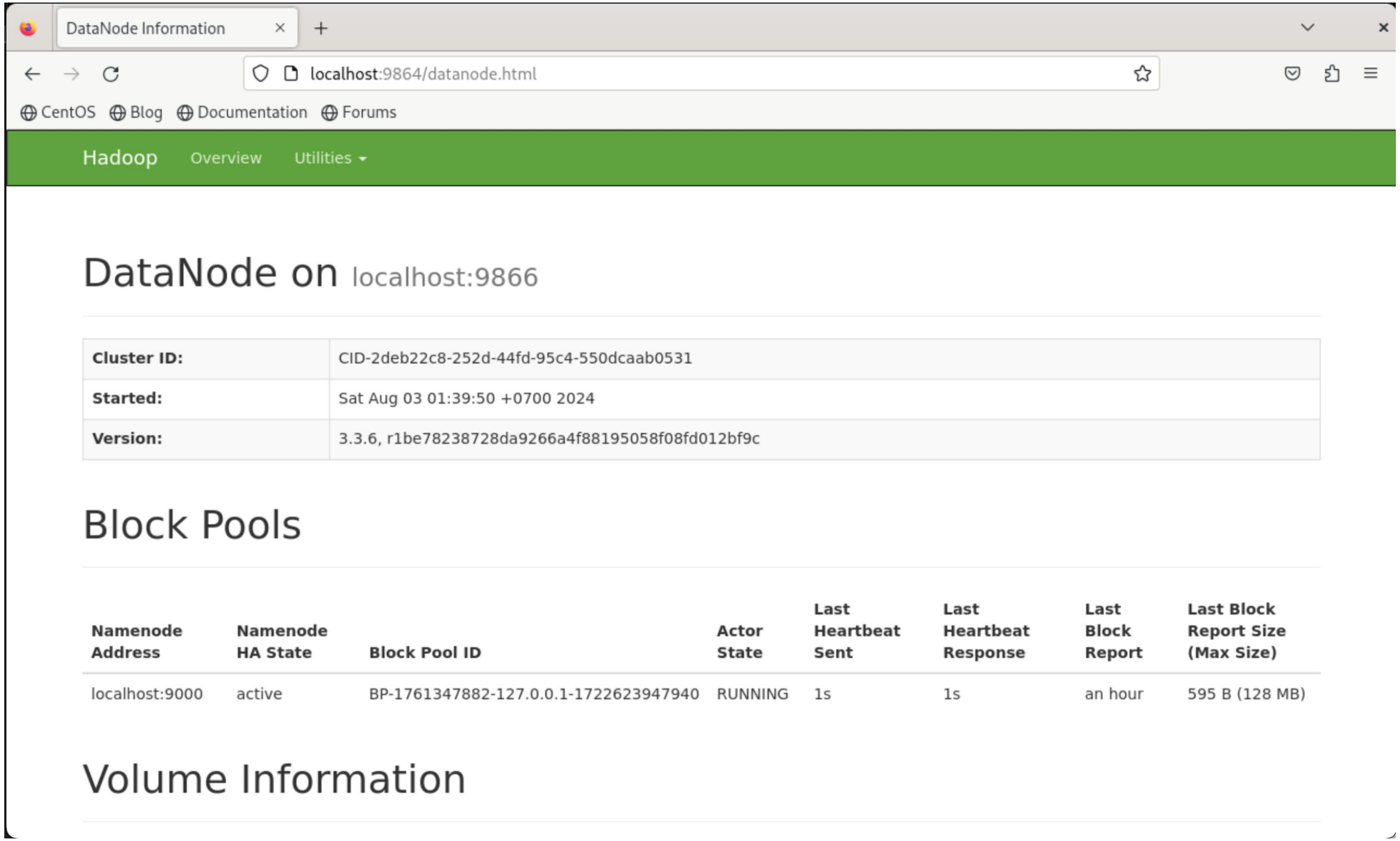
## ❖ Check Hadoop




**localhost:9870** to check namenode

# Hadoop

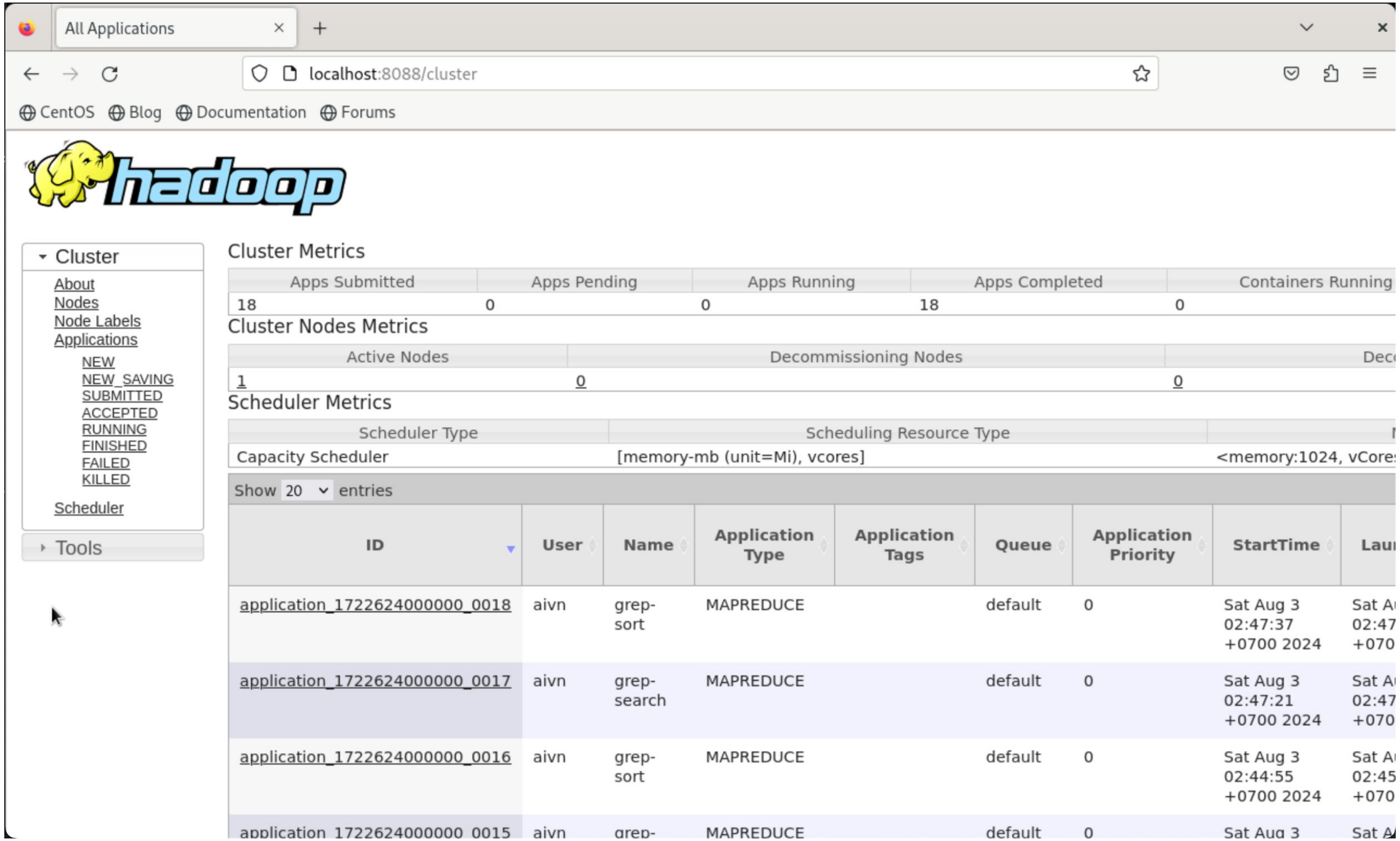
## ❖ Check Hadoop




**localhost:9864** to check datanode

# Hadoop

## ❖ Check Hadoop




**localhost:8088** to check datanode

# Hadoop

## ❖ Example: Search text

# grep Command

Last Updated: 2023-03-24

## Purpose

Searches for a pattern in a file.

## Syntax

**grep** [ **-E** | **-F** ] [ **-i** ] [ **-h** ] [ **-H** ] [ **-L** ] [ **-r** | **-R** ] [ **-s** ][ **-u** ] [ **-v** ] [ **-w** ] [ **-x** ] [ **-y** ] [ [ [ **-b** ] [ **-n** ] ] | [ **-c** | **-l** | **-q** ] ] [ **-p** [ *Separator* ] ] { [ **-e** *PatternList* ... ] [ **-f** *PatternFile* ... ] | PatternList ... } [ File ... ]

## Description

The **grep** command searches for the pattern specified by the *Pattern* parameter and writes each matching line to standard output. The patterns are limited regular expressions in the style of the **ed** or **egrep** command. The **grep** command uses a compact non-deterministic algorithm.

Pattern to search: “fs[a-z.]”

# Hadoop

## ❖ Example: Search text

**1. Initialize namenode**

```
aivn@localhost:~/hadoop
[aivn@localhost hadoop]$ ./bin/hdfs namenode -format
WARNING: /home/aivn/hadoop/logs does not exist. Creating.
2024-08-02 01:53:52,909 INFO namenode.NameNode: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = localhost.localdomain/127.0.0.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.3.6
STARTUP_MSG:   classpath = /home/aivn/hadoop/etc/hadoop:/home/aivn/hadoop/share/
hadoop/common/lib/netty-resolver-dns-native-macos-4.1.89.Final-osx-aarch_64.jar:
/home/aivn/hadoop/share/hadoop/common/lib/netty-codec-socks-4.1.89.Final.jar:/ho
me/aivn/hadoop/share/hadoop/common/lib/kerb-identity-1.0.1.jar:/home/aivn/hadoop
/share/hadoop/common/lib/kerby-asn1-1.0.1.jar:/home/aivn/hadoop/share/hadoop/com
mon/lib/commons-math3-3.1.1.jar:/home/aivn/hadoop/share/hadoop/common/lib/jetty-
xml-9.4.51.v20230217.jar:/home/aivn/hadoop/share/hadoop/common/lib/kerb-crypto-1
.0.1.jar:/home/aivn/hadoop/share/hadoop/common/lib/netty-codec-xml-4.1.89.Final.
jar:/home/aivn/hadoop/share/hadoop/common/lib/netty-transport-sctp-4.1.89.Final.
jar:/home/aivn/hadoop/share/hadoop/common/lib/commons-collections-3.2.2.jar:/hom
e/aivn/hadoop/share/hadoop/common/lib/jsp-api-2.1.jar:/home/aivn/hadoop/share/ha
doop/common/lib/commons-codec-1.15.jar:/home/aivn/hadoop/share/hadoop/common/lib
/jackson-mapper-asl-1.9.13.jar:/home/aivn/hadoop/share/hadoop/common/lib/reload4
j-1.2.22.jar:/home/aivn/hadoop/share/hadoop/common/lib/jersey-server-1.19.4.jar:
/home/aivn/hadoop/share/hadoop/common/lib/jsch-0.1.55.jar:/home/aivn/hadoop/shar
e/hadoop/common/lib/kerby-util-1.0.1.jar:/home/aivn/hadoop/share/hadoop/common/l
```

**2. Start namenode and datanode daemon**

```
aivn@localhost:~/hadoop
[aivn@localhost hadoop]$ sudo ./sbin/start-all.sh
Starting namenodes on [localhost]
Last login: Fri Aug  2 10:45:09 +07 2024 on pts/1
Starting datanodes
Last login: Fri Aug  2 10:45:15 +07 2024 on pts/1
Starting secondary namenodes [localhost.localdomain]
Last login: Fri Aug  2 10:45:17 +07 2024 on pts/1
2024-08-02 10:45:24,631 WARN util.NativeCodeLoader: Unable to load native-hadoop
 library for your platform... using builtin-java classes where applicable
Starting resourcemanager
Last login: Fri Aug  2 10:45:21 +07 2024 on pts/1
Starting nodemanagers
Last login: Fri Aug  2 10:45:25 +07 2024 on pts/1
[aivn@localhost hadoop]$
```

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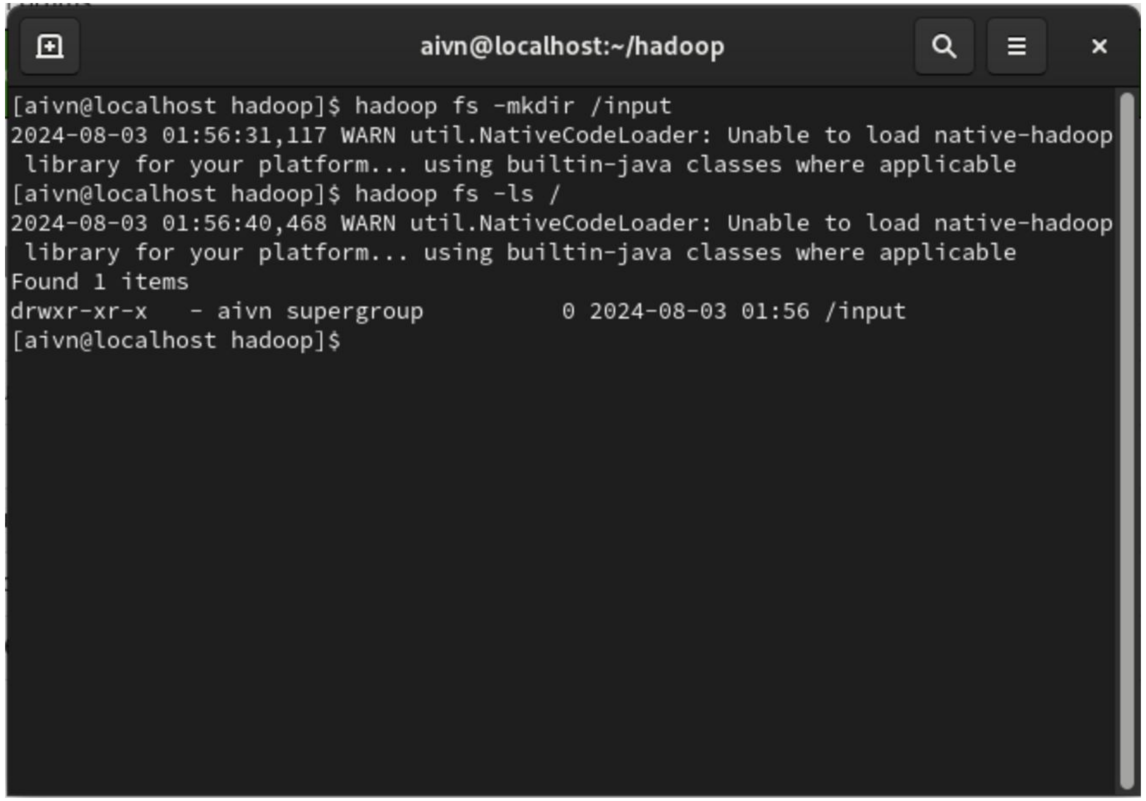
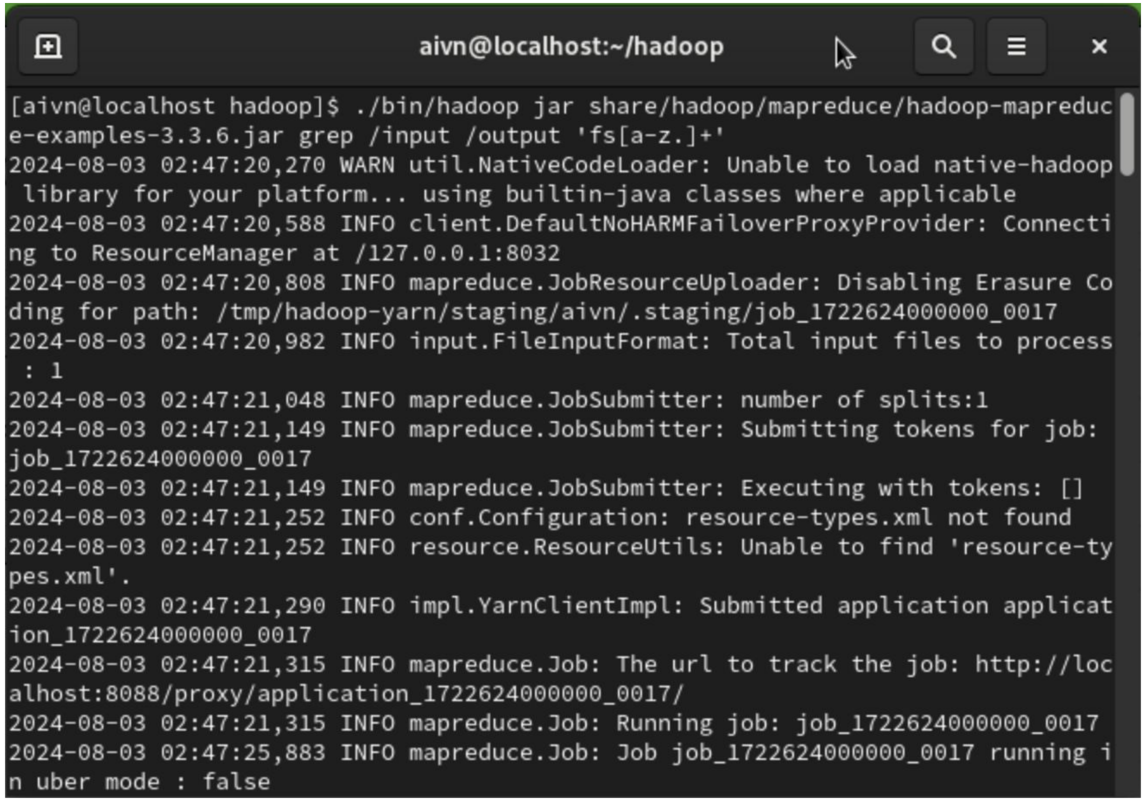
# Hadoop

## ❖ Example: Search text

**3. Create input folder and move all .xml file into input**




**4. Perform grep with string pattern "fs[a-z.]+"**



# Hadoop

## ❖ Example: Search text

aivn@localhost:~/hadoop

```
[aivn@localhost hadoop]$ hadoop fs -get /output output
2024-08-03 02:49:06,253 WARN util.NativeCodeLoader: Unable to load native-hadoop
 library for your platform... using builtin-java classes where applicable
[aivn@localhost hadoop]$ cat output/*
6       fspp
1       fstatat
[aivn@localhost hadoop]$
```

**5. Get output and print results**

# Hadoop

## ❖ Example: Other default applications

```
Valid program names are:
  aggregatewordcount: An Aggregate based map/reduce program that counts the words in the i
nput files.
  aggregatewordhist: An Aggregate based map/reduce program that computes the histogram of
the words in the input files.
  bbp: A map/reduce program that uses Bailey-Borwein-Plouffe to compute exact digits of Pi
.
  dbcount: An example job that count the pageview counts from a database.
  distbbp: A map/reduce program that uses a BBP-type formula to compute exact bits of Pi.
  grep: A map/reduce program that counts the matches of a regex in the input.
  join: A job that effects a join over sorted, equally partitioned datasets
  multifilewc: A job that counts words from several files.
  pentomino: A map/reduce tile laying program to find solutions to pentomino problems.
  pi: A map/reduce program that estimates Pi using a quasi-Monte Carlo method.
  randomtextwriter: A map/reduce program that writes 10GB of random textual data per node.
  randomwriter: A map/reduce program that writes 10GB of random data per node.
  secondarysort: An example defining a secondary sort to the reduce.
  sort: A map/reduce program that sorts the data written by the random writer.
  sudoku: A sudoku solver.
  teragen: Generate data for the terasort
  terasort: Run the terasort
  teravalidate: Checking results of terasort
  wordcount: A map/reduce program that counts the words in the input files.
  wordmean: A map/reduce program that counts the average length of the words in the input
files.
  wordmedian: A map/reduce program that counts the median length of the words in the input
 files.
  wordstandarddeviation: A map/reduce program that counts the standard deviation of the le
ngth of the words in the input files.
```

# QUIZ

# Spark

# Spark

## ❖ Introduction

**Apache Spark:** an open-source unified analytics engine for **large-scale data processing.** Spark is an alternative replacement for Hadoop MapReduce.

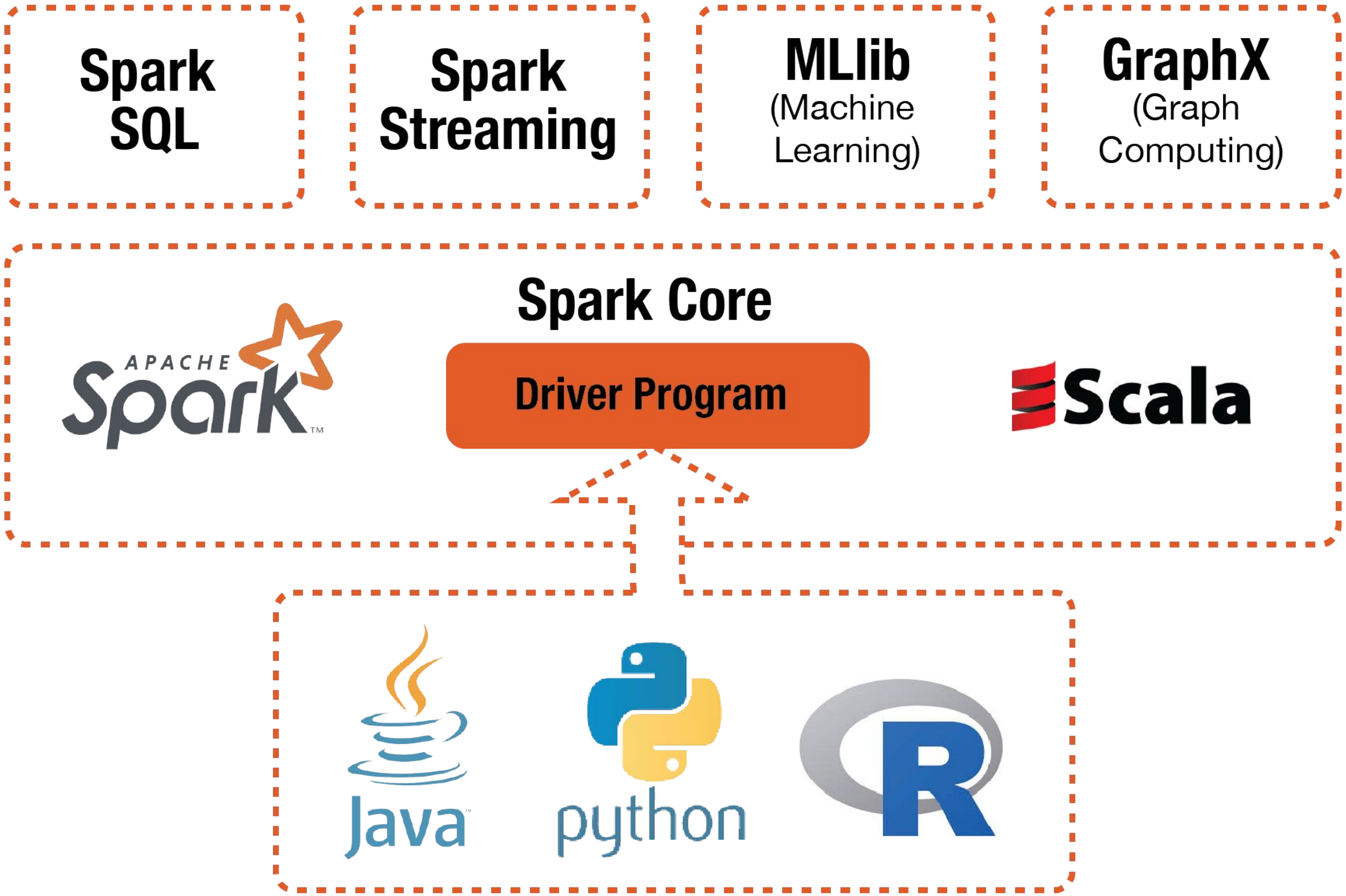
# Spark

## ❖ Why Spark?

| hadoop MapReduce | Apache Spark |
| --- | --- |
| Fast | 100x Faster than MapReduce |
| Batch Processing | Real-time Processing |
| Stores Data on Disk | Stores Data Memory |
| Written in Java | Written in Scala |
| Low Cost | High Cost |

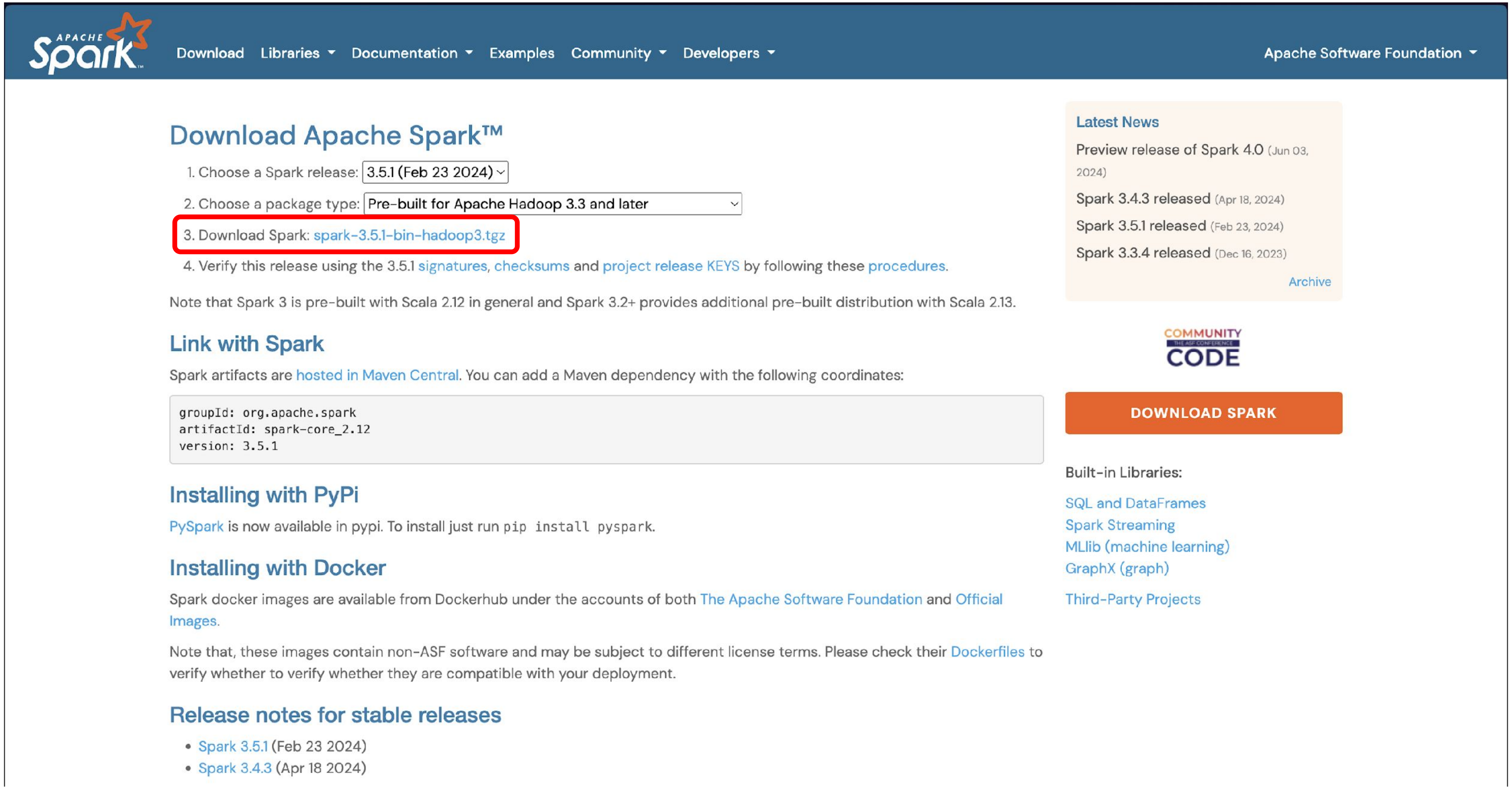
# Spark

## ❖ Spark Modules



# Spark

## ❖ Installation: Get spark



# Spark

## ❖ Installation: Get spark

Community ▾ Projects ▾ Downloads ▾ Learn ▾ Resources & Tools ▾ About ▾

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We suggest the following location for your download:

https://dlcdn.apache.org/spark/spark-3.5.1/spark-3.5.1-bin-hadoop3.tgz

Alternate download locations are suggested below.

It is essential that you verify the integrity of the downloaded file using the PGP signature ( `.asc` file) or a hash ( `.md5` or `.sha*` file).

### HTTP

https://dlcdn.apache.org/spark/spark-3.5.1/spark-3.5.1-bin-hadoop3.tgz

### BACKUP SITES

https://dlcdn.apache.org/spark/spark-3.5.1/spark-3.5.1-bin-hadoop3.tgz

### VERIFY THE INTEGRITY OF THE FILES

It is essential that you verify the integrity of the downloaded file using the PGP signature ( `.asc` file) or a hash ( `.md5` or `.sha*` file). Please read Verifying Apache Software Foundation Releases for more information on why you should verify our releases.

The PGP signature can be verified using PGP or GPG. First download the `KEYS` as well as the `asc` signature file for the relevant distribution. Make sure you get these files from the main distribution site, rather than from a mirror. Then verify the signatures using

```
% gpg --import KEYS
% gpg --verify downloaded_file.asc downloaded_file
```

# Spark

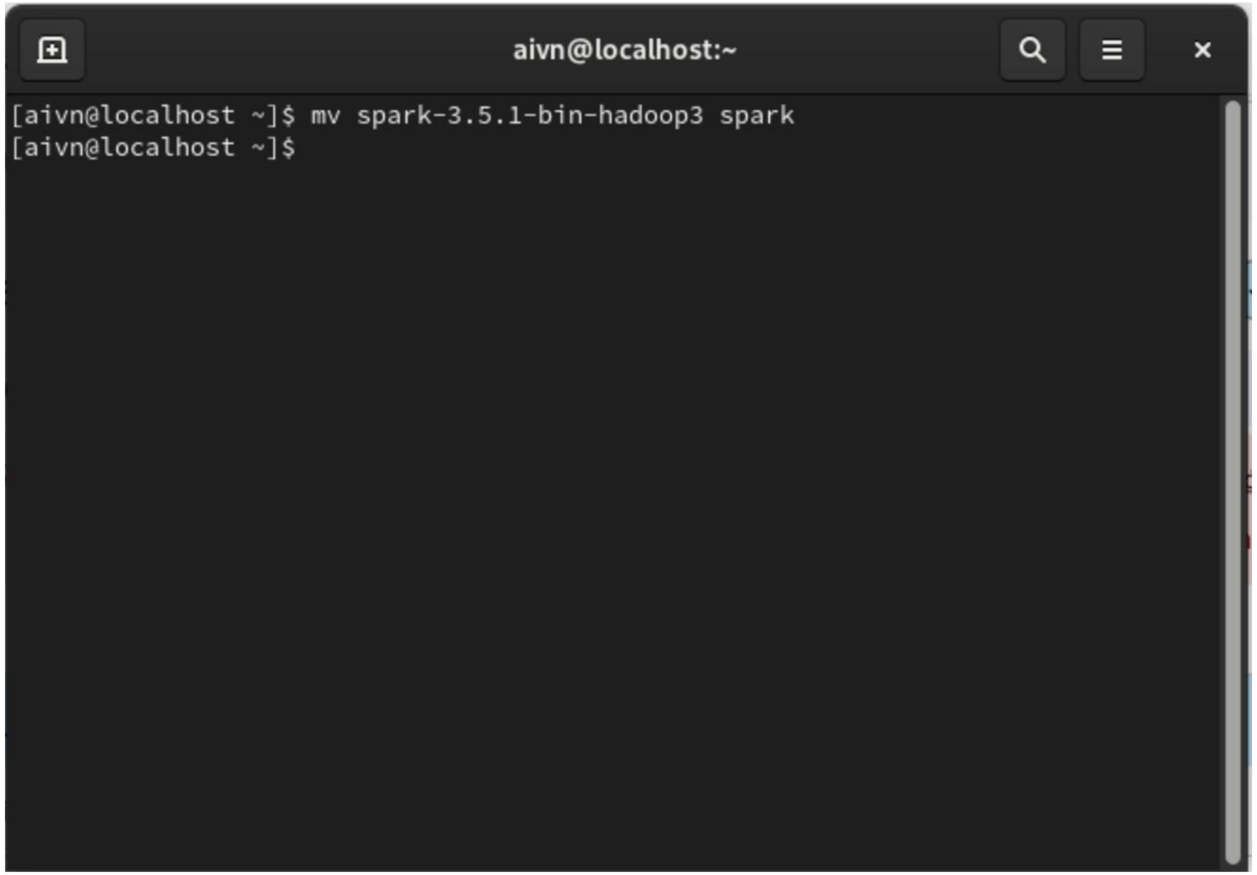
## ❖ Installation: Get spark

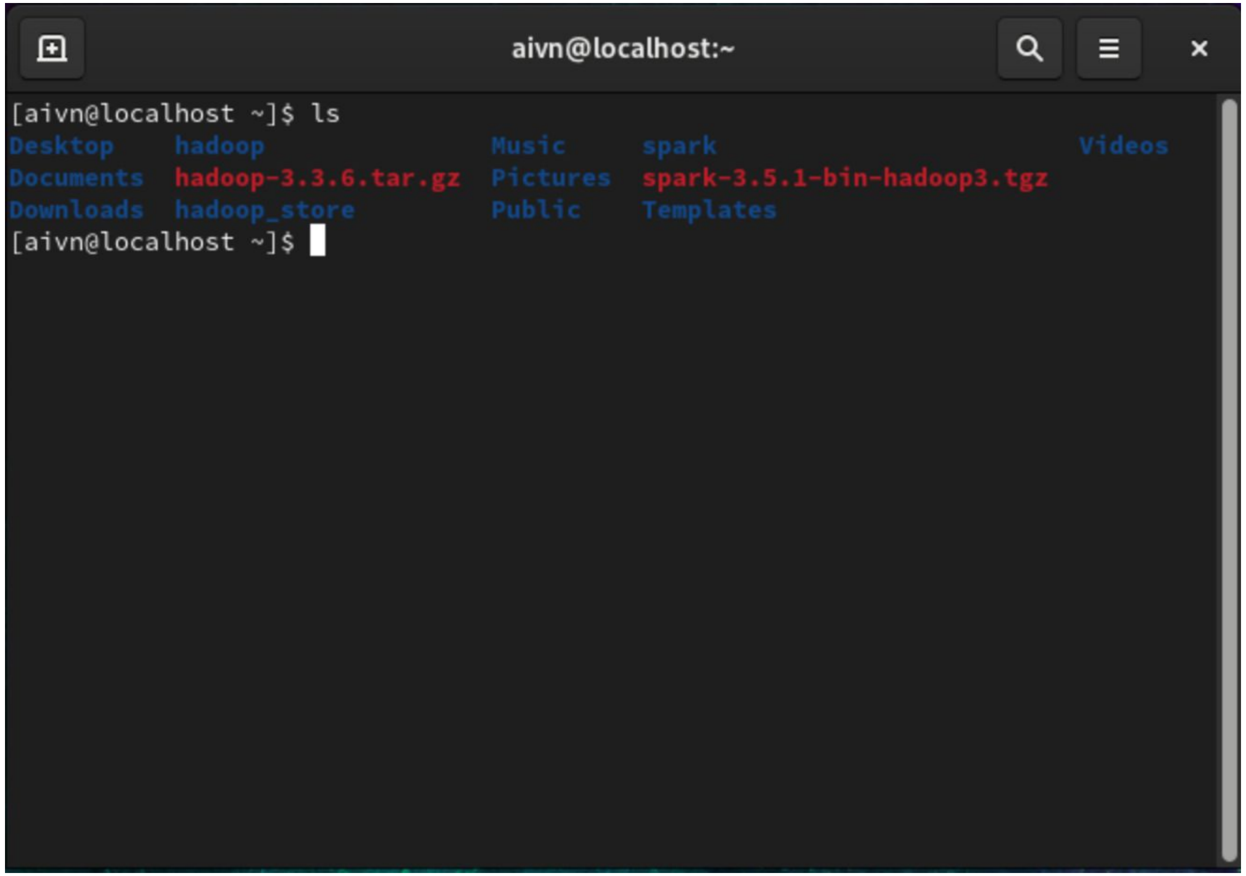
```
aivn@localhost:~
[aivn@localhost ~]$ wget https://dlcdn.apache.org/spark/spark-3.5.1/spark-3.5.1-
bin-hadoop3.tgz
--2024-08-02 22:54:27--  https://dlcdn.apache.org/spark/spark-3.5.1/spark-3.5.1-
bin-hadoop3.tgz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connecte
d.
HTTP request sent, awaiting response... 200 OK
Length: 400446614 (382M) [application/x-gzip]
Saving to: ‘spark-3.5.1-bin-hadoop3.tgz’

spark-3.5.1-bin-had 100%[===================>] 381.90M  8.90MB/s    in 41s

2024-08-02 22:55:08 (9.32 MB/s) - ‘spark-3.5.1-bin-hadoop3.tgz’ saved [400446614
/400446614]

[aivn@localhost ~]$
```

```
aivn@localhost:~
[aivn@localhost ~]$ tar -xvf spark-3.5.1-bin-hadoop3.tgz spark-3.5.1-bin-hadoop3
spark-3.5.1-bin-hadoop3/
spark-3.5.1-bin-hadoop3/sbin/
spark-3.5.1-bin-hadoop3/sbin/spark-config.sh
spark-3.5.1-bin-hadoop3/sbin/stop-slave.sh
spark-3.5.1-bin-hadoop3/sbin/stop-mesos-dispatcher.sh
spark-3.5.1-bin-hadoop3/sbin/start-workers.sh
spark-3.5.1-bin-hadoop3/sbin/start-slaves.sh
spark-3.5.1-bin-hadoop3/sbin/start-all.sh
spark-3.5.1-bin-hadoop3/sbin/stop-all.sh
spark-3.5.1-bin-hadoop3/sbin/workers.sh
spark-3.5.1-bin-hadoop3/sbin/start-mesos-dispatcher.sh
spark-3.5.1-bin-hadoop3/sbin/spark-daemon.sh
spark-3.5.1-bin-hadoop3/sbin/decommission-worker.sh
spark-3.5.1-bin-hadoop3/sbin/slaves.sh
spark-3.5.1-bin-hadoop3/sbin/stop-mesos-shuffle-service.sh
spark-3.5.1-bin-hadoop3/sbin/stop-history-server.sh
spark-3.5.1-bin-hadoop3/sbin/stop-worker.sh
spark-3.5.1-bin-hadoop3/sbin/decommission-slave.sh
spark-3.5.1-bin-hadoop3/sbin/stop-thriftserver.sh
spark-3.5.1-bin-hadoop3/sbin/start-worker.sh
spark-3.5.1-bin-hadoop3/sbin/stop-slaves.sh
spark-3.5.1-bin-hadoop3/sbin/start-connect-server.sh
spark-3.5.1-bin-hadoop3/sbin/start-thriftserver.sh
```

# Spark

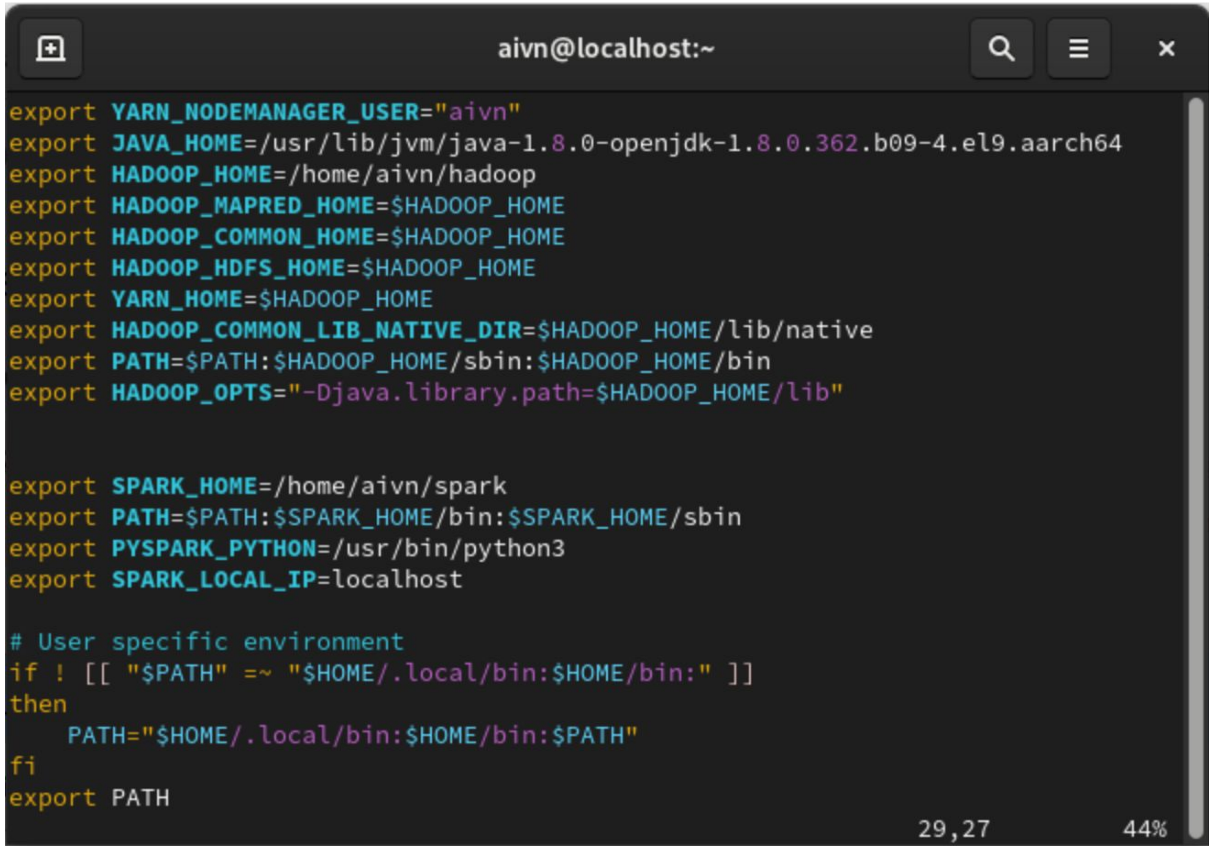
## ❖ Installation: Get spark



# Spark

## ❖ Installation: Get spark



```
aivn@localhost:~
export YARN_NODEMANAGER_USER="aivn"
export JAVA_HOME=/usr/lib/jvm/java-1.8.0-openjdk-1.8.0.362.b09-4.el9.aarch64
export HADOOP_HOME=/home/aivn/hadoop
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"


export SPARK_HOME=/home/aivn/spark
export PATH=$PATH:$SPARK_HOME/bin:$SPARK_HOME/sbin
export PYSPARK_PYTHON=/usr/bin/python3
export SPARK_LOCAL_IP=localhost

# User specific environment
if ! [[ "$PATH" =~ "$HOME/.local/bin:$HOME/bin:" ]]
then
    PATH="$HOME/.local/bin:$HOME/bin:$PATH"
fi
export PATH
                                                    29,27         44%
```

- export SPARK_HOME=/home/aivn/spark-3.5.1-bin-hadoop3
- export PATH=$PATH:$SPARK_HOME/bin:$SPARK_HOME/sbin
- export PYSPARK_PYTHON=/usr/bin/python3
- export SPARK_LOCAL_IP=localhost

# Spark

## ❖ Installation: Start spark

aivn@localhost:~

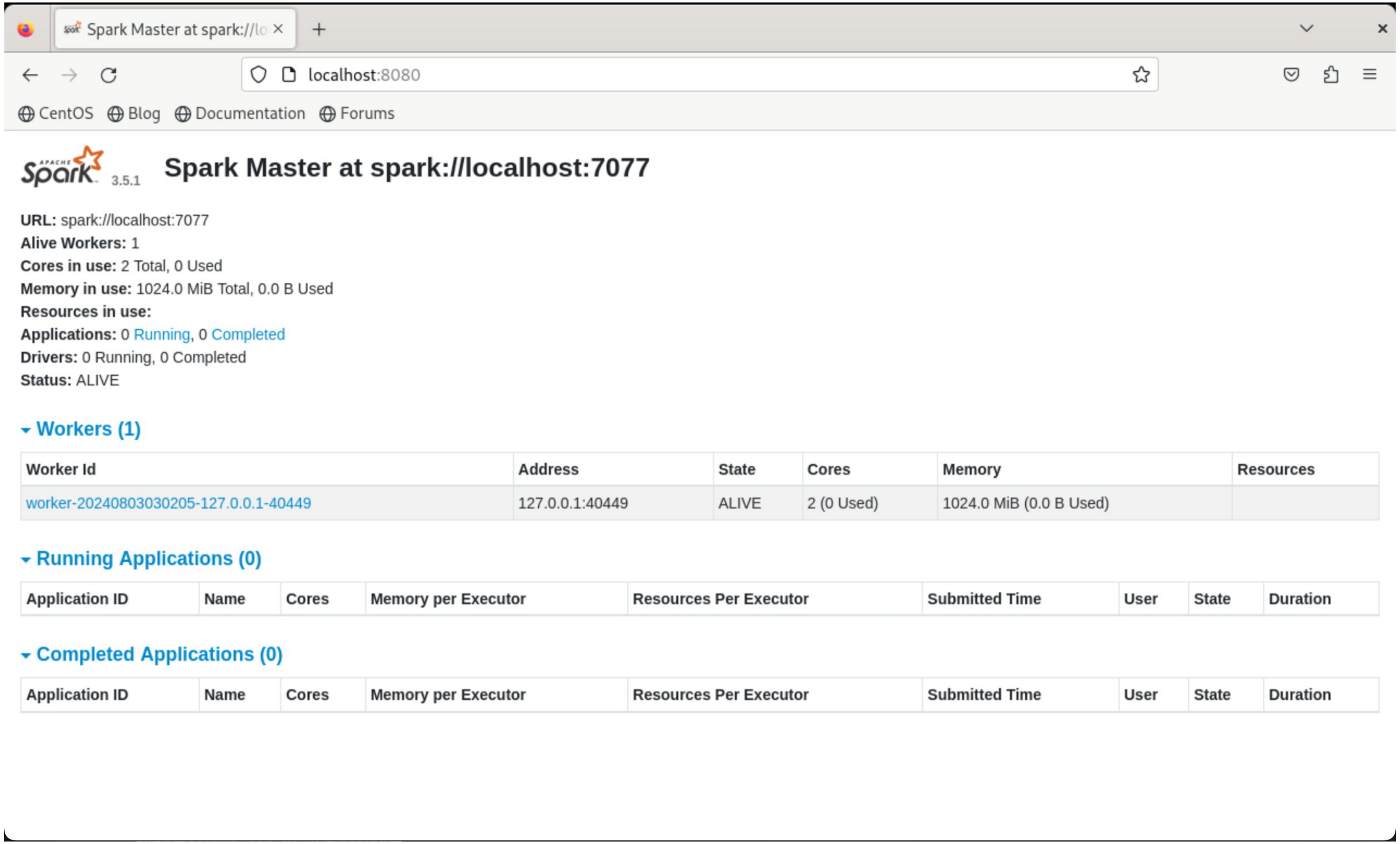
```
[aivn@localhost ~]$ sudo vi .bashrc
[aivn@localhost ~]$ source .bashrc
[aivn@localhost ~]$
```

aivn@localhost:~/spark

```
[aivn@localhost spark]$ ./sbin/start-all.sh
starting org.apache.spark.deploy.master.Master, logging to /home/aivn/spark/logs
/spark-aivn-org.apache.spark.deploy.master.Master-1-localhost.out
localhost: starting org.apache.spark.deploy.worker.Worker, logging to /home/aivn
/spark/logs/spark-aivn-org.apache.spark.deploy.worker.Worker-1-localhost.out
[aivn@localhost spark]$ jps
74657 Jps
74514 Master
57573 DataNode
58202 NodeManager
74605 Worker
57421 NameNode
57838 SecondaryNameNode
58062 ResourceManager
[aivn@localhost spark]$
```

# Spark

## ❖ Spark: localhost



# Spark

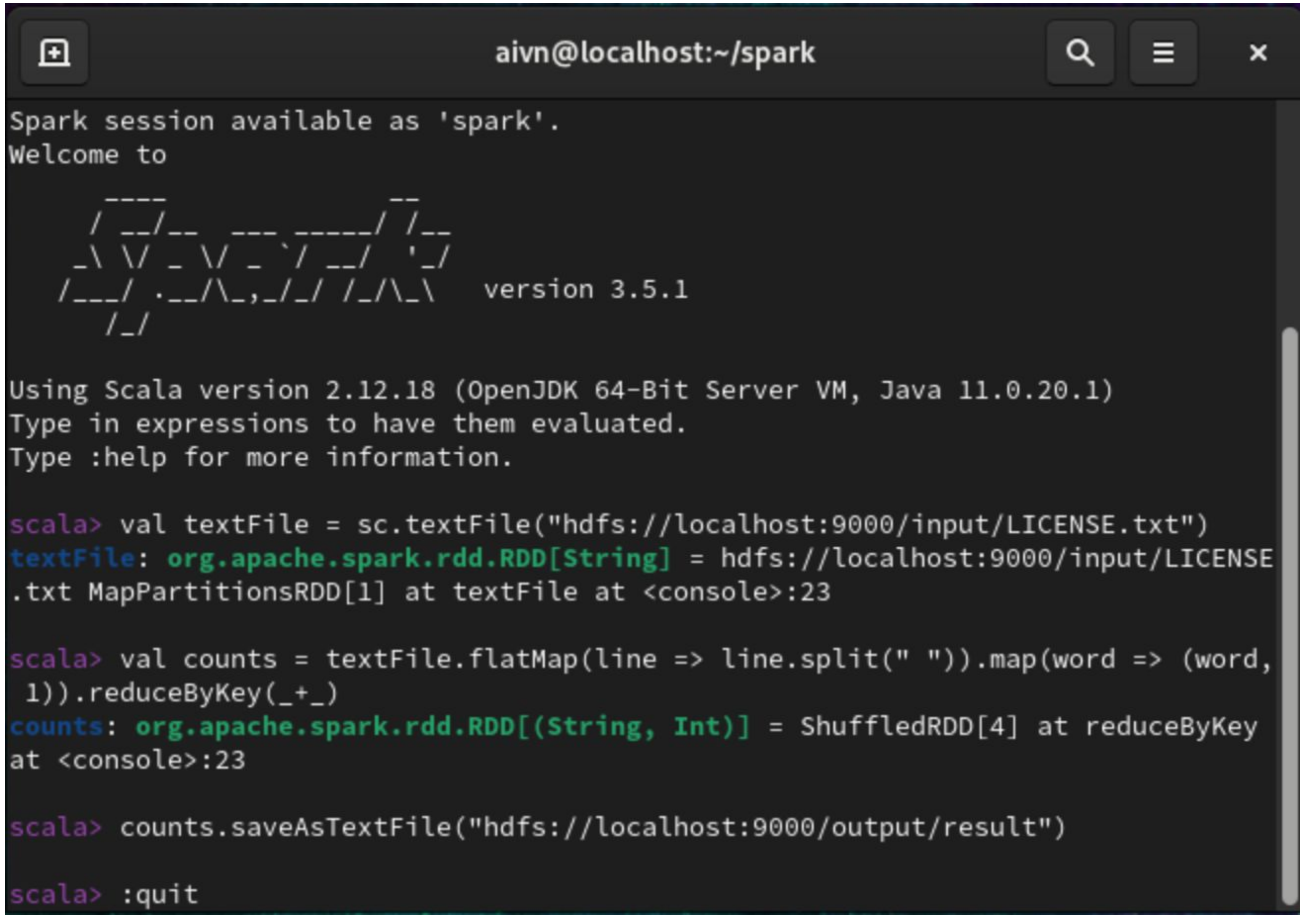
## ❖ Spark: spark session

aivn@localhost:~/spark

```
[aivn@localhost spark]$ spark-shell --master spark://localhost:7077
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLeve
l(newLevel).
24/08/03 03:04:46 WARN NativeCodeLoader: Unable to load native-hadoop library fo
r your platform... using builtin-java classes where applicable
Spark context Web UI available at http://localhost:4040
Spark context available as 'sc' (master = spark://localhost:7077, app id = app-2
0240803030448-0000).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.1
      /_/

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 11.0.20.1)
Type in expressions to have them evaluated.
Type :help for more information.

scala>
```

# Spark

## ❖ Spark: Word count




```
• val textFile = sc.textFile("hdfs://localhost:9000/input/LICENSE.txt")
• val counts = textFile.flatMap(line => line.split(" ")).map(word => (word, 1)).reduceByKey(_+_)
• counts.saveAsTextFile("hdfs://localhost:9000/output/result")
```

# Spark

## ❖ Spark: Word count

```
aivn@localhost:~/spark

[aivn@localhost spark]$ hadoop fs -ls /output/result/*
2024-08-03 03:10:47,655 WARN util.NativeCodeLoader: Unable to load native-hadoop
 library for your platform... using builtin-java classes where applicable
-rw-r--r--   3 aivn supergroup          0 2024-08-03 03:08 /output/result/_SUCCE
SS
-rw-r--r--   3 aivn supergroup       4732 2024-08-03 03:08 /output/result/part-0
0000
-rw-r--r--   3 aivn supergroup       6484 2024-08-03 03:08 /output/result/part-0
0001
^[[A[aivn@localhost spark]$ hadoop fs -cat /output/result/part-00000
2024-08-03 03:11:03,969 WARN util.NativeCodeLoader: Unable to load native-hadoop
 library for your platform... using builtin-java classes where applicable
(notice,,1)
(Unless,3)
(under,10)
(Contributions),1)
(offer,1)
(NON-INFRINGEMENT,,1)
(agree,1)
(its,3)
(event,1)
(intentionally,2)
(Grant,2)
(have,2)
```

# Spark

## ❖ Break-quiz

**1. What is Apache Spark primarily used for?**

A) Web development.

B) Real-time data processing.

C) Desktop application development.

D) Database management.

**2. Which of the following is a core component of Apache Spark?**

A) Spark SQL.

B) Spark Mapper.

C) Spark Developer.

D) Spark Coder.

**3. Which language is not supported natively by Spark?**

A) Java.

B) Python.

C) Scala.

D) Ruby.

**4. Which Spark component is used for machine learning?**

A) Spark SQL.

B) Spark Streaming.

C) Spark Mllib.

D) Spark GraphX.

# Spark

❖ **How about use Spark with Python instead of Scala ?**

Apache Spark + Python = PySpark

# Spark

## ❖ PySpark Introduction

**PySpark:** The Python API for Apache Spark. It enables to perform real-time, large-scale data processing in a distributed environment using Python.

# Spark

## ❖ PySpark Demo

```
aivn@localhost:~/spark

[aivn@localhost spark]$ pyspark
Python 3.9.19 (main, Jul 18 2024, 00:00:00)
[GCC 11.4.1 20231218 (Red Hat 11.4.1-3)] on linux
Type "help", "copyright", "credits" or "license" for more information.
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLeve
l(newLevel).
24/08/03 03:25:15 WARN NativeCodeLoader: Unable to load native-hadoop library fo
r your platform... using builtin-java classes where applicable
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.5.1
      /_/

Using Python version 3.9.19 (main, Jul 18 2024 00:00:00)
Spark context Web UI available at http://localhost:4040
Spark context available as 'sc' (master = local[*], app id = local-1722630316038
).
SparkSession available as 'spark'.
>>>
```

# Spark

## ❖ PySpark Demo

**Description:** Perform some basic operation of Spark with Flights.csv dataset. Download link: here.

aivn@localhost:~/spark

```
24/08/03 03:33:51 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cla
sses where applicable
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.5.1
      /_/

Using Python version 3.9.19 (main, Jul 18 2024 00:00:00)
Spark context Web UI available at http://localhost:4040
Spark context available as 'sc' (master = local[*], app id = local-1722630832496).
SparkSession available as 'spark'.
>>> df = spark.read.csv('hdfs://localhost:9000/input/flights_small.csv', header=True)
>>> df.show(2)
+----+-----+---+--------+---------+--------+---------+-------+-------+------+------+----+--------+--------+----+------+
|year|month|day|dep_time|dep_delay|arr_time|arr_delay|carrier|tailnum|flight|origin|dest|air_time|distance|hour|minute|
+----+-----+---+--------+---------+--------+---------+-------+-------+------+------+----+--------+--------+----+------+
|2014|   12|  8|     658|       -7|     935|       -5|     VX| N846VA|  1780|   SEA| LAX|     132|     954|   6|    58|
|2014|    1| 22|    1040|        5|    1505|        5|     AS| N559AS|   851|   SEA| HNL|     360|    2677|  10|    40|
+----+-----+---+--------+---------+--------+---------+-------+-------+------+------+----+--------+--------+----+------+
only showing top 2 rows

>>> 
```

# Spark

## ❖ PySpark Installation

### Install pyspark

```
[ ]  1 !pip install pyspark==3.4.1

Requirement already satisfied: pyspark==3.4.1 in /usr/local/lib/python3.10/dist-packages (3.4.1)
Requirement already satisfied: py4j==0.10.9.7 in /usr/local/lib/python3.10/dist-packages (from pyspark==3.4.1) (0.10.9.7)
```

# Spark

## ❖ PySpark Usage

### Init pyspark session

```
[ ]  1 from pyspark.sql import SparkSession
```

```
[ ]  1 spark = SparkSession.builder.appName('Example').getOrCreate()
```

```
[ ]  1 print(spark)
```

```
<pyspark.sql.session.SparkSession object at 0x7efffefc7190>
```

**SparkSession:** Represents the connection to a Spark cluster and provides a way to interact with various Spark APIs. It allows to create DataFrames, execute SQL queries, and perform distributed data processing tasks.

# Spark

## ❖ PySpark Usage: Create DataFrame

```python
data = [
    {'name': 'Rin', 'age': 20, 'gender': 'F'},
    {'name': 'Thang', 'age': 22, 'gender': 'M'},
    {'name': 'Saitama', 'age': 25, 'gender': 'M'},
    {'name': 'Edward', 'age': 31, 'gender': 'M'},
    {'name': 'Violet', 'age': 17, 'gender': 'F'}
]
```

```python
df = spark.createDataFrame(
    data=data
)
```

```python
df.printSchema()
```

```
root
 |-- age: long (nullable = true)
 |-- gender: string (nullable = true)
 |-- name: string (nullable = true)
```

```python
df.show()
```

```
+---+------+-------+
|age|gender|   name|
+---+------+-------+
| 20|     F|    Rin|
| 22|     M|  Thang|
| 25|     M|Saitama|
| 31|     M| Edward|
| 17|     F| Violet|
+---+------+-------+
```

# Spark

## ❖ PySpark Usage: Operations

```
1 # Select
2 df.select('age').show()
```

```
+---+
|age|
+---+
| 20|
| 22|
| 25|
| 31|
| 17|
+---+
```

```
1 # Filter
2 df.filter(df.gender == 'F').show()
```

```
+---+------+------+
|age|gender|  name|
+---+------+------+
| 20|     F|   Rin|
| 17|     F|Violet|
+---+------+------+
```

**select():** Choose columns to show in table

**filter():** Extract rows satisfied a criteria

# Spark

## ❖ PySpark Example: Working with csv file

**Description:** Perform some basic operation of Spark with Flights.csv dataset. Download link: here.

| year | month | day | dep_time | dep_delay | arr_time | arr_delay | carrier | tailnum | flight | origin | dest | air_time | distance | hour | minute |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2014 | 12 | 8 | 658 | -7 | 935 | -5 | VX | N846VA | 1780 | SEA | LAX | 132 | 954 | 6 | 58 |
| 2014 | 1 | 22 | 1040 | 5 | 1505 | 5 | AS | N559AS | 851 | SEA | HNL | 360 | 2677 | 10 | 40 |
| 2014 | 3 | 9 | 1443 | -2 | 1652 | 2 | VX | N847VA | 755 | SEA | SFO | 111 | 679 | 14 | 43 |
| 2014 | 4 | 9 | 1705 | 45 | 1839 | 34 | WN | N360SW | 344 | PDX | SJC | 83 | 569 | 17 | 5 |
| 2014 | 3 | 9 | 754 | -1 | 1015 | 1 | AS | N612AS | 522 | SEA | BUR | 127 | 937 | 7 | 54 |

# Spark

## ❖ PySpark Example: Working with csv file

```python
from pyspark.sql import SparkSession
```

```python
spark = SparkSession.builder.appName('Example').getOrCreate()
```

```python
flights_filepath = './flights_small.csv'
flights_df = spark.read.csv(flights_filepath, header=True)
```

```python
print(flights_df.printSchema())
print(flights_df.show(5))
```

```
root
 |-- year: string (nullable = true)
 |-- month: string (nullable = true)
 |-- day: string (nullable = true)
 |-- dep_time: string (nullable = true)
 |-- dep_delay: string (nullable = true)
 |-- arr_time: string (nullable = true)
 |-- arr_delay: string (nullable = true)
 |-- carrier: string (nullable = true)
 |-- tailnum: string (nullable = true)
 |-- flight: string (nullable = true)
 |-- origin: string (nullable = true)
 |-- dest: string (nullable = true)
 |-- air_time: string (nullable = true)
 |-- distance: string (nullable = true)
 |-- hour: string (nullable = true)
 |-- minute: string (nullable = true)

None
+----+-----+---+--------+---------+--------+---------+-------+-------+------+------+----+--------+--------+----+------+
|year|month|day|dep_time|dep_delay|arr_time|arr_delay|carrier|tailnum|flight|origin|dest|air_time|distance|hour|minute|
+----+-----+---+--------+---------+--------+---------+-------+-------+------+------+----+--------+--------+----+------+
|2014|   12|  8|     658|       -7|     935|       -5|     VX| N846VA|  1780|   SEA| LAX|     132|     954|   6|    58|
|2014|    1| 22|    1040|        5|    1505|        5|     AS| N559AS|   851|   SEA| HNL|     360|    2677|  10|    40|
|2014|    3|  9|    1443|       -2|    1652|        2|     VX| N847VA|   755|   SEA| SFO|     111|     679|  14|    43|
|2014|    4|  9|    1705|       45|    1839|       34|     WN| N360SW|   344|   PDX| SJC|      83|     569|  17|     5|
|2014|    3|  9|     754|       -1|    1015|        1|     AS| N612AS|   522|   SEA| BUR|     127|     937|   7|    54|
+----+-----+---+--------+---------+--------+---------+-------+-------+------+------+----+--------+--------+----+------+
only showing top 5 rows
```

# Spark

## ❖ PySpark Example: Working with csv file

Create new column: Flight Duration

```python
flights_df = flights_df.withColumn('duration_hours', flights_df.air_time / 60)
print(flights_df.show())
```

| year | month | day | dep_time | dep_delay | arr_time | arr_delay | carrier | tailnum | flight | origin | dest | air_time | distance | hour | minute | duration_hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2014 | 12 | 8 | 658 | -7 | 935 | -5 | VX | N846VA | 1780 | SEA | LAX | 132 | 954 | 6 | 58 | 2.2 |
| 2014 | 1 | 22 | 1040 | 5 | 1505 | 5 | AS | N559AS | 851 | SEA | HNL | 360 | 2677 | 10 | 40 | 6.0 |
| 2014 | 3 | 9 | 1443 | -2 | 1652 | 2 | VX | N847VA | 755 | SEA | SFO | 111 | 679 | 14 | 43 | 1.85 |
| 2014 | 4 | 9 | 1705 | 45 | 1839 | 34 | WN | N360SW | 344 | PDX | SJC | 83 | 569 | 17 | 5 | 1.3833333333333333 |
| 2014 | 3 | 9 | 754 | -1 | 1015 | 1 | AS | N612AS | 522 | SEA | BUR | 127 | 937 | 7 | 54 | 2.1166666666666667 |
| 2014 | 1 | 15 | 1037 | 7 | 1352 | 2 | WN | N646SW | 48 | PDX | DEN | 121 | 991 | 10 | 37 | 2.0166666666666666 |
| 2014 | 7 | 2 | 847 | 42 | 1041 | 51 | WN | N422WN | 1520 | PDX | OAK | 90 | 543 | 8 | 47 | 1.5 |
| 2014 | 5 | 12 | 1655 | -5 | 1842 | -18 | VX | N361VA | 755 | SEA | SFO | 98 | 679 | 16 | 55 | 1.6333333333333333 |
| 2014 | 4 | 19 | 1236 | -4 | 1508 | -7 | AS | N309AS | 490 | SEA | SAN | 135 | 1050 | 12 | 36 | 2.25 |
| 2014 | 11 | 19 | 1812 | -3 | 2352 | -4 | AS | N564AS | 26 | SEA | ORD | 198 | 1721 | 18 | 12 | 3.3 |
| 2014 | 11 | 8 | 1653 | -2 | 1924 | -1 | AS | N323AS | 448 | SEA | LAX | 130 | 954 | 16 | 53 | 2.1666666666666665 |
| 2014 | 8 | 3 | 1120 | 0 | 1415 | 2 | AS | N305AS | 656 | SEA | PHX | 154 | 1107 | 11 | 20 | 2.566666666666667 |
| 2014 | 10 | 30 | 811 | 21 | 1038 | 29 | AS | N433AS | 608 | SEA | LAS | 127 | 867 | 8 | 11 | 2.1166666666666667 |
| 2014 | 11 | 12 | 2346 | -4 | 217 | -28 | AS | N765AS | 121 | SEA | ANC | 183 | 1448 | 23 | 46 | 3.05 |
| 2014 | 10 | 31 | 1314 | 89 | 1544 | 111 | AS | N713AS | 306 | SEA | SFO | 129 | 679 | 13 | 14 | 2.15 |
| 2014 | 1 | 29 | 2009 | 3 | 2159 | 9 | UA | N27205 | 1458 | PDX | SFO | 90 | 550 | 20 | 9 | 1.5 |
| 2014 | 12 | 17 | 2015 | 50 | 2150 | 41 | AS | N626AS | 368 | SEA | SMF | 76 | 605 | 20 | 15 | 1.2666666666666666 |
| 2014 | 8 | 11 | 1017 | -3 | 1613 | -7 | WN | N8634A | 827 | SEA | MDW | 216 | 1733 | 10 | 17 | 3.6 |
| 2014 | 1 | 13 | 2156 | -9 | 607 | -15 | AS | N597AS | 24 | SEA | BOS | 290 | 2496 | 21 | 56 | 4.833333333333333 |
| 2014 | 6 | 5 | 1733 | -12 | 1945 | -10 | OO | N215AG | 3488 | PDX | BUR | 111 | 817 | 17 | 33 | 1.85 |

only showing top 20 rows

# Spark

## ❖ PySpark Example: Working with csv file

### Assign tables

```
[ ]  1 flights_df.createOrReplaceTempView('flights')
```

```
[ ]  1 spark.catalog.listDatabases()
```

```
[Database(name='default', catalog='spark_catalog', description='default database', locationUri='file:/content/spark-warehouse')]
```

```
[ ]  1 spark.catalog.listTables()
```

```
[Table(name='airports', catalog=None, namespace=[], description=None, tableType='TEMPORARY', isTemporary=True),
 Table(name='flights', catalog=None, namespace=[], description=None, tableType='TEMPORARY', isTemporary=True)]
```

**Assign DataFrame to Database**

# Spark

## ❖ PySpark Example: Working with csv file

```python
flights_table = spark.table('flights')
print(flights_table.show())
```

| year | month | day | dep_time | dep_delay | arr_time | arr_delay | carrier | tailnum | flight | origin | dest | air_time | distance | hour | minute | duration_hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2014 | 12 | 8 | 658 | -7 | 935 | -5 | VX | N846VA | 1780 | SEA | LAX | 132 | 954 | 6 | 58 | 2.2 |
| 2014 | 1 | 22 | 1040 | 5 | 1505 | 5 | AS | N559AS | 851 | SEA | HNL | 360 | 2677 | 10 | 40 | 6.0 |
| 2014 | 3 | 9 | 1443 | -2 | 1652 | 2 | VX | N847VA | 755 | SEA | SFO | 111 | 679 | 14 | 43 | 1.85 |
| 2014 | 4 | 9 | 1705 | 45 | 1839 | 34 | WN | N360SW | 344 | PDX | SJC | 83 | 569 | 17 | 5 | 1.3833333333333333 |
| 2014 | 3 | 9 | 754 | -1 | 1015 | 1 | AS | N612AS | 522 | SEA | BUR | 127 | 937 | 7 | 54 | 2.1166666666666667 |
| 2014 | 1 | 15 | 1037 | 7 | 1352 | 2 | WN | N646SW | 48 | PDX | DEN | 121 | 991 | 10 | 37 | 2.0166666666666666 |
| 2014 | 7 | 2 | 847 | 42 | 1041 | 51 | WN | N422WN | 1520 | PDX | OAK | 90 | 543 | 8 | 47 | 1.5 |
| 2014 | 5 | 12 | 1655 | -5 | 1842 | -18 | VX | N361VA | 755 | SEA | SFO | 98 | 679 | 16 | 55 | 1.6333333333333333 |
| 2014 | 4 | 19 | 1236 | -4 | 1508 | -7 | AS | N309AS | 490 | SEA | SAN | 135 | 1050 | 12 | 36 | 2.25 |
| 2014 | 11 | 19 | 1812 | -3 | 2352 | -4 | AS | N564AS | 26 | SEA | ORD | 198 | 1721 | 18 | 12 | 3.3 |
| 2014 | 11 | 8 | 1653 | -2 | 1924 | -1 | AS | N323AS | 448 | SEA | LAX | 130 | 954 | 16 | 53 | 2.1666666666666665 |
| 2014 | 8 | 3 | 1120 | 0 | 1415 | 2 | AS | N305AS | 656 | SEA | PHX | 154 | 1107 | 11 | 20 | 2.566666666666667 |
| 2014 | 10 | 30 | 811 | 21 | 1038 | 29 | AS | N433AS | 608 | SEA | LAS | 127 | 867 | 8 | 11 | 2.1166666666666667 |
| 2014 | 11 | 12 | 2346 | -4 | 217 | -28 | AS | N765AS | 121 | SEA | ANC | 183 | 1448 | 23 | 46 | 3.05 |
| 2014 | 10 | 31 | 1314 | 89 | 1544 | 111 | AS | N713AS | 306 | SEA | SFO | 129 | 679 | 13 | 14 | 2.15 |
| 2014 | 1 | 29 | 2009 | 3 | 2159 | 9 | UA | N27205 | 1458 | PDX | SFO | 90 | 550 | 20 | 9 | 1.5 |
| 2014 | 12 | 17 | 2015 | 50 | 2150 | 41 | AS | N626AS | 368 | SEA | SMF | 76 | 605 | 20 | 15 | 1.2666666666666666 |
| 2014 | 8 | 11 | 1017 | -3 | 1613 | -7 | WN | N8634A | 827 | SEA | MDW | 216 | 1733 | 10 | 17 | 3.6 |
| 2014 | 1 | 13 | 2156 | -9 | 607 | -15 | AS | N597AS | 24 | SEA | BOS | 290 | 2496 | 21 | 56 | 4.833333333333333 |
| 2014 | 6 | 5 | 1733 | -12 | 1945 | -10 | OO | N215AG | 3488 | PDX | BUR | 111 | 817 | 17 | 33 | 1.85 |

only showing top 20 rows

**Call the datatable**

# Spark

## ❖ PySpark Example: Working with csv file

```
1 flights_table.filter(flights_table.duration_hours > 5).show()
```

| year | month | day | dep_time | dep_delay | arr_time | arr_delay | carrier | tailnum | flight | origin | dest | air_time | distance | hour | minute | duration_hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2014 | 1 | 22 | 1040 | 5 | 1505 | 5 | AS | N559AS | 851 | SEA | HNL | 360 | 2677 | 10 | 40 | 6.0 |
| 2014 | 12 | 4 | 954 | -6 | 1348 | -17 | HA | N395HA | 29 | SEA | OGG | 333 | 2640 | 9 | 54 | 5.55 |
| 2014 | 6 | 7 | 1823 | -7 | 2112 | -28 | AS | N512AS | 815 | SEA | LIH | 335 | 2701 | 18 | 23 | 5.583333333333333 |
| 2014 | 4 | 30 | 801 | 1 | 1757 | 90 | AS | N407AS | 18 | SEA | MCO | 342 | 2554 | 8 | 1 | 5.7 |
| 2014 | 4 | 1 | 1010 | -5 | 1258 | -17 | HA | N381HA | 25 | PDX | HNL | 328 | 2603 | 10 | 10 | 5.466666666666667 |
| 2014 | 7 | 27 | 925 | 25 | 1232 | 27 | AS | N513AS | 875 | SEA | LIH | 343 | 2701 | 9 | 25 | 5.716666666666667 |
| 2014 | 8 | 18 | 2116 | -4 | 541 | 18 | B6 | N536JB | 264 | SEA | JFK | 304 | 2422 | 21 | 16 | 5.066666666666666 |
| 2014 | 11 | 3 | 840 | 0 | 1703 | -17 | AS | N462AS | 38 | SEA | FLL | 307 | 2717 | 8 | 40 | 5.116666666666666 |
| 2014 | 9 | 2 | 705 | -5 | 950 | -15 | AS | N442AS | 833 | PDX | HNL | 326 | 2603 | 7 | 5 | 5.433333333333334 |
| 2014 | 12 | 28 | 1705 | -10 | 2046 | -42 | AS | N517AS | 863 | PDX | OGG | 324 | 2562 | 17 | 5 | 5.4 |
| 2014 | 4 | 10 | 829 | -1 | 1128 | -7 | AS | N589AS | 861 | SEA | OGG | 332 | 2640 | 8 | 29 | 5.533333333333333 |
| 2014 | 9 | 19 | 1039 | -1 | 1335 | -6 | AS | N431AS | 843 | SEA | KOA | 332 | 2688 | 10 | 39 | 5.533333333333333 |
| 2014 | 9 | 28 | 657 | -3 | 1557 | 34 | DL | N704X | 2588 | SEA | JFK | 311 | 2422 | 6 | 57 | 5.183333333333334 |
| 2014 | 10 | 13 | 853 | -2 | 1217 | 16 | AS | N517AS | 879 | SEA | LIH | 358 | 2701 | 8 | 53 | 5.966666666666667 |
| 2014 | 5 | 2 | 1022 | -3 | 1408 | 38 | AS | N508AS | 843 | SEA | KOA | 378 | 2688 | 10 | 22 | 6.3 |
| 2014 | 3 | 30 | 1646 | 6 | 1938 | -17 | AS | N537AS | 867 | SEA | OGG | 332 | 2640 | 16 | 46 | 5.533333333333333 |
| 2014 | 5 | 29 | 705 | -3 | 1547 | 15 | DL | N3750D | 400 | PDX | JFK | 311 | 2454 | 7 | 5 | 5.183333333333334 |
| 2014 | 3 | 22 | 1007 | -8 | 1305 | -20 | HA | N389HA | 25 | PDX | HNL | 338 | 2603 | 10 | 7 | 5.633333333333334 |
| 2014 | 8 | 16 | 1032 | -8 | 1315 | -20 | HA | N395HA | 25 | PDX | HNL | 304 | 2603 | 10 | 32 | 5.066666666666666 |
| 2014 | 4 | 4 | 1008 | -7 | 1248 | -27 | HA | N385HA | 25 | PDX | HNL | 320 | 2603 | 10 | 8 | 5.333333333333333 |

only showing top 20 rows

**Perform filtering: Using function or SQL Query**

# Spark

## ❖ PySpark Example: Working with csv file

```
spark.sql('SELECT * FROM flights WHERE duration_hours > 5').show()
```

| year | month | day | dep_time | dep_delay | arr_time | arr_delay | carrier | tailnum | flight | origin | dest | air_time | distance | hour | minute | duration_hours |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2014 | 1 | 22 | 1040 | 5 | 1505 | 5 | AS | N559AS | 851 | SEA | HNL | 360 | 2677 | 10 | 40 | 6.0 |
| 2014 | 12 | 4 | 954 | -6 | 1348 | -17 | HA | N395HA | 29 | SEA | OGG | 333 | 2640 | 9 | 54 | 5.55 |
| 2014 | 6 | 7 | 1823 | -7 | 2112 | -28 | AS | N512AS | 815 | SEA | LIH | 335 | 2701 | 18 | 23 | 5.583333333333333 |
| 2014 | 4 | 30 | 801 | 1 | 1757 | 90 | AS | N407AS | 18 | SEA | MCO | 342 | 2554 | 8 | 1 | 5.7 |
| 2014 | 4 | 1 | 1010 | -5 | 1258 | -17 | HA | N381HA | 25 | PDX | HNL | 328 | 2603 | 10 | 10 | 5.466666666666667 |
| 2014 | 7 | 27 | 925 | 25 | 1232 | 27 | AS | N513AS | 875 | SEA | LIH | 343 | 2701 | 9 | 25 | 5.716666666666667 |
| 2014 | 8 | 18 | 2116 | -4 | 541 | 18 | B6 | N536JB | 264 | SEA | JFK | 304 | 2422 | 21 | 16 | 5.066666666666666 |
| 2014 | 11 | 3 | 840 | 0 | 1703 | -17 | AS | N462AS | 38 | SEA | FLL | 307 | 2717 | 8 | 40 | 5.116666666666666 |
| 2014 | 9 | 2 | 705 | -5 | 950 | -15 | AS | N442AS | 833 | PDX | HNL | 326 | 2603 | 7 | 5 | 5.433333333333334 |
| 2014 | 12 | 28 | 1705 | -10 | 2046 | -42 | AS | N517AS | 863 | PDX | OGG | 324 | 2562 | 17 | 5 | 5.4 |
| 2014 | 4 | 10 | 829 | -1 | 1128 | -7 | AS | N589AS | 861 | SEA | OGG | 332 | 2640 | 8 | 29 | 5.533333333333333 |
| 2014 | 9 | 19 | 1039 | -1 | 1335 | -6 | AS | N431AS | 843 | SEA | KOA | 332 | 2688 | 10 | 39 | 5.533333333333333 |
| 2014 | 9 | 28 | 657 | -3 | 1557 | 34 | DL | N704X | 2588 | SEA | JFK | 311 | 2422 | 6 | 57 | 5.183333333333334 |
| 2014 | 10 | 13 | 853 | -2 | 1217 | 16 | AS | N517AS | 879 | SEA | LIH | 358 | 2701 | 8 | 53 | 5.966666666666667 |
| 2014 | 5 | 2 | 1022 | -3 | 1408 | 38 | AS | N508AS | 843 | SEA | KOA | 378 | 2688 | 10 | 22 | 6.3 |
| 2014 | 3 | 30 | 1646 | 6 | 1938 | -17 | AS | N537AS | 867 | SEA | OGG | 332 | 2640 | 16 | 46 | 5.533333333333333 |
| 2014 | 5 | 29 | 705 | -3 | 1547 | 15 | DL | N3750D | 400 | PDX | JFK | 311 | 2454 | 7 | 5 | 5.183333333333334 |
| 2014 | 3 | 22 | 1007 | -8 | 1305 | -20 | HA | N389HA | 25 | PDX | HNL | 338 | 2603 | 10 | 7 | 5.633333333333334 |
| 2014 | 8 | 16 | 1032 | -8 | 1315 | -20 | HA | N395HA | 25 | PDX | HNL | 304 | 2603 | 10 | 32 | 5.066666666666666 |
| 2014 | 4 | 4 | 1008 | -7 | 1248 | -27 | HA | N385HA | 25 | PDX | HNL | 320 | 2603 | 10 | 8 | 5.333333333333333 |

only showing top 20 rows

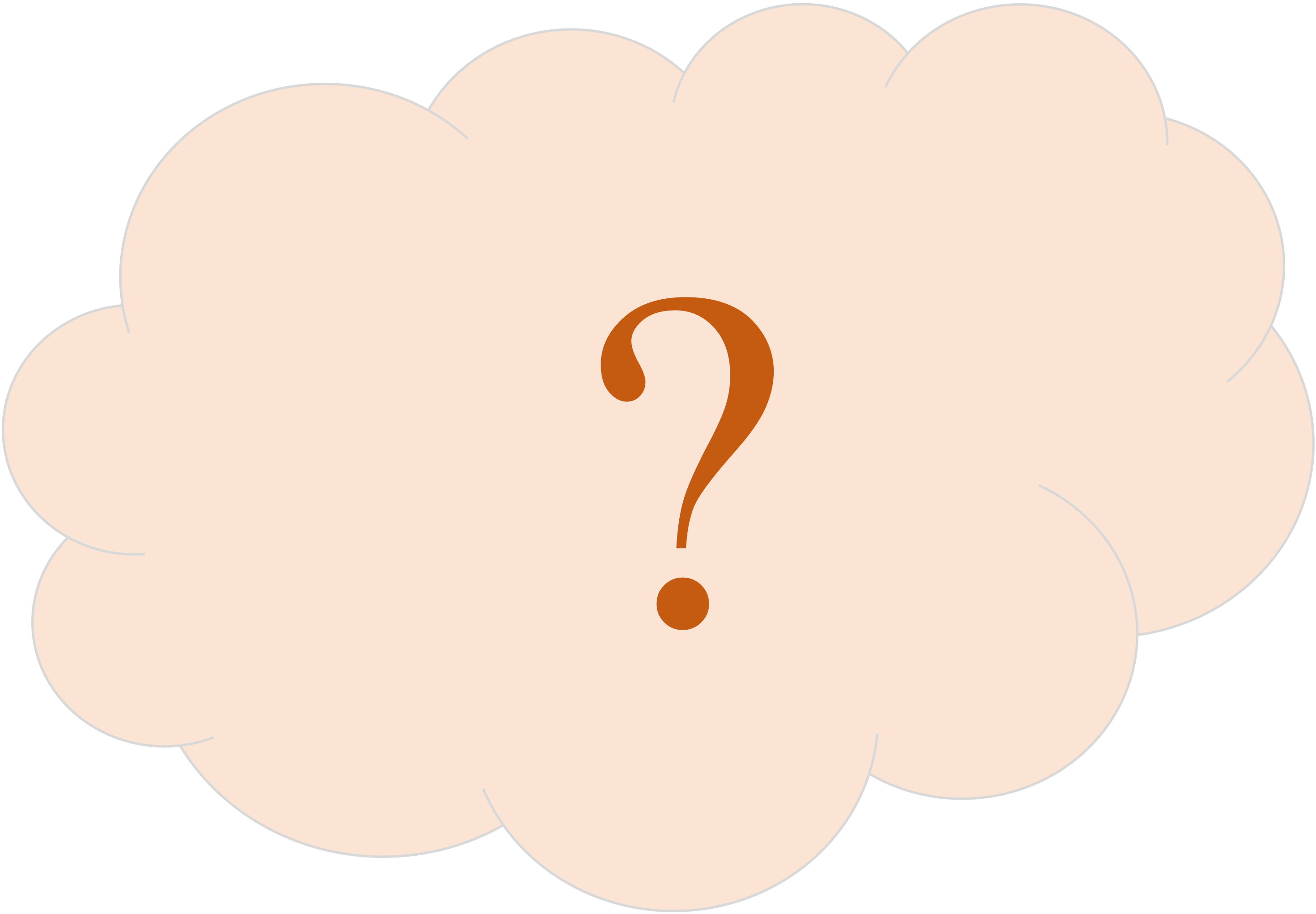
**Perform filtering: Using function or SQL Query**

# Summarization

**In this session, we will discuss about:**

- Introduction to big data.
- Introduction to Hadoop.
- How to install virtual machine.
- How to install and use Hadoop on virtual machine.
- Introduction to spark and pyspark.

# Question




Thank you!